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Nivolumab Products (Opdivo, Opdivo Qvantig, and Opdualag)

Policy History

Last Review 01/14/2025
Effective: 03/06/2015
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Review History

Definitions

Additional Information

Clinical Policy Bulletin Notes

Number: 0892

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As defined in Aetna commercial policies, health care services are not medically necessary when they are more costly than alternative services that are at least as likely to produce equivalent therapeutic or

diagnostic results. Opdivo Qvantig (nivolumab and hyaluronidase-nvhy) is more costly to Aetna than Opdivo IV (intravenous nivolumab). There is a lack of reliable evidence that Opdivo Qvantig (nivolumab and hyaluronidase-nvhy) is superior to the lower cost Opdivo IV (intravenous nivolumab). Therefore, Aetna considers Opdivo Qvantig (nivolumab and hyaluronidase-nvhy) to be medically necessary only for members who have a contraindication or intolerance to the available equivalent alternative: Opdivo IV (intravenous nivolumab).

Policy

Scope of Policy

This Clinical Policy Bulletin addresses nivolumab (Opdivo), nivolumab and hyaluronidase-nvhy (Opdivo Qvantig), and nivolumab and relatlimab-rmbw (Opdualag) for commercial medical plans. For Medicare criteria, see Medicare Part B Criteria.

Note: Requires Precertification:

Precertification of nivolumab (Opdivo), nivolumab and hyaluronidase-nvhy (Opdivo Qvantig), and nivolumab and relatlimab-rmbw (Opdualag) is required of all Aetna participating providers and members in applicable plan designs. For precertification of nivolumab (Opdivo), nivolumab and hyaluronidase-nvhy (Opdivo Qvantig), and nivolumab and relatlimab-rmbw (Opdualag), call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification.

Note: Site of Care Utilization Management Policy applies for nivolumab (Opdivo), nivolumab and hyaluronidase-nvhy (Opdivo Qvantig), and nivolumab and relatlimab-rmbw (Opdualag). For information on site of service, see Utilization Management Policy on Site of Care for Specialty Drug Infusions.

Nivolumab (Opdivo)

I. *Exclusions*

Aetna will not provide coverage for members who have experienced disease progression while on programmed death receptor-1 (PD-1) or programmed death ligand 1 (PD-L1) inhibitor therapy (other

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than when used as second-line or subsequent therapy for metastatic or unresectable melanoma in combination with ipilimumab following progression on single agent anti-PD-1 immunotherapy).

II. *Criteria for Initial Approval*

Aetna considers nivolumab (Opdivo) medically necessary for the treatment of following indications:

A. Ampullary adenocarcinoma - in combination with ipilumab (4 doses of ipilimumab, followed by Opdivo as a single agent) for treatment of progressive, unresectable, or metastatic microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) ampullary adenocarcinoma;
B. Anal carcinoma - as a single agent for subsequent treatment of metastatic anal carcinoma;
C. Anaplastic thyroid carcinoma - as a single agent for treatment of stage IVC anaplastic thyroid carcinoma;
D. Biliary tract cancers (cholangiocarcinoma and gallbladder cancer) - as subsequent treatment in combination with ipilimumab for unresectable or resected gross residual (R2) disease, or metastatic disease that is tumor mutation burden-high (TMB-H);
E. Bone cancer - in combination with ipilimumab for unresectable or metastatic disease when *all* of the following are met:

    1. Disease has tumor mutation burden-high (TMB-H) [greater than or equal to 10 mutations/megabase (mut/Mb)] tumors; and
    2. Disease has progressed following prior treatment and has no satisfactory alternative treatment options;

F. Central nervous system (CNS) brain metastases - when *either* of the following criteria are met:

    1. The requested medication will be used as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) in members with melanoma; *or*
    2. The requested medication will be used as a single agent in members with PD-L1 positive non-small cell lung cancer;

G. Cervical cancer - for subsequent treatment of recurrent or metastatic cervical cancer as a single agent if PD-L1 positive (combined positive score [CPS] greater than or equal to 1);
H. Classical Hodgkin lymphoma (cHL) - when *either* of the following criteria is met:

    1. The requested medication will be used as single agent subsequent therapy and the member has relapsed or refractory disease that was either heavily pretreated or there was a decrease in cardiac function;
    2. The requested medication will be used in combination with brentuximab vedotin or in combination with ICE (ifosfamide, carboplatin, etoposide) for relapsed or refractory disease;
    3. The requested medication will be used as a single agent for disease refractory to at least three lines of prior therapy;

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I. Colorectal cancer, including appendiceal adenocarcinoma and anal adenocarcinoma - for microsatellite-instability high (MSI-H) or mismatch repair deficient (dMMR) or polymerase epsilon/delta (POLE/POLD1 tumors when used as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent);

J. Cutaneous melanoma - in *any* of the following settings:

1. The requested medication will be used as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) for locally recurrent, unresectable, or metastatic disease; *or*
2. The requested medication will be used as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) as adjuvant treatment of stage III or IV disease following complete resection or no evidence of disease; *or*
3. The requested medication will be used as a single agent as adjuvant treatment of stage IIB and IIC disease following complete resection; or
4. The requested medication will be used as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) as neoadjuvant treatment of resectable disease;

K. Endometrial carcinoma - as single agent for subsequent treatment of recurrent or metastatic microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) endometrial carcinoma;

L. Esophageal and esophagogastric junction cancer -

1. For treatment of esophageal or esophagogastric junction cancer in members who are not surgical candidates or have unresectable locally advanced, recurrent, or metastatic disease, when the requested medication will be used in combination with ipilimumab or chemotherapy or will be used as subsequent therapy;
2. For adjuvant treatment of completely resected esophageal or esophagogastric junction cancer with residual pathologic disease;
3. As a single agent or in combination with ipilimumab for neoadjuvant or perioperative treatment of esophageal or esophagogastric junction adenocarcinoma if tumor is microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) and member is medically fit for surgery.

M. Extranodal NK/T-Cell lymphoma - relapsed or refractory disease;

N. Gastric cancer - for treatment of gastric cancer in *any* of the following settings:

1. When the requested medication is being used in members who are not surgical candidates or have unresectable, recurrent, or metastatic disease, when the requested medication will be used in combination with ipilimumab or chemotherapy; *or*
2. When the requested medication will be used as a single agent or in combination with ipilimumab for neoadjuvant or perioperative treatment of gastric adenocarcinoma if tumor is

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ipilimumab for neoadjuvant or perioperative treatment of gastric adenocarcinoma if tumor is microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) and member is medically fit for surgery; *or*

3. When the requested medication will be used in combination with ipilimumab or chemotherapy in members with early stage microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) tumors and completed endoscopic resection; *or*
4. When the requested medication will be used in combination with chemotherapy in members with early stage HER-2 overexpression negative disease and completed endoscopic resection;

O. Gestational trophoblastic neoplasia - as a single agent for treatment of gestational trophoblastic neoplasia for multiagent chemotherapy-resistant disease when *either* of the following criteria is met:

1. Member has recurrent or progressive intermediate trophoblastic tumor (placental site trophoblastic tumor or epithelioid trophoblastic tumor); *or*
2. Member has high-risk disease;

P. Head and neck cancers - for treatment of head and neck cancers in members who meet *either* of the following criteria:

1. For unresectable, recurrent, persistent or metastatic disease; *or*
2. For nasopharyngeal cancer in combination with cisplatin and gemcitabine for unresectable recurrent, persistent or metastatic disease;

Q. Hepatocellular carcinoma - as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) for treatment of hepatocellular carcinoma;

R. Kaposi sarcoma - in combination with ipilimumab for subsequent treatment of relapsed/refractory classic Kaposi sarcoma;

S. Peural or peritoneal mesothelioma - for the treatment of pleural or peritoneal mesothelioma, including pericardial mesothelioma and tunica vaginalis testis mesothelioma, in *either* of the following settings:

1. The requested medication will be used as first line therapy in combination with ipilimumab; *or*
2. The requested medication will be used as subsequent therapy as a single agent or in combination with ipilimumab;

T. Merkel cell carcinoma - for treatment of Merkel cell carcinoma in *any* of the following settings:

1. Metastatic disease; or
2. Neoadjuvant treatment of node positive disease and node negative locally advanced disease when used as a single agent; *or*
3. Unresectable, recurrent, or stage IV disease when used in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent);

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U. Non-small cell lung cancer (NSCLC)

1. For treatment of recurrent, advanced or metastatic non-small cell lung cancer if *either* of the following criteria are met:

   a. There are no EGFR exon 19 deletions or exon 21 L858R mutations or ALK rearrangements (unless testing is not feasible due to insufficient tissue) and the requested medication will be used in a regimen containing ipilimumab; *or*
   b. The requested medication will be used as single agent subsequent therapy; *or*

2. For neoadjuvant treatment (for up to 3 cycles total) of resectable non-small cell lung cancer (NSCLC) in combination with platinum-doublet chemotherapy.

V. Pediatric diffuse high-grade gliomas - for hypermutant tumor pediatric diffuse high-grade glioma as adjuvant treatment or for recurrent or progressive disease;

W. Pediatric primary mediastinal large B-cell lymphoma - as a single agent or in combination with brentuximab vedotin for treatment of relapsed or refractory pediatric primary mediastinal large B-cell lymphoma;

X. Renal cell carcinoma (RCC) - for treatment of relapsed, advanced, or stage IV renal cell carcinoma, in *any* of the following settings:

1. The requested medication will be used as a single agent for clear cell histology as subsequent therapy; *or*
2. The requested medication will be used as a single agent for non-clear cell histology; *or*
3. The requested medication will be used in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) for disease with clear cell histology; *or*

4. The requested medication will be used in combination with cabozantinib;

Y. Small bowel adenocarcinoma - as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) for treatment of advanced or metastatic small bowel adenocarcinoma for microsatellite-instability high (MSI-H), mismatch repair deficient (dMMR) or polymerase epsilon/delta (POLE/POLD1) tumors;

Z. Small cell lung cancer - for subsequent treatment of relapsed or progressive small cell lung cancer as a single agent;

AA. Soft tissue sarcoma - for treatment of soft tissue sarcoma in the following settings:

1. The requested medication will be used as a single agent or in combination with ipilimumab for treatment of extremity/body wall sarcomas, head/neck sarcomas and retroperitoneal/intra-abdominal sarcomas and rhabdomyosarcoma;
2. The requested medication will be used in combination with ipilimumab for the treatment of angiosarcoma;

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AB. Urothelial carcinoma - Bladder cancer

1. In combination with gemcitabine and cisplatin for up to 6 cycles followed by nivolumab maintenance therapy as first line treatment of bladder cancer; or
2. As a single agent for treatment of bladder cancer when *either* of the following conditions are met:

    a. As subsequent therapy for locally advanced, recurrent, persistent, or metastatic disease; *or*
    b. As adjuvant therapy in members who are at high risk of recurrence after undergoing resection;

AC. Urothelial carcinoma - Primary carcinoma of the urethra

1. In combination with gemcitabine and cisplatin for up to 6 cycles followed by nivolumab maintenance therapy as first line treatment of primary carcinoma of the urethra; *or*
2. As a single agent for treatment of primary carcinoma of the urethra when *either* of the following are met:
    a. As subsequent therapy for recurrent, locally advanced or metastatic disease; *or*
    b. As adjuvant therapy in members who are at high risk of recurrence after undergoing resection;

AD. Urothelial carcinoma - Upper genitourinary (GU) tract tumors or urothelial carcinoma of the prostate -

1. In combination with gemcitabine and cisplatin for up to 6 cycles followed by nivolumab maintenance therapy as first line treatment of metastatic upper genitourinary (GU) tract tumors or urothelial carcinoma of the prostate; *or*
2. As a single agent for treatment of upper genitourinary (GU) tract tumors or urothelial carcinoma of the prostate when *either* of the following are met:

    a. As subsequent therapy for locally advanced or metastatic disease; *or*
    b. As adjuvant therapy in members who are at high risk of recurrence after undergoing resection;

AE. Uveal melanoma - as a single agent or in combination with ipilimumab (4 doses of ipilimumab, followed by Opdivo as a single agent) for treatment of uveal melanoma for unresectable or metastatic disease;

AF. Vulvar cancer - for treatment of HPV-related advanced, recurrent, or metastatic vulvar cancer as subsequent therapy as a single agent.

Aetna considers all other indications as not medically necessary; experimental, investigational, or unproven.

III. *Continuation of Therapy*

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III. *Continuation of Therapy*

Aetna considers continuation of nivolumab (Opdivo) therapy medically necessary for the following indications:

A. Adjuvant treatment of melanoma - for continued treatment (up to 12 months total) in members requesting reauthorization for cutaneous melanoma who have not experienced disease recurrence or an unacceptable toxicity; *or*
B. Gastric Cancer, Esophageal Cancer, and Esophagogastric Junction Carcinoma - for continued treatment in members requesting reauthorization for gastric cancer, esophageal cancer, and esophagogastric junction carcinoma when there is no evidence of unacceptable toxicity or disease progression while on the current regimen for the following durations of therapy:

    1. Esophageal squamous cell carcinoma in combination with ipilimumab or chemotherapy for up to 24 months;
    2. Unresectable advanced, recurrent or metastatic esophageal squamous cell carcinoma as a single agent until disease progression or unacceptable toxicity;
    3. Adjuvant treatment of resected esophageal or esophagogastric junction cancer as a single agent for up to 12 months;
    4. Gastric cancer, esophagogastric junction cancer, and esophageal adenocarcinoma in combination with chemotherapy for up to 24 months;
    5. Gastric cancer in members who have completed endoscopic resection for up to 24 months;

C. Non-small cell lung cancer (NSCLC) or pleural mesothelioma - for continued treatment (up to 24 months total when used in combination with ipilimumab) in members requesting reauthorization for non-small cell lung cancer or pleural mesothelioma, including pericardial mesothelioma and tunica vaginalis testis mesothelioma subtypes, when there is no evidence of unacceptable toxicity or disease progression while on the current regimen. Neoadjuvant treatment of NSCLC will be approved for a total of 3 months of therapy; *or*
D. Renal cell carcinoma - for continued treatment (up to 24 months total when used in combination with cabozantinib) in members requesting reauthorization for renal cell carcinoma when there is no evidence of unacceptable toxicity or disease progression while on the current regimen; *or*
E. Urothelial carcinoma -

    1. For continued treatment (up to 12 months total) in members requesting reauthorization for adjuvant treatment of urothelial carcinoma who have not experienced disease recurrence or an unacceptable toxicity; *or*
    2. For continued treatment (up to 24 months total) in members requesting reauthorization for urothelial carcinoma when the requested medication is used in combination with gemcitabine and cisplatin for up to 6 cycles followed by nivolumab maintenance therapy when the member has not experienced disease progression or an unacceptable toxicity.

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F. All other indications - for continued treatment in members requesting reauthorization for all other indications listed in Section II when there is no evidence of unacceptable toxicity or disease progression while on the current regimen.

IV. *Related Policies*

- CPB 0815 - Ipilimumab (Yervoy)
- CPB 0715 - Pharmacogenetic and Pharmacodynamic Testing
- CPB 0890 - Pembrolizumab (Keytruda)

Nivolumab and Hyaluronidase-nvhy (Opdivo Qvantig)

I. *Exclusions*

Aetna considers members not eligible for Opdivo Qvantiq when they have experienced disease progression while on programmed death receptor-1 (PD-1) or programmed death ligand 1 (PD-L1) inhibitor therapy.

II. *Criteria for Initial Approval*

Aetna considers nivolumab and hyaluronidase-nvhy (Opdivo Qvantig) medically necessary for the treatment of the following indications:

A. Colorectal Cancer - As a single agent for treatment of microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) metastatic colorectal cancer when the disease has progressed following treatment with a fluoropyrimidine, oxaliplatin, and irinotecan;

B. Esophageal Carcinoma

1. As a single agent for adjuvant treatment of completely resected esophageal or gastroesophageal junction cancer with residual pathologic disease in members who have received neoadjuvant chemoradiotherapy (CRT); *or*
2. As a single agent for treatment of unresectable advanced, recurrent, or metastatic esophageal squamous cell carcinoma (ESCC) after prior fluoropyrimidine- and platinum-based chemotherapy; *or*
3. In combination with fluoropyrimidine- and platinum-containing chemotherapy, for first-line treatment of unresectable advanced or metastatic ESCC;

C. Gastric Cancer, Gastroesophageal Junction Cancer, Esophageal Adenocarcinoma - In combination with fluoropyrimidine- and platinum-containing chemotherapy for treatment of advanced or metastatic gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma;

D. Head and Neck Cancer - for treatment of squamous cell head and neck carcinoma when *all* of the following criteria are met:

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1. The member has recurrent or metastatic disease; *and*

2. The member had disease progression on or after platinum based chemotherapy; *and*
3. The requested medication will be used as a single agent.

E. Hepatocellular Carcinoma - As a single agent for treatment of hepatocellular carcinoma when member has been previously treated with sorafenib and following treatment with intravenous nivolumab and ipilimumab;

F. Melanoma - for treatment of melanoma when the requested medication will be used in *either* of the following settings:

1. As a single agent for unresectable or metastatic disease; *or*
2. As a single agent for adjuvant treatment of completely resected Stage IIB, Stage IIC, Stage III, or Stage IV disease;

G. Non-Small Cell Lung Cancer (NSCLC)

1. For treatment of metastatic non-small cell lung cancer when *all* of the following criteria are met:

   a. If member has EGFR mutations or ALK rearrangements, disease has progressed on FDA-approved targeted therapy; *and*
   b. The member had disease progression on or after platinum based chemotherapy; *and*
   c. The requested medication will be used as a single agent;

2. For neoadjuvant treatment (for up to 3 cycles total) of resectable non-small cell lung cancer (NSCLC) in combination with platinum based chemotherapy when there are no EGFR mutations or ALK rearrangements;
3. For neoadjuvant and adjuvant treatment of resectable non-small cell lung cancer (NSCLC) when *both* of the following criteria are met:

   a. There are no EGFR mutations or ALK rearrangements; *and*
   b. The requested medication is used in combination with platinum based chemotherapy (for up to 4 cycles total), followed by single agent adjuvant therapy (for up to 13 cycles);

H. Renal Cell Carcinoma - for treatment of renal cell carcinoma when the requested medication will be used in *any* of the following settings:

1. As a single agent for the first-line treatment of intermediate or poor risk advanced renal cell carcinoma (RCC) following treatment with intravenous nivolumab and ipilimumab combination therapy; *or*
2. As a single agent for clear cell histology as subsequent therapy; *or*
3. As a single agent for non-clear cell histology; *or*

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3. As a single agent for non-clear cell histology; *or*
4. In combination with cabozantinib;

I. Urothelial Carcinoma

1. In combination with gemcitabine and cisplatin for up to 6 cycles for first-line treatment of unresectable or metastatic urothelial carinoma; *or*
2. As a single agent for treatment of urothelial carcinoma when *any* of the following conditions are met:

   a. The requested medication will be used to treat locally advanced or metastatic disease in members who:

      i. have disease progression during or following platinum-containing chemotherapy; *or*
      ii. have disease progression within 12 months of neoadjuvant or adjuvant treatment with platinum-containing chemotherapy;

   b. The requested medication will be used as adjuvant therapy in members who are at high risk of recurrence after undergoing radical resection.

Aetna considers all other indications experimental, investigational, or unproven.

III. *Continuation of Therapy*

Aetna considers continuation of nivolumab and hyaluronidase-nvhy (Opdivo Qvantig) therapy medically necessary for the following indications:

A. Adjuvant Treatment of Melanoma - for continued treatment (up to 12 months total) in members requesting reauthorization for cutaneous melanoma who have not experienced disease recurrence or unacceptable toxicity;
B. Esophageal Squamous Cell Carcinoma, Gastric Cancer, Gastroesophageal Junction Cancer, Esophageal Adenocarcinoma

   1. For continued treatment (up to 24 months total) in members requesting reauthorization for esophageal squamous cell carcinoma in combination with fluoropyrimidine- and platinum-containing chemotherapy when there is no evidence of unacceptable toxicity or disease progression while on the current regimen;
   2. For continued treatment (up to 24 months total) in members requesting reauthorization for gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma when there is no evidence of unacceptable toxicity or disease progression while on the current regimen. Adjuvant treatment of resected esophageal or gastroesophageal junction cancer will be limited to 12 months total therapy;

C. Non-Small Cell Lung Cancer

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C. Non-Small Cell Lung Cancer

1. For continued treatment in members requesting reauthorization for metastatic non-small cell lung cancer (NSCLC) when there is no evidence of unacceptable toxicity or disease progression while on the current regimen;
2. For neoadjuvant treatment (up to 3 cycles total) of resectable NSCLC when there is no evidence of unacceptable toxicity or disease progression while on the current regimen;
3. For neoadjuvant and adjuvant treatment of resectable NSCLC (up to 4 cycles in combination with chemotherapy, followed by single agent adjuvant treatment up to 13 cycles) when there is no evidence of unacceptable toxicity or disease progression while on the current regimen;

D. Renal Cell Carcinoma - for continued treatment (up to 24 months total when used as a single agent) in members requesting reauthorization for renal cell carcinoma when there is no evidence of unacceptable toxicity or disease progression while on the current regimen;

E. Urothelial Carcinoma

1. For continued treatment (up to 12 months total) in members requesting reauthorization for adjuvant treatment of urothelial carcinoma who have not experienced disease recurrence or unacceptable toxicity;
2. For continued treatment (up to 24 months total) in members requesting reauthorization for first line treatment of urothelial carcinoma when the requested medication is used in combination with gemcitabine and cisplatin for up to 6 cycles followed by nivolumab and and hyaluronidase-nvhy maintenance therapy when the member has not experienced disease progression or unacceptable toxicity.

F. All Other Indications - for continued treatment in members requesting reauthorization for all other indications listed in Section II when there is no evidence of unacceptable toxicity or disease progression while on the current regimen.

Nivolumab and Relatlimab-rmbw (Opdualag)

I. *Criteria for Initial Approval*

Melanoma

Aetna considers nivolumab and relatlimab-rmbw (Opdualag) medically necessary for treatment of adult members and children, 12 years of age and older weighing at least 40 kg, with unresectable or metastatic melanoma.

Aetna considers all other indications as not medically necessary; experimental, investigational, or unproven.

II. *Continuation of Therapy*

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II. *Continuation of Therapy*

Aetna considers continuation of nivolumab and relatlimab-rmbw (Opdualag) therapy medically necessary in members requesting reauthorization for an indication listed in Section I when there is no evidence of unacceptable toxicity or disease progression while on the current regimen.

Dosage and Administration

Nivolumab (Opdivo)

Nivolumab is available as Opdivo injection and supplied as 40 mg/4 mL (10 mg/mL), 100 mg/10 mL (10 mg/mL), and 240 mg/24 mL (10 mg/mL), solution in a single-dose vial.

The FDA-approved labeling of Opdivo recommends the following dosages administered as an intravenous infusion over 30 minutes until disease progression or unacceptable toxicity:

- Adjuvant treatment of resected esophageal or gastroesophageal junction cancer: 240 mg every 2 weeks or 480 mg every 4 weeks until disease progression or unacceptable toxicity for a total treatment duration of 1 year;
- Classic Hodgkin lymphoma (cHL): Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;
- Colorectal cancer, microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) metastatic:

    - Adult and pediatric persons age 12 years and older weighing 40 kg or greater: 240 mg every 2 weeks or 480 mg every 4 weeks;
    - Pediatrics age 12 years and older weighing less than 40 kg: Opdivo 3 mg/kg every 2 weeks;
    - Adult and pediatric persons 40 kg or greater: Opdivo 3 mg/kg followed by ipilimumab 1 mg/kg on the same day every 3 weeks for 4 doses, then 240 mg every 2 weeks or 480 mg every 4 weeks;

- Esophageal squamous cell carcinoma:

    - Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks up to 2 years;
    - Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks up to 2 years in combination with fluoropyrimidne- and platinum-containing chemotherapy (Opdivo up to 2 years; chemotherapy until disease progression or unacceptable toxicity);
    - Opdivo 3 mg/kg every 2 weeks or 360 mg every 3 weeks with ipilimumab 1 mg/kg every 6 weeks up to 2 years;

- Gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma: 360 mg every 3

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- Gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma: 360 mg every 3 weeks with fluoropyrimidine- and platinum-containing chemotherapy every 3 weeks or 240 mg

  every 2 weeks with fluoropyrimidine- and platinum-containing chemotherapy every 2 weeks up to 2 years;
- Hepatocellular carcinoma:

  - Opdivo with Yervoy (ipilimumab): Opdivo 1 mg/kg, followed by ipilimumab 3 mg/kg on the same day, every 3 weeks for 4 doses, then Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;

- Malignant pleural mesothelioma: Opdivo 360 mg every 3 weeks with ipilimumab 1 mg/kg every 6 weeks. Duration of therapy: in combination with ipilimumab until disease progression, unacceptable toxicity, or up to 2 years in persons without disease progression;
- Melanoma:

  - Adjuvant treatment:

    - Adult and pediatric persons weighing 40 kg or greater: Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks for a maximum of 1 year;
    - Pediatric persons weighing less than 40 kg: 3 mg/kg every 2 weeks or 6 mg/kg every 4 weeks for a maximum of 1 year;

  - Unresectable or metastatic melanoma:

    - Adult and pediatric persons weighing 40 kg or greater: Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;
    - Pediatric persons weighing less than 40 kg: 3 mg/kg every 2 weeks or 6 mg/kg every 4 weeks;
    - Adult and pediatric patients weighing 40 kg or greater: Opdivo 1 mg/kg followed by ipilimumab 3 mg/kg on the same day every 3 weeks for 4 doses, then 240 mg every 2 weeks or 480 mg every 4 weeks;
    - Pediatric persons weighing less than 40 kg: 1 mg/kg followed by ipilimumab 3 mg/kg on the same day every 3 weeks for 4 doses, then 3 mg/kg every 2 weeks or 6 mg/kg every 4 weeks;

- Neoadjuvant treatment of resectable non-small cell lung cancer:

  - Opdivo 360 mg every 3 weeks with platinum-doublet chemotherapy on the same day every 3 weeks. Duration of therapy: in combination with platinum-doublet chemotherapy for 3 cycles.

- Non-small cell lung cancer (NSCLC):

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    - Metastatic NSCLC: Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;

    - Metastatic or recurrent NSCLC: Opdivo 360 mg every 3 weeks with ipilimumab 1 mg/kg every 6 weeks and 2 cycles of histology-based platinum-doublet chemotherapy every 3 weeks up to 2 years in individuals without disease progression;
    - Metastatic NSCLC expressing PD-L1: Opdivo 3 mg/kg every 2 weeks with ipilimumab 1 mg/kg every 6 weeks. Duration of therapy: in combination with ipilimumab until disease progression, unacceptable toxicity, or up to 2 years in persons without disease progression;

- Renal cell carcinoma, advanced:

    - Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;
    - Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks administered in combination with cabozantinib 40 mg orally once daily without food. Duration of therapy for Opdivo: up to 2 years;
    - Opdivo 3 mg/kg followed by ipilimumab 1 mg/kg on the same day, every 3 weeks for a maximum of 4 doses, then Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;

- Squamous cell carcinoma of the head and neck, recurrent or metastatic: Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;
- Urothelial carcinoma:

    - First-line unresectable or metastatic urothelial carcinoma: 360 mg every 3 weeks with cisplatin and gemcitabine on the same day for up to 6 cycles, then 240 mg every 2 weeks or 480 mg every 4 weeks;
    - Previously treated locally advanced or metastatic urothelial carcinoma: Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks;
    - Adjuvant treatment of urothelial carcinoma (UC): Opdivo 240 mg every 2 weeks or 480 mg every 4 weeks for up to 1 year.

Source: Bristol-Myers Squibb, 2024a

Nivolumab and Hyaluronidase-nvhy (Opdivo Qvantig)

Nivolumab and hyaluronidase-nvhy is supplied as Opdivo Qvantig consisting of 600 mg nivolumab and 10,000 units hyaluronidase per 5 mL (120 mg/2,000 units per mL) in a single-dose vial for injection for subcutaneous use only. Administer by subcutaneous injection over 3 to 5 minutes. Opdivo Qvantig is to be administered by a healthcare professional only.

- Renal cell carcinoma

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- 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.

- 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks administered in combination with cabozantinib 40 mg once daily without food.

- Melanoma
  - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- Neoadjuvant and adjuvant treatment of resectable (tumors ≥4 cm or node positive) non-small cell lung cancer
  - 900 mg/15,000 units with platinum-doublet chemotherapy on the same day every 3 weeks for 3 cycles, then single-agent Opdivo Qvantig 1,200 mg/20,000 units every 4 weeks after surgery for up to 13 cycles.
- Metastatic non-small cell lung cancer
  - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- Squamous cell carcinoma of the head and neck
  - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- Urothelial carcinoma
  - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- First-line unresectable or metastatic urothelial carcinoma
  - 900 mg/15,000 units every 3 weeks with cisplatin and gemcitabine on the- same day for up to 6 cycles, then 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- Colorectal cancer
  - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- Hepatocellular carcinoma
  - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
- Esophageal cancer

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- Esophageal cancer
    - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks.
    - 600 mg/10,000 units every 2 weeks or 1,200 mg/20,000 units every 4 weeks administered in combination with fluoropyrimidine- and platinumcontaining chemotherapy.
- Gastric Cancer, Gastroesophageal Junction Cancer, and Esophageal Adenocarcinoma
    - 600 mg/10,000 units every 2 weeks in combination with fluoropyrimidine- and platinum-containing chemotherapy every 2 weeks.
    - 900 mg/15,000 units every 3 weeks with fluoropyrimidine- and platinumcontaining chemotherapy every 3 weeks.
- See full prescribing information for preparation and administration instructions and dosage modifications for adverse reactions.

Source: Bristol-Myers Squibb, 2024b

Nivolumab and Relatlimab-rmbw (Opdualag)

Nivolumab and relatlimab-rmbw is available as Opdualag and supplied as 240 mg of nivolumab and 80 mg of relatlimab per 20 mL (12 mg and 4 mg per mL) in a single-dose vial.

The FDA-approved labeling of Opdualag recommends the following dosages administered as an intravenous infusion over 30 minutes until disease progression or unacceptable toxicity:

- Melanoma: 480 mg nivolumab and 160 mg relatlimab intravenously every 4 weeks.

Source: Bristol-Myers Squibb; 2023

Not Medically Necessary

Aetna considers nivolumab (Opdivo) not medically necessary for members who have experienced disease progression while on programmed death receptor-1 (PD-1) or programmed death ligand 1 (PD-L1) inhibitor therapy (other than when used as second-line or subsequent therapy for metastatic or unresectable melanoma in combination with ipilimumab following progression on single agent anti-PD-1 immunotherapy).

Experimental, Investigational, or Unproven

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Aetna considers nivolumab experimental, investigational, or unproven for all other indications including the following (not an all-inclusive list):

- Astrocytoma
- Breast cancer
- Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma
- Glioblastoma
- Non-Hodgkin lymphoma (excluding extranodal NK/T-cell lymphoma (nasal type) and pediatric primary mediastinal large B-cell lymphoma)
- Ovarian cancer
- Pancreatic cancer
- Pancreatic neuroendocrine cancer.

Aetna considers nivolumab plus ipilimumab experimental, investigational, or unproven for the treatment of urethral/urothelial carcinoma of the penis.

Aetna considers nivolumab plus regorafenib experimental, investigational, or unproven for the treatment of metastatic rectal adenocarcinoma.

CPT Codes / HCPCS Codes / ICD-10 Codes

Nivolumab (Opdivo):

Code	Code Description
Other CPT codes related to the CPB:	
***RET and ROS1 gene testing*- no specific code**	
81210	BRAF (B-Raf proto-oncogene, serine/threonine kinase) (eg, colon cancer, melanoma), gene analysis, V600 variant(s)
81235	EGFR (epidermal growth factor receptor) (eg, non-small cell lung cancer) gene analysis, common variants (eg, exon 19 LREA deletion, L858R, T790M, G719A, G719S, L861Q)
96413	Chemotherapy administration, intravenous infusion technique; up to 1 hour, single or initial substance/drug
HCPCS codes covered if selection criteria are met:	
J9299	Injection, nivolumab, 1 mg
Other HCPCS codes related to the CPB:	
***Ibrutinib, Cabozantinib, Regorafenib, Fluoropyrimidine* - no specific code**	

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C9257	Injection, bevacizumab, 0.25 mg
J8520	Capecitabine, oral, 150 mg

Code	Code Description
J8521	Capecitabine, oral, 500 mg
J9035	Injection, bevacizumab, 10 mg
J9042	Injection, brentuximab vedotin, 1 mg
J9045	Injection, carboplatin, 50 mg
J9060	Injection, cisplatin, powder or solution, 10 mg
J9181	Injection, etoposide, 10 mg
J9190	Injection, fluorouracil, 500 mg
J9196	Injection, gemcitabine hydrochloride (accord), not therapeutically equivalent to j9201, 200 mg
J9200	Injection, floxuridine, 500 mg
J9201	Injection, gemcitabine hydrochloride, not otherwise specified, 200 mg
J9206	Injection, irinotecan, 20 mg
J9208	Injection, ifosfamide, 1 gram
J9228	Injection, ipilimumab, 1 mg
J9263	Injection, oxaliplatin, 0.5 mg
Q5107	Injection, bevacizumab-awwb, biosimilar, (mvasi), 10 mg
ICD-10 codes covered if selection criteria are met:	
C00.0 - C05.1, C06.0 - C06.9, C09.0 - C09.9, C10.3, C11.0 - C11.9, C12, C13.0 - C13.9, C30.0, C31.0 - C31.9, C32.0- C32.9	Malignant neoplasms of lip, tongue, gum, floor of mouth, palate, other and unspecified parts of mouth, posterior was of oropharynx, nasopharynx, pyriform sinus, hypopharynx, nasal cavity, accessary sinuses, larynx
C15.3 - C15.9	Malignant neoplasm of esophagus
C16.0 - C16.9	Malignant neoplasm of stomach
C17.0 - C17.9	Malignant neoplasm of small intestine
C18.0 - C20	Malignant neoplasm of colon and rectum
C21.0 - C21.8	Malignant neoplasm of anus and anal canal [adenocarcinoma]
C22.0 - C22.9	Malignant neoplasm of liver and intrahepatic bile ducts
C23	Malignant neoplasm of gallbladder

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C24.1	Malignant neoplasm of ampulla of Vater [tunica vaginalis mesothelioma]
C33 - C34.92	Malignant neoplasm of trachea, bronchus and lung

Code	Code Description
C4A.0 - C4A.9	Merkel cell carcinoma
C40.00 - C41.9	Malignant neoplasm of bone and articular cartilage
C43.0 - C43.9	Malignant melanoma of skin
C44.42	Squamous cell carcinoma of skin of scalp and neck
C44.520	Squamous cell carcinoma of anal skin
C45.0	Mesothelioma of pleura
C45.1	Mesothelioma of peritoneum [tunica vaginalis mesothelioma]
C45.2	Mesothelioma of pericardium [tunica vaginalis mesothelioma]
C45.7	Mesothelioma of other sites [tunica vaginalis mesothelioma]
C49.1 - C49.9	Malignant neoplasm of other connective and soft tissue
C51.0 - C51.9	Malignant neoplasm of vulva
C53.0 - C53.9	Malignant neoplasm of cervix uteri
C54.1	Malignant neoplasm of endometrium
C58.0	Malignant neoplasm of placenta
C61	Malignant neoplasm of prostate
C64.1 - C64.9	Malignant neoplasm of kidney, except pelvis
C65.1 - C65.9	Malignant neoplasm of renal pelvis
C66.1 - C66.9	Malignant neoplasm of ureter
C67.0 - C67.9	Malignant neoplasm of bladder
C68.0	Malignant neoplasm of urethra
C69.30 - C69.32	Malignant neoplasm of choroid [uveal melanoma]
C73	Malignant neoplasm of thyroid gland [Anaplastic thyroid carcinoma]
C76.0	Malignant neoplasm of head, face, and neck
C77.0 - C77.9	Secondary and unspecified malignant neoplasm of lymph nodes
C79.31 - C79.32	Secondary malignant neoplasm of brain and cerebral meninges
C81.10 - C81.79	Hodgkin lymphoma [adults age 18 and older only]
C83.00 - C83.09	Small cell B-cell lymphoma [small lymphocytic lymphoma]
C84.90 - C84.99	Mature T/NK-cell lymphomas, unspecified
C84.Z0 - C84.Z9	Other mature T/NK-cell lymphomas
C85.20 - C85.29	Mediastinal (thymic) large B-cell lymphoma
C86.0	Extranodal NK/T-cell lymphoma, nasal type

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C86.0	Extranodal NK/T-cell lymphoma, nasal type
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	

Code	Code Description
C25.0 - C25.9	Malignant neoplasm of pancreas
C41.0 - C41.9, C44.00 - C46.9	Malignant neoplasm of bone of skull and face
C50.011 - C50.929	Malignant neoplasm of breast
C56.1 - C56.9	Malignant neoplasm of ovary
C60.9	Malignant neoplasm of penis, unspecified [Urethral/urothelial carcinoma of the penis]
C71.0 - C71.9	Malignant neoplasm of brain [glioblastoma][Astrocytoma]
C78.5	Secondary malignant neoplasm of large intestine and rectum
C82.00 - C91.92	Malignant neoplasm of lymphoid, hematopoietic and related tissue [non-Hodgkin's lymphoma]
Nivolumab and Relatlimab-rmbw (Opdualag):	
Other CPT codes related to the CPB:	
96365 - 96368	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug)
96413 - 96417	Chemotherapy administration, intravenous infusion technique
HCPCS codes covered if selection criteria are met:	
J9298	Injection, nivolumab and relatlimab-rmbw, 3 mg/1 mg
ICD-10 codes covered if selection criteria are met:	
C43.0 - C43.9	Malignant melanoma of skin

Background

Nivolumab (Opdivo)

U.S. Food and Drug Administration (FDA)-Approved Indications

- Unresectable or Metastatic Melanoma: Opdivo, as a single agent or in combination with ipilimumab, is indicated for the treatment of adult and pediatric patients 12 years and older with unresectable or metastatic melanoma.
- Adjuvant Treatment of Melanoma: Opdivo is indicated for the adjuvant treatment of adult and pediatric patients 12 years and older with completely resected stage IIB, stage IIC, stage III, or

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stage IV melanoma.

- Metastatic Non-Small Cell Lung Cancer:

  - Opdivo, in combination with ipilimumab, is indicated for the first-line treatment of adult patients with metastatic non-small cell lung cancer (NSCLC) whose tumors express PD-L1 (≥1%) as determined by an FDA-approved test, with no EGFR or ALK genomic tumor aberrations;
  - Opdivo, in combination with ipilimumab and 2 cycles of platinum-doublet chemotherapy, is indicated for the first-line treatment of adult patients with metastatic or recurrent NSCLC, with no EGFR or ALK genomic tumor aberrations;
  - Opdivo is indicated for the treatment of adult patients with metastatic NSCLC with progression on or after platinum-based chemotherapy. Patients with EGFR or ALK genomic tumor aberrations should have disease progression on FDA-approved therapy for these aberrations prior to receiving Opdivo.

- Neoadjuvant Treatment of Resectable Non-Small Cell Lung Cancer: Opdivo, in combination with platinum-doublet chemotherapy, is indicated as neoadjuvant treatment of adult patients with resectable (tumors ≥4 cm or node positive) non-small cell lung cancer (NSCLC).
- Malignant Pleural Mesothelioma: Opdivo, in combination with ipilimumab, is indicated for the first-line treatment of adult patients with unresectable malignant pleural mesothelioma.
- Advanced Renal Cell Carcinoma:

  - Opdivo as a single agent is indicated for the treatment of adult patients with advanced renal cell carcinoma (RCC) who have received prior anti-angiogenic therapy;
  - Opdivo, in combination with ipilimumab, is indicated for the first-line treatment of adult patients with intermediate or poor risk advanced RCC;
  - Opdivo, in combination with cabozantinib, is indicated for the first-line treatment of adult patients with advanced RCC.

- Classical Hodgkin Lymphoma: Opdivo is indicated for the treatment of adult patients with classical Hodgkin lymphoma (cHL) that has relapsed or progressed after:

  - Autologous hematopoietic stem cell transplantation (HSCT) and brentuximab vedotin; or
  - Three or more lines of systemic therapy that includes autologous HSCT.

- Squamous Cell Carcinoma of the Head and Neck: Opdivo is indicated for the treatment of adult patients with recurrent or metastatic squamous cell carcinoma of the head and neck (SCCHN) with disease progression on or after platinum-based therapy.
- Urothelial Carcinoma:

  - Opdivo is indicated for the adjuvant treatment of adult patients with urothelial carcinoma

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- Opdivo is indicated for the adjuvant treatment of adult patients with urothelial carcinoma (UC) who are at high risk of recurrence after undergoing radical resection of UC.

- Opdivo, in combination with cisplatin and gemcitabine, is indicated for the first-line treatment of adult patients with unresectable or metastatic urothelial carcinoma.
- Opdivo is indicated for the treatment of adult patients with locally advanced or metastatic urothelial carcinoma who:

  - Have disease progression during or following platinum-containing chemotherapy;
  - Have disease progression within 12 months of neoadjuvant or adjuvant treatment with platinum-containing chemotherapy.

- Microsatellite Instability-High or Mismatch Repair Deficient Metastatic Colorectal Cancer: Opdivo, as a single agent or in combination with ipilimumab, is indicated for the treatment of adult and pediatric patients 12 years and older with microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) metastatic colorectal cancer (CRC) that has progressed following treatment with a fluoropyrimidine, oxaliplatin, and irinotecan.
- Hepatocellular Carcinoma: Opdivo, in combination with ipilimumab, is indicated for the treatment of adult patients with hepatocellular carcinoma (HCC) who have been previously treated with sorafenib.
- Esophageal Carcinoma:

  - Opdivo is indicated for the adjuvant treatment of completely resected esophageal or gastroesophageal junction cancer with residual pathologic disease in adult patients who have received neoadjuvant chemoradiotherapy (CRT);
  - Opdivo, in combination with fluoropyrimidine- and platinum-containing chemotherapy, is indicated for the first-line treatment of adult patients with unresectable advanced or metastatic esophageal squamous cell carcinoma (ESCC).
  - Opdivo, in combination with ipilimumab, is indicated for the first-line treatment of adult patients with unresectable advanced or metastatic esophageal squamous cell carcinoma (ESCC).
  - Opdivo is indicated for the treatment of adult patients with unresectable advanced, recurrent or metastatic esophageal squamous cell carcinoma (ESCC) after prior fluoropyrimidine- and platinum-based chemotherapy.

- Gastric Cancer, Gastroesophageal Junction Cancer, Esophageal Adenocarcinoma: Opdivo, in combination with fluoropyrimidine- and platinum-containing chemotherapy, is indicated for the treatment of adult patients with advanced or metastatic gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma.

Compendial Uses

- Cutaneous melanoma

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- Cutaneous melanoma
- Non-small cell lung cancer
- Renal cell carcinoma
- Classical Hodgkin lymphoma
- Head and neck cancers
- Urothelial carcinoma (bladder cancer, primary carcinoma of the urethra, upper genitourinary tract tumors, and urothelial carcinoma of the prostate)
- Colorectal cancer, including appendiceal adenocarcinoma and anal adenocarcinoma
- Hepatocellular carcinoma
- Uveal melanoma
- Anal carcinoma
- Merkel cell carcinoma
- Central nervous system (CNS) brain metastases
- Gestational trophoblastic neoplasia
- Pleural mesothelioma
- Peritoneal mesothelioma
- Small bowel adenocarcinoma
- Ampullary adenocarcinoma
- Extranodal NK/T-cell lymphoma
- Endometrial carcinoma
- Vulvar cancer
- Gastric cancer
- Esophageal/esophagogastric junction cancers
- Small cell lung cancer
- Cervical cancer
- Pediatric diffuse high-grade gliomas
- Pediatric primary mediastinal large B-cell lymphoma
- Kaposi sarcoma
- Bone cancer
- Biliary tract cancers
    - Cholangiocarcinoma
    - Gallbladder cancer
- Soft tissue sarcoma
    - Extremity/body wall sarcoma
    - Head/neck sarcoma
    - Retroperitoneal/intra-abdominal sarcoma
    - Rhabdomyosarcoma
    - Angiosarcoma
- Anaplastic thyroid carcinoma

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- Anaplastic thyroid carcinoma

Opdivo (nivolumab) is a programmed death receptor-1 (PD-1) blocking antibody. Binding of the PD-1 ligands, PD-L1 and PD-L2, to the PD-1 receptor found on T cells, inhibits T-cell proliferation and cytokine production. Upregulation of PD-1 ligands occurs in some tumors and signaling through this pathway can contribute to inhibition of active T-cell immune surveillance of tumors. Nivolumab is a human immunoglobulin G4 (IgG4) monoclonal antibody that binds to the PD-1 receptor and blocks its interaction with PD-L1 and PD-L2, releasing PD-1 pathway-mediated inhibition of the immune response, including the anti-tumor immune response. In syngeneic mouse tumor models, blocking PD-1 activity resulted in decreased tumor growth (BMS, 2021).

Labeled warnings and precautions include immune-mediated adverse reactions, infusion-related reactions, complications of allogeneic HSCT and embryo-fetal toxicity. Treatment of patients with multiple myeloma with a PD-1 or PD-L1 blocking antibody in combination with a thalidomide analogue plus dexamethasone is not recommended outside of controlled clinical trials (BMS, 2021).

The most common adverse reactions, greater or equal to 20 percent, include:

- As a single agent: fatigue, rash, musculoskeletal pain, pruritus, diarrhea, nausea, asthenia, cough, dyspnea, constipation, decreased appetite, back pain, arthralgia, upper respiratory tract infection, pyrexia, headache, abdominal pain, and vomiting;
- In combination with ipilimumab: fatigue, diarrhea, rash, pruritus, nausea, musculoskeletal pain, pyrexia, cough, decreased appetite, vomiting, abdominal pain, dyspnea, upper respiratory tract infection, arthralgia, headache, hypothyroidism, decreased weight, and dizziness;
- In combination with ipilimumab and platinum-doublet chemotherapy: fatigue, musculoskeletal pain, nausea, diarrhea, rash, decreased appetite, constipation, and pruritus;
- In combination with cabozantinib: diarrhea, fatigue, hepatotoxicity, palmar-plantar erythrodysaesthesia syndrome, stomatitis, rash, hypertension, hypothyroidism, musculoskeletal pain, decreased appetite, nausea, dysgeusia, abdominal pain, cough, and upper respiratory tract infection.

Anal Cell Carcinoma

Morris et al (2017) stated squamous cell carcinoma of the anal canal (SCCA) is a rare malignancy associated with infection by human papillomavirus (HPV). No consensus treatment approach exists for the treatment of metastatic disease. Because intratumoral HPV oncoproteins upregulate immune checkpoint proteins such as PD-1 to evade immune-mediated cytotoxicity, this was a trial of the anti-PD-1 antibody nivolumab for patients with metastatic SCCA. This was a single-arm, multicenter, phase 2 trial at ten academic centers in the USA. The study enrolled patients with treatment-refractory metastatic SCCA, who were given nivolumab every 2 weeks (3 mg/kg). The primary endpoint was response according to Response Evaluation Criteria in Solid Tumors, version 1.1, in the intention-to-treat population. At the time of data cutoff, the study was ongoing, with patients continuing to receive

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treat population. At the time of data cutoff, the study was ongoing, with patients continuing to receive treatment. The study screened 39 patients, of whom 37 were enrolled and received at least one dose of nivolumab. Among the 37 patients, nine (24% [95% CI 15-33]) had responses. There were two complete responses and seven partial responses. Grade 3 adverse events were anemia (n=2), fatigue (n=1), rash (n=1), and hypothyroidism (n=1). No serious adverse events were reported. The authors stated to their knowledge, this is the first completed phase 2 trial of immunotherapy for SCCA. Nivolumab is well tolerated and effective as a monotherapy for patients with metastatic SCCA. Immune checkpoint blockade appears to be a promising approach for patients with this orphan disease.

Breast Cancer

Janakiram et al (2012) noted that co-stimulation and co-inhibition generated by the B7 family and their receptor CD28 family have key roles in regulating T-lymphocyte activation and tolerance. These pathways are attractive therapeutic targets for human cancers including breast cancer. Gene polymorphisms of B7x (B7-H4/B7S1), PD-1 (CD279), and CTLA-4 (CD152) are associated with increased risk of developing breast cancer although the underlying mechanisms are unclear. In human breast cancer microenvironment, up-regulation of co-inhibitory B7/CD28 members B7x, B7-H3 (CD276), and PD-L1 (B7-H1/CD274) on tumor cells as well as PD-1 and PD-L1 on tumor-infiltrating immune cells are emerging as immune evasion pathways. Chemotherapy can affect the expression of these molecules, and therefore may dampen the immune response against breast cancer. Immunotherapy targeting T-cell co-inhibition as monotherapy or combined with standard therapies are in early stages of clinical development, but hold great promise for treatment of human breast cancer.

Central Nervous System - Brain Metastases in Patients With Melanoma

Tawbi et al (2018) stated brain metastases are a common cause of disabling neurologic complications and death in patients with metastatic melanoma. Previous studies of nivolumab combined with ipilimumab in metastatic melanoma have excluded patients with untreated brain metastases. The authors evaluated the efficacy and safety of nivolumab plus ipilimumab in patients with melanoma who had untreated brain metastases. In this open-label, multicenter, phase 2 study, patients with metastatic melanoma and at least one measurable, nonirradiated brain metastasis (tumor diameter, 0.5 to 3 cm) and no neurologic symptoms received nivolumab (1 mg per kilogram of body weight) plus ipilimumab (3 mg per kilogram) every 3 weeks for up to four doses, followed by nivolumab (3 mg per kilogram) every 2 weeks until progression or unacceptable toxic effects. The primary end point was the rate of intracranial clinical benefit, defined as the percentage of patients who had stable disease for at least 6 months, complete response, or partial response. Among 94 patients with a median follow-up of 14.0 months, the rate of intracranial clinical benefit was 57% (95% confidence interval [CI], 47 to 68); the rate of complete response was 26%, the rate of partial response was 30%, and the rate of stable disease for at least 6 months was 2%. The rate of extracranial clinical benefit was 56% (95% CI, 46 to 67). Treatment-related grade 3 or 4 adverse events were reported in 55% of patients, including events involving the central nervous system in 7%. One patient died from immune-related myocarditis. The safety profile of the regimen was similar to that reported in patients with melanoma who do not have brain metastases. The authors concluded that nivolumab combined with ipilimumab had clinically

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brain metastases. The authors concluded that nivolumab combined with ipilimumab had clinically meaningful intracranial efficacy, concordant with extracranial activity, in patients with melanoma who had untreated brain metastases. (CheckMate 204 ClinicalTrials.gov number, NCT02320058).
Long et al (2018) stated nivolumab monotherapy and combination nivolumab plus ipilimumab increase proportions of patients achieving a response and survival versus ipilimumab in patients with metastatic melanoma; however, efficacy in active brain metastases is unknown. The authors aimed to establish the efficacy and safety of nivolumab alone or in combination with ipilimumab in patients with active melanoma brain metastases. This multicentre open-label randomised phase 2 trial was done at four sites in Australia, in three cohorts of immunotherapy-naive patients aged 18 years or older with melanoma brain metastases. Patients with asymptomatic brain metastases with no previous local brain therapy were randomly assigned using the biased coin minimization method, stratified by site, in a 30:24 ratio (after a safety run-in of six patients) to cohort A (nivolumab plus ipilimumab) or cohort B (nivolumab). Patients with brain metastases in whom local therapy had failed, or who had neurological symptoms, or leptomeningeal disease were enrolled in non-randomised cohort C (nivolumab). Patients in cohort A received intravenous nivolumab 1 mg/kg combined with ipilimumab 3 mg/kg every 3 weeks for four doses, then nivolumab 3 mg/kg every 2 weeks; patients in cohort B or cohort C received intravenous nivolumab 3 mg/kg every 2 weeks. The primary endpoint was intracranial response from week 12. Primary and safety analyses were done on an intention-to-treat basis in all patients who received at least one dose of the study drug. Between Nov 4, 2014, and April 21, 2017, 79 patients were enrolled; 36 in cohort A, 27 in cohort B, and 16 in cohort C. One patient in cohort A and two in cohort B were found to be ineligible and excluded from the study before receiving the study drug. At the data cutoff (Aug 28, 2017), with a median follow up of 17 months (IQR 8-25), intracranial responses were achieved by 16 (46%; 95% CI 29-63) of 35 patients in cohort A, five (20%; 7-41) of 25 in cohort B, and one (6%; 0-30) of 16 in cohort C. Intracranial complete responses occurred in six (17%) patients in cohort A, three (12%) in cohort B, and none in cohort C. Treatment-related adverse events occurred in 34 (97%) of 35 patients in cohort A, 17 (68%) of 25 in cohort B, and eight (50%) of 16 in cohort C. Grade 3 or 4 treatment-related adverse events occurred in 19 (54%) patients in cohort A, four (16%) in cohort B, and two (13%) in cohort C. No treatment-related deaths occurred. The authors concluded that nivolumab combined with ipilimumab and nivolumab monotherapy are active in melanoma brain metastases. A high proportion of patients achieved an intracranial response with the combination. Thus, nivolumab combined with ipilimumab should be considered as a first-line therapy for patients with asymptomatic untreated brain metastases. This study is ongoing for the final survival analysis (ClinicalTrials.gov number NCT02374242).

Cervical Cancer

There are currently 2 clinical trials (one is a phase I/II; the other is a phase II) on the use of nivolumab for the treatment of cervical cancer.

A phase I/II study -- “A Study to Investigate the Safety and Efficacy of Nivolumab in Virus-associated Tumors (CheckMate358)” (NCT02488759) is not yet open for participant recruitment. The purpose of this study to investigate the safety and efficacy of using nivolumab to treat subjects who have virus-associated tumors. Certain viruses that infect human cells have been known to play a role in tumor

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associated tumors. Certain viruses that infect human cells have been known to play a role in tumor formation and growth. This study will investigate the effects of the study drug nivolumab in human subjects who have the following types of viral-associated tumors: Gastric Cancer, nasopharyngeal carcinoma, cervical cancer, vaginal cancer, vulvar cancer, squamous cell carcinoma of the head and neck, and Merkel cell carcinoma. (Last verified June 2015).

A phase II study -- "Nivolumab in Treating Patients with Persistent, Recurrent, or Metastatic Cervical Cancer" (NCT02257528) is currently recruiting participants. This phase II trial studies the side effects and how well nivolumab works in treating patients with cervical cancer that has grown, come back, or spread to other places in the body. Monoclonal antibodies, such as nivolumab, may block tumor growth in different ways by targeting certain cells. (Last verified June 2015).

Colorectal Cancer

On July 31, 2017, the U.S. Food and Drug Administration granted accelerated approval to nivolumab for the treatment of patients 12 years and older with mismatch repair deficient (dMMR) and microsatellite instability high (MSI-H) metastatic colorectal cancer that has progressed following treatment with a fluoropyrimidine, oxaliplatin, and irinotecan. The approval was based on data from Study CA209142 (CHECKMATE 142; NCT 02060188), a multicenter, open-label, single arm study conducted in 53 patients with locally determined dMMR or MSI-H metastatic colorectal cancer (CRC) who had disease progression during, after, or were intolerant to prior treatment with fluoropyrimidine-, oxaliplatin-, and irinotecan-based chemotherapy. This was a subset of the 74 patients who received at least one prior regimen for treatment of metastatic disease containing a fluoropyrimidine with oxaliplatin or irinotecan for treatment of metastatic disease. All patients received nivolumab 3 mg/kg by intravenous infusion every 2 weeks until unacceptable toxicity or radiographic progression. The objective response rate (ORR) as assessed by independent radiographic review committee using RECIST 1.1 was 28% (n=15) (95% CI: 17, 42) in the 53 patients who received prior fluoropyrimidine, oxaliplatin, and irinotecan. Responses lasted 6 or more months for 67% (95% CI: 38, 88) of patients. There was 1 complete response and 14 partial responses. The ORR was 32% (n=24) (95% CI: 22, 44) among the 74 patients in the overall population. Trials of nivolumab have not been conducted in pediatric patients. Efficacy for adolescent patients (12 years and older) with MSI-H or dMMR metastatic CRC is extrapolated from the results in the respective adult population. The most common adverse reactions (≥20%) to nivolumab as a single agent include fatigue, rash, musculoskeletal pain, pruritus, diarrhea, nausea, asthenia, cough, dyspnea, constipation, decreased appetite, back pain, arthralgia, upper respiratory tract infection, pyrexia. The recommended nivolumab dose for this indication is 240 mg every 2 weeks.

Colorectal Cancer (dMMR/MSI-H) - Combined Ipilimumab and Nivolumab

On July 11, 2018, Bristol-Meyers Squibb announced the U.S. Food and Drug Administration (FDA) approval of Opdivo (nivolumab) 3 mg/kg plus low-dose Yervoy (ipilimumab) 1 mg/kg (injections for intravenous use) for the treatment of patients 12 years and older with microsatellite instability high (MSI-H) or mismatch repair deficient (dMMR) metastatic colorectal cancer (mCRC) that has progressed following treatment with a fluoropyrimidine, oxaliplatin and irinotecan (BMS, 2018).

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following treatment with a fluoropyrimidine, oxaliplatin and irinotecan (BMS, 2018).

Approval for this indication has been granted under the FDA's accelerated approval which was based on overall response rate (ORR) and duration of response (DOR) from the ongoing Phase 2, multicenter, non-randomized, multiple-parallel cohort, open-label CheckMate -142 trial, which demonstrated an ORR of 46% (95% CI: 35-58; n = 38/82). The CheckMate -142 trial enrolled MSI-H/dMMR mCRC patients who had received at least one prior line of therapy for metastatic disease, and efficacy was analyzed for both patients who had received prior treatment with a fluoropyrimidine, oxaliplatin and irinotecan (82 of the total 119 patients) as well as for all enrolled patients.

Among all enrolled patients, 49% (95% CI: 39-58; n = 58/119) responded to treatment with Opdivo + Yervoy; 4.2% (n = 5/119) experienced a complete response, while 45% (n = 53/119) experienced a partial response. Among the 58 responders, the median DOR was not reached (range: 1.9-23.2+ months); 83% of those patients had responses of six months or longer, and 19% had responses of 12 months or longer. In the combination cohort, 51 of 58 responders were ongoing at the time of database lock; 78% of these ongoing responders had not reached 12 months of follow-up from the date of onset of response. The application was granted Priority Review and Breakthrough Therapy Designation by the FDA. Continued approval for this indication may be contingent upon verification and description of clinical benefit in confirmatory trials (BMS, 2018).

Overman et al. (2018) state nivolumab plus ipilimumab may provide clinical benefit in previously treated patients with DNA mismatch repair-deficient (dMMR)/microsatellite instability-high (MSI-H) metastatic colorectal cancer (mCRC). In the CheckMate-142 trial, 119 patients received nivolumab 3 mg/kg plus ipilimumab 1 mg/kg once every 3 weeks (four doses) followed by nivolumab 3 mg/kg once every 2 weeks. Primary end point was investigator-assessed ORR. Out of 119 patients, 76% had received ≥ two prior systemic therapies. At median follow-up of 13.4 months, investigator-assessed ORR was 55% and disease control rate for ≥ 12 weeks was achieved in 80% of patients. Median duration of response was not reached; most responses (94%) were ongoing at data cutoff. Progression-free survival rates were 76% (9 months) and 71% (12 months); respective OS rates were 87% and 85%. Statistically significant and clinically meaningful improvements were observed in patient-reported outcomes, including functioning, symptoms, and quality of life. The authors concluded that nivolumab plus ipilimumab demonstrated high response rates, encouraging progression-free survival and OS at 12 months, manageable safety, and meaningful improvements in key patient-reported outcomes. Indirect comparisons suggest combination therapy provides improved efficacy relative to anti-programmed death-1 monotherapy and has a favorable benefit-risk profile. Nivolumab plus ipilimumab provides a promising new treatment option for patients with dMMR/MSI-H mCRC. Evaluation of nivolumab plus ipilimumab as a first line therapy (phase II) in patients with dMMR/MSI-H mCRC is ongoing.

The most frequent serious adverse reactions (greater than 2% of patients) were colitis/diarrhea, hepatic events, abdominal pain, acute kidney injury, pyrexia, and dehydration. The most common adverse reactions (greater than 20% of patients) were fatigue (49%), diarrhea (45%), pyrexia (36%), musculoskeletal pain (36%), abdominal pain (30%), pruritus (28%), nausea (26%), rash (25%)

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musculoskeletal pain (30%), abdominal pain (30%), pruritus (28%), nausea (26%), rash (25%), decreased appetite (20%), and vomiting (20%) (BMS, 2018).

Esophageal Cancer

ATTRACTION-3 (NCT02569242) was a multicenter, randomized (1:1), active-controlled, open-label trial in patients with unresectable advanced, recurrent, or metastatic esophageal squamous cell cancer (ESCC), who were refractory or intolerant to at least one fluoropyrimidine- and platinum-based regimen. The trial enrolled patients regardless of PD-L1 status, but tumor specimens were evaluated prospectively using the PD-L1 IHC 28-8 pharmDx assay at a central laboratory. The trial excluded patients who were refractory or intolerant to taxane therapy, had brain metastases that were symptomatic or required treatment, had autoimmune disease, used systemic corticosteroids or immunosuppressants, or had apparent tumor invasion of organs adjacent to the esophageal tumor or had stents in the esophagus or respiratory tract. Patients were randomized to receive Opdivo 240 mg by intravenous infusion over 30 minutes every 2 weeks or investigator's choice of taxane chemotherapy consisting of docetaxel (75 mg/m2 intravenously every 3 weeks) or paclitaxel (100 mg/m2 intravenously once a week for 6 weeks followed by 1 week off). Randomization was stratified by region (Japan vs. Rest of World), number of organs with metastases (≤1 vs. ≥2), and PD-L1 status (≥1% vs. <1% or indeterminate). Patients were treated until disease progression, assessed by the investigator per RECIST v1.1, or unacceptable toxicity. The tumor assessments were conducted every 6 weeks for 1 year, and every 12 weeks thereafter. The major efficacy outcome measure was overall survival (OS). Additional efficacy outcome measures were overall response rate (ORR) and progression free survival (PFS) as assessed by the investigator using RECIST v1.1 and DOR. A total of 419 patients were randomized; 210 to the Opdivo arm and 209 to the investigator's choice arm (docetaxel: 31%, paclitaxel: 69%). The trial population characteristics were: median age 65 years (range: 33 to 87), 53% were ≥65 years of age, 87% were male, 96% were Asian and 4% were White. Sixty-seven percent of patients had received one prior systemic therapy regimen and 26% had received two prior systemic therapy regimens prior to enrolling in ATTRACTION-3. Baseline ECOG performance status was 0 (50%) or 1 (50%).

ATTRACTION-3 demonstrated a statistically significant improvement in OS for patients randomized to Opdivo as compared with investigator's choice of taxane chemotherapy. OS benefit was observed regardless of PD-L1 expression level. The minimum follow-up was 17.6 months. Median OS for patients receiving nivolumab was 10.9 months (95% CI: 9.2, 13.3) compared with 8.4 months (95% CI: 7.2, 9.9) for patients receiving investigator's choice of taxane chemotherapy (HR: 0.77; 95% CI: 0.62, 0.96; p=0.0189). OS benefit was observed regardless of tumor PD-L1 expression level. The ORR was 19.3% (95% CI: 13.7, 26) in the nivolumab arm versus 21.5% (95% CI: 15.4, 28.8) in the taxane chemotherapy arm, with median response duration of 6.9 months (95% CI: 5.4, 11.1) and 3.9 months (95% CI: 2.8, 4.2), respectively. The trial did not demonstrate an improvement in PFS (HR: 1.1; 95% CI: 0.9, 1.3). The most common adverse reactions in 10% patients receiving nivolumab were rash, decreased appetite, diarrhea, constipation, musculoskeletal pain, upper respiratory tract infection, cough, pyrexia, pneumonia, anemia, fatigue, pruritus, nausea, and hypothyroidism. The recommended nivolumab dose for ESCC is 240 mg every 2 weeks or 480 mg every 4 weeks.

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On May 20, 2021, the U.S. Food and Drug Administration (FDA) approved Opdivo (nivolumab, injection for intravenous use) for the adjuvant treatment of completely resected esophageal or gastroesophageal junction (GEJ) cancer with residual pathologic disease in patients who have received neoadjuvant chemoradiotherapy (CRT). Prior to FDA approval, a review was conducted using the FDA's Real-Time Oncology Review (RTOR) pilot program in an effort to ensure that safe and effective treatments are available to patients as early as possible. The FDA approval was based on supporting data from the Phase 3 CheckMate-577 study (BMS, 2021b).

Kelly and colleagues (2021) conducted the CheckMate-577 study which consisted of a global, randomized, double-blind, placebo-controlled phase 3 trial evaluating Opdivo (nivolumab) as adjuvant therapy in 794 patients with completely resected (negative margins) esophageal or gastroesophageal junction cancer who had residual pathologic disease following neoadjuvant chemoradiotherapy (CRT). Patients were randomly assigned in a 2:1 ratio to receive Opdivo (at a dose of 240 mg every 2 weeks for 16 weeks, followed by Opdivo at a dose of 480 mg every 4 weeks) or matching placebo. Treatment continued until disease recurrence, unacceptable toxicity, or for a maximum duration of 1 year. The primary end point was disease-free survival. in the 532 patients receiving Opdivo, the median disease-free survival was 22.4 months (95% confidence interval [CI], 16.6 to 34.0) in comparison to 11.0 months (95% CI, 8.3 to 14.3) in the 262 patients who received placebo (hazard ration for disease recurrence or death, 0.69; 96.4% CI, 0.56 to 0.86; $p<0.001$). Grade 3 or 4 adverse events occurred in 13% (71/532) of patients in the Opdivo group and 6% (15/260) of patients in the placebo group.

On May 27, 2022, the U.S. Food and Drug Administration (FDA) approved nivolumab (Opdivo) in combination with fluoropyrimidine- and platinum-based chemotherapy and nivolumab (Opdivo) in combination with ipilimumab (Yervoy) for the first-line treatment of patients with advanced or metastatic esophageal squamous cell carcinoma (ESCC). The FDA approval was based on supporting data from the CHECKMATE-648 study (FDA, 2022b).

In the CHECKMATE-648 study, a randomized, active-controlled, open-label phase 3 trial, Doki and colleagues (2022) evaluated 970 adult patients with previously untreated unresectable advanced, recurrent, or metastatic ESCC. Patients were randomized 1:1:1 to receive nivolumab plus chemotherapy [nivolumab 240 mg on days 1 and 15, fluorouracil 800 mg/m2/day intravenously on days 1through 5 (for 5 days), and cisplatin 80 mg/m2 intravenously on day 1 (of a 4-week cycle)] , nivolumab plus ipilimumab [nivolumab 3 mg/kg every 2 weeks in combination with ipilimumab 1 mg/kg every 6 weeks], or chemotherapy [fluorouracil 800 mg/m2/day intravenously on days 1 through 5 (for 5 days), and cisplatin80 mg/m2 intravenously on day 1 (of a 4-week cycle)]. The primary efficacy outcome measures were overall survival and progression-free survival. The following were noted at a 13-month minimum follow-up, overall survival was significantly longer with nivolumab plus chemotherapy than with chemotherapy alone, both among patients with tumor-cell PD-L1 expression of 1% or greater (median, 15.4 vs. 9.1 months; hazard ratio, 0.54; 99.5% confidence interval [CI], 0.37 to 0.80; $p < 0.001$) and in the overall population(median, 13.2 vs. 10.7 months; hazard ratio, 0.74; 99.1% CI, 0.58 to 0.96; $p = 0.002$) and overall survival was also significantly longer with nivolumab plus ipilimumab than with chemotherapy among patients with tumor-cell PD-L1 expression of 1% or greater (median, 13.7 vs

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with chemotherapy among patients with tumor-cell PD-L1 expression of 1% or greater (median, 15.7 vs. 9.1 months; hazard ratio, 0.64; 98.6% CI,0.46 to 0.90; p = 0.001) and in the overall population (median, 12.7 vs. 10.7 months; hazard ratio, 0.78; 98.2% CI, 0.62 to 0.98; p = 0.01). Notably among patients with tumor-cell PD-L1 expression of 1% or greater, a significant progression-free survival benefit was observed with nivolumab plus chemotherapy over chemotherapy alone (hazard ratio for disease progression or death, 0.65; 98.5% CI, 0.46 to 0.92; p = 0.002) but not with nivolumab plus ipilimumab as compared with chemotherapy. Grad 3 or 4 treatment-related adverse events occurrence was 47% with nivolumab plus chemotherapy, 32% with nivolumab plus ipilimumab, and 36% with chemotherapy alone. The investigators concluded that both first-line treatment with nivolumab plus chemotherapy and first-line treatment with nivolumab plus ipilimumab showed significantly longer overall survival than chemotherapy alone in patients with advanced ESCC with no new safety signals identified.

Gastric Cancer

Venerito et al (2014) noted that gastric cancer (GC) continues to be an important health threat as the third leading cause of cancer related death in both sexes worldwide. In a recent analysis, the mortality trends for the time period from 1980 till 2011 were significantly downward in all countries, but the declines in the USA, EU and several other major countries were of low magnitude when compared with the past. Furthermore, the relative contribution of cardia cancers compared with non-cardia cancers increased among countries with higher GC rates. With respect to pre-neoplastic changes of the gastric mucosa, a large population-based study suggested that Helicobacter pylori (H. pylori) infection and anti-gastric parietal cell antibodies-mediated autoimmune response might, for the most part, be independent and follow distinct pathways rather than causally related pathways leading to chronic atrophic gastritis. A large prospective, randomized, open-label Korean trial questioned the role of H. pylori eradication for the prevention of metachronous lesions after endoscopic resection of early GC. A review of 1,258 Japanese cases undergoing curative endoscopic submucosa dissection for early GC showed that scheduled follow-up endoscopy is mandatory for detecting metachronous lesions at an early stage, where they can be treated by endoscopic resection. Ramucirumab, a vascular endothelial growth factor (VEGF) receptor-2 antagonist, is the first biological treatment that provides survival benefits to patients with advanced GC in progress after first-line chemotherapy. Rilotumumab is currently being evaluated in patients with advanced GC over-expressing the HGF/c-MET signaling pathway. In the near future, ipilimumab and nivolumab, two immune-stimulatory monoclonal antibodies with anti-neoplastic effects, might offer new therapeutic options for patients with advanced GC.

On April 16, 2021, the U.S. Food and Drug Administration (FDA) approved Opdivo (nivolumab, injection for intravenous use), in combination with fluoropyrimidine- and platinum-containing chemotherapy, for the treatment of patients with advanced or metastatic gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma, regardless of PD-L1 expression. Prior to FDA approval, a review was conducted using the FDA's Real-Time Oncology Review (RTOR) pilot program in an effort to ensure that safe and effective treatments are available to patients as early as possible. The FDA approval was supported by data from the CheckMate-649 study (BMS, 2021c).

The CheckMae-649 study was a randomized, multicenter, open-label Phase 3 clinical trial in patients

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The CheckMate-649 study was a randomized, multicenter, open-label Phase 3 clinical trial in patients (n=1581) with previously untreated advanced or metastatic gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma. Patients were enrolled regardless of PD-L1 status. In addition, patients who were known human epidermal growth factor receptor 2 (HER2) positive, or had untreated CNS metastases. Patient randomization consisted of receiving Opdivo in combination with chemotherapy (n=789) or chemotherapy (n=792). Patients received one of the following treatments: a) Opdivo 240 mg in combination with mFOLFOX6 (fluorouracil, leucovorin and oxaliplatin) every 2 weeks or mFOLFOX6 every 2 weeks OR b) Opdivo 360 mg in combination with CapeOX (capecitabine and oxaliplatin) every 3 weeks or CapeOX every 3 weeks. Treatment continued until disease progression, unacceptable toxicity, or up to 2 years. The major efficacy endpoint measures, assessed in patients with PD-L1 CPS ≥5, were progression-free survival (PFS) and overall survival (OS). Additionally, OS and PFS in patients with PD-L1 CPS ≥1 and in all randomized patients, and overall response rate (ORR) and duration of response (DOR) in patients with PD-L1 CPS ≥1 ad ≥5, in all randomized patients. In patients with PD-L1 CPS<1 (n=265), the median OS was 13.1 months (95% Confidence Interval [CI]: 9.8, 16.7) for Opdivo and chemotherapy arm and 12.5 months (95% CI: 10.1, 13.8) for the chemotherapy arm. In patients with PD-L1 CPS<5 (n=606), the median OS was 12.4 months (95% CI: 11.0, 13.2) for the chemotherapy arm. In conclusion, the CHECKMATE-649 study demonstrated a statistically significant improvement in OS and PFS for patients with PD-L1 CPS ≥5. Statistically significant improvement in OS was also apparent for all randomized patients (BMS, 2021b).

Gestational Trophoblastic Neoplasia

The Gestational Trophoblastic Neoplasia NCCN guidelines (Version 1.2024) state PD1-/PD-L1 inhibitors (e.g., nivolumab) has shown to have some activity in treating resistant gestational trophoblastic neoplasia.

Veras et al (2017) stated one of the major immune checkpoints responsible for immune evasion in cancer cells is the interaction between programmed cell death-1 (PD-1) and its ligand (PD-L1). As human trophoblastic cells display many of the features of malignant cells such as the ability to invade normal tissue including blood vessels and are apparently not eradicated by the host immune system, we undertook the present study to determine whether PD-L1 was upregulated in different types of trophoblastic cells during normal pregnancy and in gestational trophoblastic diseases. Immunohistochemistry using an anti-PD-L1-specific antibody demonstrated that in early and term normal placentas, PD-L1 was highly expressed in syncytiotrophoblast and to a much lower extent in intermediate trophoblastic cells located in the chorion laeve and implantation site. PD-L1 immunoreactivity was undetectable in cytotrophoblastic cells. This staining pattern in normal placenta was recapitulated in various types of gestational trophoblastic disease. PD-L1 was highly expressed by syncytiotrophoblast in complete moles and choriocarcinomas. The intermediate trophoblastic neoplasms, placental site trophoblastic tumors, and epithelioid trophoblastic tumors showed variable PD-L1 immunoreactivity but at a lower intensity than in the neoplastic syncytiotrophoblast in choriocarcinoma. In addition, we observed PD-1-positive lymphocytes located within the implantation site and in trophoblastic tumors. In summary, this study describes a novel mechanism for trophoblastic cells to create a tolerogenic feto-maternal interface by upregulating PD-L1 in syncytiotrophoblast and in

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create a tolerogenic feto-maternal interface by upregulating PD-L1 in syncytiotrophoblast and in intermediate trophoblast. Trophoblastic tumors may also use PD-L1 expression to evade the host immune response thereby promoting their survival.

Head and Neck Cancer

Swanson and Sinha (2015) noted that targeted immunotherapy promoting anti-tumor T-cell activity has shown improved survival and durable objective responses in advanced melanoma patients. Data are mounting that concurrent use of ipilimumab and nivolumab has a more pronounced effect than either as monotherapy. Pre-clinical data suggested these therapies would be beneficial in head and neck malignancies as well.

The FDA approved nivolumab for the treatment of patients with recurrent or metastatic squamous cell carcinoma of the head and neck (SCCHN) with disease progression on or after platinum-based therapy. The approval was based on results from the Phase 3, CheckMate -141 trial in which nivolumab demonstrated statistically significant overall survival versus the comparator arm (investigator's choice of methotrexate, docetaxel or cetuximab), with a 30% reduction in the risk of death (HR=0.70 [95% CI: 0.53-0.92; p=0.0101]).1 The median OS was 7.5 months (95% CI: 5.5-9.1) for nivolumab compared to 5.1 months (95% CI: 4.0-6.0) for investigator's choice.

CheckMate -141 was a phase 3, open-label, randomized, trial evaluating nivolumab versus investigator's choice of therapy in patients with recurrent or metastatic SCCHN who had tumor progression during or within six months of receiving platinum-based therapy administered in the adjuvant, neo-adjuvant, primary (unresectable locally advanced) or metastatic setting. Patients were included regardless of their HPV or PD-L1 status. Patients were randomized 2:1 to receive nivolumab 3 mg/kg intravenously over 60 minutes every two weeks (n=240), or investigator's choice (n=121) of: methotrexate 40 to 60 mg/m2 intravenously weekly, docetaxel 30 to 40 mg/m2 intravenously weekly, or cetuximab 400 mg/m2 intravenously once then 250 mg/m2 weekly. The primary endpoint was overall survival. The trial's secondary endpoints included progression-free survival (PFS) and objective response rate (ORR). Nivolumab demonstrated statistically significant superior overall survival with a 30% reduction in the risk of death (HR=0.70 [95% CI: 0.53-0.92; p=0.0101]), and a median OS of 7.5 months (95% CI: 5.5-9.1) for nivolumab compared to 5.1 months (95% CI: 4.0-6.0) for the investigator's choice arm. There were no statistically significant differences between the two arms for PFS (HR=0.89; 95% CI: 0.70, 1.13) or ORR (13.3% [95% CI: 9.3, 18.3] vs 5.8% [95% CI: 2.4, 11.6] for nivolumab and investigator's choice, respectively.

In CheckMate -141, nivolumab was discontinued in 14% of patients and was delayed in 24% of patients for an adverse reaction. Serious adverse reactions occurred in 49% of patients receiving nivolumab. The most frequent serious adverse reactions reported in at least 2% of patients receiving nivolumab were pneumonia, dyspnea, aspiration pneumonia, respiratory failure, respiratory tract infection, and sepsis.

Hepatocellular Carcinoma

On September 22, 2017, the U.S. Food and Drug Administration granted accelerated approval to

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On September 22, 2017, the U.S. Food and Drug Administration granted accelerated approval to nivolumab (Opdivo) for the treatment of hepatocellular carcinoma (HCC) in patients who have been previously treated with sorafenib.
FDA-approval was based on a 154-patient subgroup (median age 63 years; range 19-81 years) of CHECKMATE-040 (NCT 01658878), a multicenter, open-label trial conducted in patients with hepatocellular carcinoma (HCC) and Child-Pugh A cirrhosis who progressed on or were intolerant to sorafenib. In addition to including patients without active hepatitis viral infection, the trial enrolled patients with either active HBV (31%) or HCV (21%) but not those with active co-infection with HBV and HCV or with hepatitis D virus infection. Patients received nivolumab 3 mg/kg by intravenous infusion every 2 weeks. The confirmed overall response rate, as assessed by blinded independent central review using RECIST 1.1, was 14.3% (95% CI: 9.2, 20.8), with 3 complete responses and 19 partial responses. Response duration ranged from 3.2 to 38.2+ months; 91% of responders had responses lasting 6 months or longer and 55% had responses lasting 12 months or longer (FDA, 2017).

On March 10, 2020, the Food and Drug Administration granted accelerated approval to the combination of nivolumab (Opdivo) and ipilimumab (Yervoy) for patients with hepatocellular carcinoma (HCC) who have been previously treated with sorafenib (CHECKMATE-040; NCT01658878). The CHECKMATE-40 trial was a multicenter, multiple cohort, open-label trial conducted in patients with HCC who progressed on or were intolerant to sorafenib. Additional eligibility criteria included histologic confirmation of HCC and Child-Pugh Class A cirrhosis. The trial excluded patients with active autoimmune disease, brain metastasis, a history of hepatic encephalopathy, clinically significant ascites, infection with HIV, or active co-infection with hepatitis B virus (HBV) and hepatitis C virus (HCV) or HBV and hepatitis D virus (HDV); however, patients with only active HBV or HCV were eligible. The efficacy of ipilimumab 3 mg/kg in combination with nivolumab 1 mg/kg was evaluated in Cohort 4 of CHECKMATE-040. A total of 49 patients received the combination regimen, which was administered every 3 weeks for four doses, followed by single-agent nivolumab at 240 mg every 2 weeks until disease progression or unacceptable toxicity. The median age was 60 years (range: 18 to 80); 88% were male; 74% were Asian, and 25% were White. Baseline ECOG performance status was 0 (61%) or 1 (39%). Fifty-seven percent (57%) of patients had active HBV infection, 8% had active HCV infection, and 35% had no evidence of active HBV or HCV. The etiology for HCC was alcoholic liver disease in 16% and non-alcoholic liver disease in 6% of patients. Child-Pugh class and score was A5 for 82% and A6 for 18%; 80% of patients had extrahepatic spread; 35% had vascular invasion; and 51% had alfa-fetoprotein (AFP) levels ≥400 µg/L. Prior treatment history included surgery (74%), radiotherapy (29%), or local treatment (59%). All patients had received prior sorafenib, of whom 10% were unable to tolerate sorafenib; 29% of patients had received 2 or more prior systemic therapies. The main efficacy outcome measures were overall response rate and duration of response as determined by blinded independent central review (BICR) using RECIST v1.1. ORR was 33% (n=16; 95% CI: 20, 48), with 4 complete responses and 12 partial responses. Response duration ranged from 4.6 to 30.5+ months, with 31% of responses lasting at least 24 months.

Most common adverse reactions (≥20%) with ipilimumab in combination with nivolumab are fatigue, rash, pruritus, diarrhea, musculoskeletal pain, cough, pyrexia, decreased appetite, nausea, abdominal pain, arthralgia, headache, vomiting, dyspnea, dizziness, hypothyroidism, and decreased weight

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pain, arthralgia, headache, vomiting, dyspnea, dizziness, hypothyroidism, and decreased weight.

For HCC, the recommended doses are nivolumab 1 mg/kg followed by ipilimumab 3 mg/kg on the same day every 3 weeks for 4 doses, then nivolumab 240 mg every 2 weeks or 480 mg every 4 weeks.

On July 23, 2021, Bristol Myers Squibb, in consultation with the U.S. Food and Drug Administration (FDA), voluntarily withdrew the indication for Opdivo (nivolumab) as a single agent for patients with hepatocellular carcinoma (HCC) who were previously treated with sorafenib from the U.S. market. The initial approval for this indication was granted in 2017 under the FDA's accelerated approval based on results from the Phase 1/2 CheckMate -040 trial. However, the confirmatory CheckMate -459 randomized study of Opdivo compared to sorafenib in the first-line setting, did not result in statistical significance for the primary endpoint of overall survival per the pre-specified analysis (BMS, 2021a).

Hodgkin's Lymphoma

On May 17, 2016, the FDA granted accelerated approval to nivolumab for the treatment of patients with classical Hodgkin lymphoma that has relapsed or progressed after autologous hematopoietic stem cell transplantation and post-transplantation brentuximab vedotin (Adcetris).

Ansell et al (2015) stated that pre-clinical studies suggested that Reed-Sternberg cells exploit the PD-1 pathway to evade immune detection. In classic Hodgkin's lymphoma, alterations in chromosome 9p24.1 increase the abundance of the PD-1 ligands, PD-L1 and PD-L2, and promote their induction through Janus kinase (JAK)-signal transducer and activator of transcription (STAT) signaling. These investigators hypothesized that nivolumab could inhibit tumor immune evasion in patients with relapsed or refractory Hodgkin's lymphoma. In this ongoing study, 23 patients with relapsed or refractory Hodgkin's lymphoma that had already been heavily treated received nivolumab (at a dose of 3 mg/kg of body weight) every 2 weeks until they had a complete response, tumor progression, or excessive toxic effects. Study objectives were measurement of safety and efficacy and assessment of the PDL1 and PDL2 (also called CD274 and PDCD1LG2, respectively) loci and PD-L1 and PD-L2 protein expression. Of the 23 participants, 78 % were enrolled in the study after a relapse following autologous stem-cell transplantation and 78 % after a relapse following the receipt of brentuximab vedotin. Drug-related AEs of any grade and of grade 3 occurred in 78 % and 22 % of patients, respectively. An objective response was reported in 20 patients (87 %), including 17 % with a complete response (CR) and 70 % with a partial response (PR); the remaining 3 patients (13 %) had stable disease (SD). The rate of PFS at 24 weeks was 86 %; 11 patients were continuing to participate in the study. Reasons for discontinuation included: stem-cell transplantation (in 6 patients), disease progression (in 4 patients), and drug toxicity (in 2 patients). Analyses of pre-treatment tumor specimens from 10 patients revealed copy-number gains in PDL1 and PDL2 and increased expression of these ligands. Reed-Sternberg cells showed nuclear positivity of phosphorylated STAT3, indicative of active JAK-STAT signaling. The authors concluded that nivolumab had substantial therapeutic activity and an acceptable safety profile in patients with previously heavily treated relapsed or refractory Hodgkin's lymphoma.

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Kidney Cancer (Renal Cell Carcinoma)

In a phase II clinical trial, Motzer et al (2015) evaluated the anti-tumor activity, dose-response relationship, and safety of nivolumab in patients with metastatic RCC (mRCC). Patients with clear-cell mRCC previously treated with agents targeting the VEGF pathway were randomly assigned (blinded ratio of 1:1:1) to nivolumab 0.3, 2, or 10 mg/kg intravenously once every 3 weeks. The primary objective was to evaluate the dose-response relationship as measured by progression-free survival (PFS); secondary end-points included ORR, OS, and safety. A total of 168 patients were randomly assigned to the nivolumab 0.3- (n = 60), 2- (n = 54), and 10-mg/kg (n = 54) cohorts; 118 patients (70 %) had received more than 1 prior systemic regimen. Median PFS was 2.7, 4.0, and 4.2 months, respectively (p = 0.9). Respective ORRs were 20 %, 22 %, and 20 %. Median OS was 18.2 months (80 % confidence interval [CI]: 16.2 to 24.0 months), 25.5 months (80 % CI: 19.8 to 28.8 months), and 24.7 months (80 % CI: 15.3 to 26.0 months), respectively. The most common treatment-related adverse event (AE) was fatigue (24 %, 22 %, and 35 %, respectively); 19 patients (11 %) experienced grade 3 to 4 treatment-related AEs. The authors concluded that nivolumab demonstrated anti-tumor activity with a manageable safety profile across the 3 doses studied in mRCC. No dose-response relationship was detected as measured by PFS. These efficacy and safety results in mRCC support study in the phase III setting.

In a phase I clinical trial with expansion cohorts, McDermott et al (2015) reported clinical activity, survival, and long-term safety in patients with advanced RCC treated with nivolumab. A total of 34 patients with previously treated advanced RCC, enrolled between 2008 and 2012, received intravenous nivolumab (1 or 10 mg/kg) in an outpatient setting once every 2 weeks for up to 96 weeks and were observed for survival and duration of response after treatment discontinuation. Ten patients (29 %) achieved objective responses (according to RECIST [version 1.0]), with median response duration of 12.9 months; 9 additional patients (27 %) demonstrated stable disease lasting greater than 24 weeks; 3 of 5 patients who stopped treatment while in response continued to respond for greater than or equal to 45 weeks. Median OS in all patients (71 % with 2 to 5 prior systemic therapies) was 22.4 months; 1-, 2-, and 3-year survival rates were 71 %, 48 %, and 44 %, respectively. Grade 3 to 4 treatment-related AEs occurred in 18 % of patients; all were reversible. The authors concluded that patients with advanced treatment-refractory RCC treated with nivolumab demonstrated durable responses that in some responders persisted after drug discontinuation. They stated that OS is encouraging, and toxicities were generally manageable. Moreover, they stated that ongoing randomized clinical trials will further assess the impact of nivolumab on OS in patients with advanced RCC.

Motzer, et al. (2015) reported that, among patients with previously treated advanced renal-cell carcinoma, overall survival was longer and fewer grade 3 or 4 adverse events occurred with nivolumab than with everolimus. A randomized, open-label, phase 3 study (Checkmate 025) compared nivolumab with everolimus in patients with renal-cell carcinoma who had received previous treatment. Investigators randomly assigned 821 patients with advanced clear-cell renal-cell carcinoma for which they had received previous treatment with one or two regimens of antiangiogenic therapy to receive 3 mg of nivolumab per kilogram of body weight intravenously every 2 weeks or a 10-mg everolimus tablet orally once daily. The primary end point was overall survival. The secondary end

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everolimus tablet orally once daily. The primary end point was overall survival. The secondary end points included the objective response rate and safety. The median overall survival was 25.0 months (95% confidence interval [CI], 21.8 to not estimable) with nivolumab and 19.6 months (95% CI, 17.6 to 23.1) with everolimus. The hazard ratio for death with nivolumab versus everolimus was 0.73 (98.5% CI, 0.57 to 0.93; P=0.002), which met the prespecified criterion for superiority (P≤0.0148). The objective response rate was greater with nivolumab than with everolimus (25% vs. 5%; odds ratio, 5.98 [95% CI, 3.68 to 9.72]; P<0.001). The median progression-free survival was 4.6 months (95% CI, 3.7 to 5.4) with nivolumab and 4.4 months (95% CI, 3.7 to 5.5) with everolimus (hazard ratio, 0.88; 95% CI, 0.75 to 1.03; P=0.11). Grade 3 or 4 treatment-related adverse events occurred in 19% of the patients receiving nivolumab and in 37% of the patients receiving everolimus; the most common event with nivolumab was fatigue (in 2% of the patients), and the most common event with everolimus was anemia (in 8%). The investigators concluded that, among patients with previously treated advanced renal-cell carcinoma, overall survival was longer and fewer grade 3 or 4 adverse events occurred with nivolumab than with everolimus.

An accompanying editorial (Quinn & Lara, 2015) stated that nivolumab is the choice for patients who have disease progression while they are receiving VEGF-targeted therapy given the overall survival advantage it confers and its relatively good side-effect profile. However, "for all the success reported here, many questions remain." The editorialist stated that complete remissions remained "disappointingly elusive'; the complete remission rate was 1%." Although it is possible that complete remission with nivolumab may be unnecessary to achieve a long-term benefit, the lack of profound responses begs for selection or combination approaches that expand the benefit spectrum." In other tumors, PD-1 ligand 1 (PD-L1) expression, in tumor cells or infiltrating immune cells, is associated with benefit from PD-1 or PD-L1 inhibitors. Unfortunately, PD-L1 expression in renal-cell cancer tissue did not delineate the patients who were more likely to benefit. In addition, the most effective duration of therapy with nivolumab and whether the therapy should continue beyond progression remains unknown.

Dreicer, et al. (2015) commented that, nivolumab, by virtue of its clear survival benefit and favorable effects profile in Checkmate 025, is now the treatment of choice for patients with disease progression on VEGF-targeted therapy. He observed that, unlike other neoplasms, PD-L1 expression, although likely prognostic regarding disease biology, is not a predictive biomarker for the selection anti-PD1 directed therapy in advanced renal cancer.

Clinical trials are ongoing on the use of nivolumab for the treatment of other malignancies (e.g., Hodgkin's lymphoma and non-Hodgkin lymphoma) and solid tumors (e.g., bladder cancer, breast cancer, colon cancer, gastric cancer, glioblastoma, head and neck cancer, and pancreatic cancer).

Taube and colleagues (2014) stated that immune-modulatory drugs differ in mechanism-of-action from directly cytotoxic cancer therapies. Identifying factors predicting clinical response could guide patient selection and therapeutic optimization. Patients (n = 41) with melanoma, NSCLC, RCC, colorectal carcinoma, or castration-resistant prostate cancer were treated on an early-phase trial of nivolumab at one institution and had evaluable pre-treatment tumor specimens. Immuno-architectural features, including PD-1, PD-L1, and PD-L2 expression, patterns of immune cell infiltration, and lymphocyte

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including PD-1, PD-L1, and PD-L2 expression, patterns of immune cell infiltration, and lymphocyte subpopulations, were assessed for inter-relationships and potential correlations with clinical outcomes. Membranous PD-L1 expression by tumor cells and immune infiltrates varied significantly by tumor type and was most abundant in melanoma, NSCLC, and RCC. In the overall cohort, PD-L1 expression was geographically associated with infiltrating immune cells ($p < 0.001$), although lymphocyte-rich regions were not always associated with PD-L1 expression. Expression of PD-L1 by tumor cells and immune infiltrates was significantly associated with expression of PD-1 on lymphocytes. PD-L2, the second ligand for PD-1, was associated with PD-L1 expression. Tumor cell PD-L1 expression correlated with objective response to anti-PD-1 therapy, when analyzing either the specimen obtained closest to therapy or the highest scoring sample among multiple biopsies from individual patients. These correlations were stronger than borderline associations of PD-1 expression or the presence of intra-tumoral immune cell infiltrates with response. The authors concluded that tumor PD-L1 expression reflects an immune-active microenvironment and, while associated other immunosuppressive molecules, including PD-1 and PD-L2, is the single factor most closely correlated with response to anti-PD-1 blockade.

Ohaegbulam et al (2015) noted that PD-1 receptor and its ligands PD-L1 and PD-L2, members of the CD28 and B7 families, play critical roles in T cell co-inhibition and exhaustion. Over-expression of PD-L1 and PD-1 on tumor cells and tumor-infiltrating lymphocytes, respectively, correlates with poor disease outcome in some human cancers. Monoclonal antibodies (mAbs) blockading the PD-1/PD-L1 pathway have been developed for cancer immunotherapy via enhancing T cell functions. Clinical trials with mAbs to PD-1 and PD-L1 have shown impressive response rates in patients, particularly for melanoma, NSCLC, RCC, and bladder cancer. The authors concluded that further studies are needed to dissect the mechanisms of variable response rate, to identify biomarkers for clinical response, to develop small-molecule inhibitors, and to combine these treatments with other therapies.

Gunturi and McDermott (2015) noted that one mechanism by which tumor cells are thought to evade the host's immune system is by inducing negative signals that cause T-cell suppression. An important interaction that results in this phenomenon is the one between PD-1 on the T cell and its ligand PD-L1 on the tumor cell. PD-1 pathway blocking agents (e.g., nivolumab) are therefore capable of reversing T-cell suppression and ultimately induce anti-tumor responses. These researchers summarized investigations related to the safety and efficacy of nivolumab in a variety of malignancies thus far, including advanced melanoma, NSCLC, and RCC. The results have been promising with a large number of objective responses and favorable safety profiles inspiring several phase III trials in these settings. More recent studies are exploring the role of this drug in the treatment of various other cancers. Combination therapies involving nivolumab are also being studied and are yielding interesting results. Finally, the role of tumor PD-L1 expression as a predictive biomarker remains to be ascertained. Thus, with rational refinement through biomarker and combination clinical trials, nivolumab and other PD-1 blocking agents will likely lead to significant improvements in cancer therapeutics.

In a phase III trial (CheckMate 214), the combination of ipilimumab plus nivolumab (N + I) resulted in significantly improved overall survival and response rates compared with sunitinib (S) in previously untreated intermediate- and poor-risk patients with advanced clear cell RCC. Escudier et al (2017)

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untreated intermediate- and poor-risk patients with advanced clear cell RCC. Escudier, et al.(2017) randomized adults with clear-cell mRCC to nivolumab 3 mg/kg plus ipilimumab 1 mg/kg every 3 weeks for 4 doses followed by nivolumab 3 mg/kg every 2 weeks, or sunitinib 50 mg daily orally for 4 weeks (6-week cycles). Co-primary endpoints were overall response rate (ORR), progression-free survival (PFS) per independent committee (IRRC) and overall survival (OS) all in intermediate/poor risk patients. The investigators randomized 1096 patients. With approximately 17.5 month minimum follow-up, confirmed ORR in intermediate/poor risk patients was 41.6% (9.4% complete response [CR]) versus 26.5% (1.2% CR) for N+I versus S (P<0.0001); median duration of response was not reached (NR; 95% CI 21.82-NR) versus 18.2 months (95% CI 14.82-NR), respectively; median PFS was 11.6 versus 8.4 months (HR 0.82, P=0.0331), respectively. In all treated patients, drug-related adverse events occurred in 509/547 (93% any grade, 46% grade 3-4) with N+I versus 521/535 (97% any grade, 63% grade 3-5) with S, including 22% versus 12% with adverse events leading to discontinuation. Death occurred in 159 N+I patients (7 [1%] drug-related) and 202 S patients (4 [1%] drug-related). The authors concluded that this study showed higher ORR and longer PFS for N+I compared with S in intermediate/poor risk mRCC, particularly in patients with tumor PD-L1 expression ≥1%, with a manageable safety profile. These results support the use of N+I as a potential first-line treatment for these patients. Based upon the results of Checkmate 214, the European Association of Urology revised its guidelines to recommend this combination as first-line treatment.

In January 2021, the FDA approved combination of nivolumab (Opdivo, Bristol-Myers Squibb Co.) and cabozantinib (Cabometyx, Exelixis) as first-line treatment for patients with advanced renal cell carcinoma (RCC). FDA approval was based on the results of a randomized, open-label trial [CHECKMATE-9ER (NCT03141177)] which evaluated the efficacy of nivolumab combined with cabozantinib, versus sunitinib, in patients with previously untreated advanced RCC. CHECKMATE-9ER excluded patients with autoimmune disease or other medical conditions requiring systemic immunosuppression. Patients were randomized to nivolumab 240 mg intravenously every 2 weeks and cabozantinib 40 mg orally daily (n=323), or sunitinib 50 mg orally daily for the first 4 weeks of a 6-week cycle (4 weeks on treatment followed by 2 weeks off) (n=328). Treatment continued until disease progression per RECIST v1.1 or unacceptable toxicity. Treatment beyond RECIST-defined disease progression was permitted if the patient was clinically stable and considered to be deriving clinical benefit by the investigator. Tumor assessments were performed at baseline, after randomization at Week 12, then every 6 weeks until Week 60, and then every 12 weeks thereafter. The primary efficacy outcome measure was progression free survival (PFS). Secondary efficacy outcome measures were overall survival (OS) and confirmed overall response rate (ORR). The trial demonstrated a statistically significant improvement in PFS, OS, and ORR for patients randomized to nivolumab and cabozantinib compared with sunitinib. Median PFS per blinded independent central review (BICR) was 16.6 months versus 8.3 months; HR 0.51 (95% CI: 0.41, 0.64). Median OS was not reached in either arm; HR 0.60 (95% CI: 0.40, 0.89). Confirmed ORR per BICR was 55.7% and 27.1% in the nivolumab plus cabozantinib and sunitinib arms, respectively (BMS, 2021; FDA, 2021b).

Lung Cancer - Non-Small Cell Lung Cancer

Lung cancer is one of the leading causes of cancer deaths in the United States. There are two main

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Lung cancer is one of the leading causes of cancer deaths in the United States. There are two main categories of lung cancer: non-small cell lung cancer and small cell lung cancer. Non-small cell lung cancer (NSCLC) accounts for approximately 85 percent of cases of lung cancer (FDA, 2015; BMS, 2015). Squamous cell NSCLC accounts for approximately 25 to 30 percent of all lung cancers. Survival rates vary depending on the stage and type of the cancer and when it is diagnosed.

Shimanovsky and Dasanu (2014) stated that immunotherapy has made significant progress in patients with NSCLC in the last years. Early tumor vaccine studies showed trends toward better clinical outcomes, and larger trial results are currently being awaited. Immune check-point inhibitors are promising therapeutic agents in advanced NSCLC. While ipilimumab has clearly improved outcomes in metastatic malignant melanoma, its safety and efficacy in NSCLC are not yet known. The authors noted that PD-1 and PD-1 ligand inhibitors such as nivolumab, MK3475 and MPDL3280 have demonstrated clinical efficacy in patients with advanced/metastatic NSCLC in early clinical trials. Their validation in larger phase III trials is being awaited. Furthermore, exploring efficacy of these molecules in patients with early stages of lung cancer is also needed.

The FDA approved nivolumab for the treatment of patients with metastatic squamous non-small cell lung cancer (NSCLC) with progression on or after platinum-based chemotherapy (FDA, 2015).

The approval was based on the results of CheckMate -017 and CheckMate -063 (BMS, 2015). Nivolumab demonstrated significantly superior overall survival (OS) vs. docetaxel, with a 41% reduction in the risk of death (hazard ratio: 0.59 [95% CI: 0.44, 0.79; p=0.00025]), in a prespecified interim analysis of a Phase III clinical trial. The median OS was 9.2 months in the Opdivo arm (95% CI: 7.3, 13.3) and 6 months in the docetaxel arm (95% CI: 5.1, 7.3).

CheckMate -017 was a Phase III, open-label, randomized, multinational, multicenter clinical trial that evaluated nivolumab (3 mg/kg intravenously over 60 minutes every two weeks) (n=135) versus standard of care, docetaxel (75 mg/m2 intravenously administered every 3 weeks) (n=137) (BMS, 2015), in patients with metastatic squamous NSCLC who had progressed during or after prior platinum doublet-based chemotherapy regimen. This trial included patients regardless of their PD-L1 (programmed death ligand-1) status. The primary endpoint of this trial was overall survival (OS). In January, the trial was stopped based on an assessment conducted by the independent Data Monitoring Committee (DMC), which concluded that the study met its endpoint, demonstrating superior OS in patients receiving nivolumab compared to docetaxel. The prespecified interim analysis was conducted when 199 events (86% of the planned number of events for final analysis) were observed (86 in the nivolumab arm and 113 in the docetaxel arm). The median OS was 9.2 months in the nivolumab arm (95% CI: 7.3, 13.3) and 6 months in the docetaxel arm (95% CI: 5.1, 7.3). The hazard ratio was 0.59 (95% CI: 0.44, 0.79; p=0.00025).

The safety profile of nivolumab in squamous NSCLC was addressed in CheckMate -063, a phase II single-arm, open-label, multinational, multicenter trial of nivolumab, administered as a single agent in patients with metastatic squamous NSCLC who have progressed after receiving a platinum-based therapy and at least 1 additional systemic treatment regimen (n = 117) (BMS, 2015; Rizvi et al, 2015)

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therapy and at least 1 additional systemic treatment regimen (n = 117) (BMS, 2015; Rizvi et al, 2015). Patients received 3 mg/kg of nivolumab administered intravenously over 60 minutes every 2 weeks. This trial included patients regardless of their PD-L1 status. The most common adverse reactions (reported in greater than or equal to 20 % of patients) were fatigue (50 %), dyspnea (38 %), musculoskeletal pain (36 %), decreased appetite (35 %), cough (32 %), nausea (29 %), and constipation (24 %). Serious adverse reactions occurred in 59 % of patients receiving nivolumab. The most frequent serious adverse reactions reported in greater than or equal to 2 % of patients were dyspnea, pneumonia, chronic obstructive pulmonary disease exacerbation, pneumonitis, hypercalcemia, pleural effusion, hemoptysis, and pain. Nivolumab was discontinued due to adverse reactions in 27 % of patients; 29 % of patients receiving nivolumab had a drug delay for an adverse reaction. With at least 10 months of minimum follow-up for all patients, the confirmed objective response rate (ORR), the study's primary end-point, was 15 % (17/117) (95 % CI: 9 to 22) of which all were partial responses (BMS, 2015). The median time to onset of response was 3.3 months (range of 1.7 to 8.8 months) after the start of nivolumab treatment; 76 % of nivolumab responders (13/17 patients) had ongoing responses with durability of response ranging from 1.9+ to 11.5+ months; 10 of these 17 (59 %) patients had durable responses of 6 months or longer.

Nivolumab is associated with immune-mediated: pneumonitis, colitis, hepatitis, nephritis and renal dysfunction, hypothyroidism and hyperthyroidism, other adverse reactions; and embryofetal toxicity (BMS, 2015).

In Trial 3, serious adverse reactions occurred in 59 % of patients receiving nivolumab (BMS, 2015). The most frequent serious adverse drug reactions reported in greater than or equal to 2 % of patients were dyspnea, pneumonia, chronic obstructive pulmonary disease exacerbation, pneumonitis, hypercalcemia, pleural effusion, hemoptysis, and pain.

The most common adverse reactions (greater than or equal to 20 %) reported with nivolumab in Trial 3 were fatigue (50 %), dyspnea (38 %), musculoskeletal pain (36 %), decreased appetite (35 %), cough (32 %), nausea (29 %), and constipation (24 %) (BMS, 2015).

Severe pneumonitis or interstitial lung disease, including fatal cases, occurred with nivolumab treatment (BMS, 2015). Across the clinical trial experience in 691 patients with solid tumors, fatal immune-mediated pneumonitis occurred in 0.7 % (5/691) of patients receiving nivolumab; no cases occurred in Trial 3. In Trial 3, immune-mediated pneumonitis occurred in 6 % (7/117) of patients receiving nivolumab including 5 Grade 3 and 2 Grade 2 cases. The labeling recommends that patients be monitored for signs and symptoms of pneumonitis. Corticosteroids should be administered for Grade 2 or greater pneumonitis. The labeling recommends withholding nivolumab for Grade 2 pneumonitis until resolution, and permanently discontinuing nivolumab for Grade 3 or 4 pneumonitis.

In Trial 3, diarrhea occurred in 21 % (24/117) of patients receiving nivolumab (BMS, 2015). Grade 3 immune-mediated colitis occurred in 0.9 % (1/117) of patients. The labeling states that patients should be monitored for immune-mediated colitis. Corticosteroids should be administered for Grade 2 (of more than 5 days duration), 3, or 4 colitis. Nivolumab should be withheld for Grade 2 or 3 colitis. Nivolumab

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than 5 days duration), 3, or 4 colitis. Nivolumab should be withheld for Grade 2 or 3 colitis. Nivolumab should be permanently discontinued for Grade 4 colitis or recurrent colitis upon restarting nivolumab.

In Trial 3, the incidences of increased liver test values were AST (16 %), alkaline phosphatase (14 %), ALT (12 %), and total bilirubin (2.7 %) (BMS, 2015). The labeling recommends monitoring patients for abnormal liver tests prior to and periodically during treatment. Corticosteroids should be administered for Grade 2 or greater transaminase elevations. Nivolumab should be withheld for Grade 2 and permanently discontinued for Grade 3 or 4 immune-mediated hepatitis.

In Trial 3, the incidence of elevated creatinine was 22 %. Immune-mediated renal dysfunction (Grade 2) occurred in 0.9 % (1/117) of patients (BMS, 2015). The labeling states that patients should be monitored for elevated serum creatinine prior to and periodically during treatment. For Grade 2 or 3 serum creatinine elevation, nivolumab should be withheld and corticosteroids administered; if worsening or no improvement occurs, nivolumab should be permanently discontinued. Corticosteroids should be administered for Grade 4 serum creatinine elevation and nivolumab permanently discontinued.

In Trial 3, hypothyroidism occurred in 4.3 % (5/117) of patients receiving nivolumab (BMS, 2015). Hyperthyroidism occurred in 1.7 % (2/117) of patients including 1 Grade 2 case. The labeling recommends monitoring thyroid function prior to and periodically during treatment. Hormone replacement therapy should be administered for hypothyroidism. Medical management should be initiated for control of hyperthyroidism.

The following clinically significant immune-mediated adverse reactions occurred in less than 2 % of nivolumab-treated patients: adrenal insufficiency, uveitis, pancreatitis, facial and abducens nerve paresis, demyelination, autoimmune neuropathy, motor dysfunction and vasculitis (BMS, 2015). Across clinical trials of nivolumab administered at doses 3 mg/kg and 10 mg/kg, additional clinically significant, immune-mediated adverse reactions were identified: hypophysitis, diabetic ketoacidosis, hypopituitarism, Guillain-Barre syndrome, and myasthenic syndrome. Based on the severity of adverse reaction, nivolumab should be withheld, high-dose corticosteroids administered, and, if appropriate, hormone-replacement therapy should be administered.

Based on its mechanism of action, nivolumab can cause fetal harm when administered to a pregnant woman (BMS, 2015). The labeling recommends advising pregnant women of the potential risk to a fetus. Females of reproductive potential should be advised to use effective contraception during treatment with nivolumab and for at least 5 days after the last dose of nivolumab.

It is not known whether nivolumab is present in human milk (BMS, 2015). The labeling states, because many drugs, including antibodies, are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from nivolumab, women should be advised to discontinue breastfeeding during treatment.

Nivolumab for squamous cell lung cancer is administered 3 mg/kg as an intravenous infusion over 60 minutes every 2 weeks (BMS, 2015).

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minutes every 2 weeks (BMS, 2015).

The FDA approved nivolumab injection for the treatment of patients with metastatic non-small cell lung cancer (NSCLC) with progression on or after platinum-based chemotherapy (BMS, 2015). The labeled indication states that patients with EGFR mutation or ALK translocation should have disease progression on appropriate targeted therapy prior to receiving nivolumab.

The FDA approval was based on the results of the CheckMate -057 trial, a Phase 3 trial which demonstrated superior OS benefit for nivolumab versus docetaxel in previously treated metastatic NSCLC (BMS, 2015). CheckMate -057 was a Phase 3, open-label, randomized clinical trial that evaluated nivolumab (3 mg/kg administered intravenously every two weeks) (n=292) versus docetaxel (75 mg/m2 administered intravenously every three weeks) (n=290), in patients with metastatic non-squamous NSCLC who had experienced disease progression during or after one prior platinum doublet-based chemotherapy regimen. Appropriate targeted therapy may have been given to patients with known EGFR mutation or ALK translocation. This study included patients regardless of their PD-L1 expression status. The primary endpoint of this trial was OS.

The median OS was 12.2 months in the nivolumab arm (95% CI: 9.7, 15.0) and 9.4 months in the docetaxel arm (95% CI: 8.0, 10.7) (BMS, 2015). The hazard ratio (HR) was 0.73 (95% CI: 0.60, 0.89; p=0.0015), which translates to a 27% reduction in the risk of death with nivolumab compared to docetaxel. The prespecified interim analysis was conducted when 413 events were observed (93% of the planned number of events for final analysis). Additional secondary endpoints included investigator-assessed objective response rate (ORR) and progression-free survival (PFS). The ORR in the nivolumab arm was 19% (56/292; 4 complete responses, 52 partial responses) (95% CI: 15, 24) and 12% with docetaxel (36/290; 1 complete response, 35 partial responses) (95% CI: 9, 17) p=0.02. The median duration of response was 17 months in the nivolumab arm and 6 months in the docetaxel arm. Median PFS was 2.3 months in the nivolumab arm versus 4.2 months with docetaxel; HR=0.92 (95% CI:0.77, 1.11, p=0.39).

The manufacturer stated that the safety profile of nivolumab in CheckMate -057 was consistent with prior studies (BMS, 2015). Serious adverse reactions occurred in 47% of patients receiving nivolumab. The most frequent serious adverse reactions in at least 2% of patients receiving nivolumab were pneumonia, pulmonary embolism, dyspnea, pleural effusions and respiratory failure. Nivolumab was discontinued in 13% of patients and was delayed in 29% of patients for an adverse reaction. The most common adverse reactions (reported in ≥20% of patients) were fatigue (49%), musculoskeletal pain (36%), cough (30%), decreased appetite (29%) and constipation (23%).

In Checkmate 057, immune-mediated pneumonitis, including interstitial lung disease, occurred in 3.4% (10/287) of patients receiving nivolumab including five Grade 3, two Grade 2, and three Grade 1 cases (BMS, 2015). Diarrhea or colitis occurred in 17% (50/287) of patients receiving nivolumab. Immune-mediated colitis occurred in 2.4% (7/287) of patients including three Grade 3, two Grade 2, and two Grade 1 cases. One patient (0.3%) developed immune-mediated hepatitis

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Grade 1 cases. One patient (0.5%) developed immune-mediated hepatitis.

In Checkmate 057, Grade 1 or 2 hypothyroidism, including thyroiditis, occurred in 7% (20/287) and elevated TSH occurred in 17% of patients receiving nivolumab (BMS, 2015). Grade 1 or 2 hyperthyroidism occurred in 1.4% (4/287) of patients. Grade 2 immune-mediated renal dysfunction occurred in 0.3% (1/287) of patients. Immune-mediated rash occurred in 6% (17/287) of patients receiving nivolumab including 4 Grade 3 cases. Fatal limbic encephalitis occurred in one patient (0.3%) receiving nivolumab . Grade 2 infusion reactions occurred in 1% (3/287) of patients receiving nivolumab. .

In Checkmate 057, serious adverse reactions occurred in 47% of patients receiving nivolumab (BMS, 2015). The most frequent serious adverse reactions reported in ≥2% of patients were pneumonia, pulmonary embolism, dyspnea, pleural effusion, and respiratory failure. The most common adverse reactions (≥20%) reported with nivolumab were fatigue (49%), musculoskeletal pain (36%), cough (30%), decreased appetite (29%), and constipation (23%).

Nivolumab for metastatic non-small cell lung cancer is administered 3 mg/kg as an intravenous infusion over 60 minutes every 2 weeks (BMS, 2015).

Based upon data from the CheckMate-57 phase III randomized trial for patients receiving nivolumab, the median OS was 12.2 months compared with 9.4 months for docetaxel (HR 0.73, 95 % CI: 0.59 to 0.89, p = 0.0015). Fewer grade 3 to 5 adverse events were reported for nivolumab (10 %) when compared to docetaxel (54 %) in the CheckMate-057 trial (Paz-Ares et al, 2015).

In the CheckMate-017 trial, the median OS was 9.2 months with nivolumab compared with 6.0 months for docetaxel (Brahmer et al, 2015). There were fewer grade 3 to 4 adverse events with nivolumab (7 %) when compared with docetaxel (55 %).

Hellman et al (2019; CHECKMATE-227 Part 1a, NCT02477826) stated that in an early-phase study involving patients with advanced non-small-cell lung cancer (NSCLC), the response rate was better with nivolumab plus ipilimumab than with nivolumab monotherapy, particularly among patients with tumors that expressed programmed death ligand 1 (PD-L1). Data are needed to assess the long-term benefit of nivolumab plus ipilimumab in patients with stage IV or recurrent NSCLC. In this open-label, phase 3 trial, the authors screened pretreatment tumor tissue (freshly collected or archived ≤6 months before enrollment) for tumor PD-L1 expression. Patients who had PD-L1 expression in 1% or more of tumor cells were enrolled in Part 1a of the trial, and those with a PD-L1 expression level of less than 1% were enrolled in Part 1b. In Part 1a, patients were randomly assigned in a 1:1:1 ratio to receive nivolumab (at a dose of 3 mg per kilogram of body weight every 2 weeks) plus ipilimumab (at a dose of 1 mg per kilogram every 6 weeks), nivolumab monotherapy (240 mg every 2 weeks), or platinum-doublet chemotherapy every 3 weeks for up to four cycles. In Part 1b, patients were randomly assigned in a 1:1:1 ratio to receive nivolumab plus ipilimumab, nivolumab (360 mg every 3 weeks) plus platinum-doublet chemotherapy (every 3 weeks for up to four cycles), or platinum-doublet chemotherapy alone

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doublet chemotherapy (every 3 weeks for up to four cycles), or platinum-doublet chemotherapy alone (every 3 weeks for up to four cycles). In both portions of the trial, patients were stratified according to tumor histologic features (squamous vs. nonsquamous). All the patients had received no previous chemotherapy. Treatment continued until disease progression or unacceptable toxicity or, for the immunotherapy regimens, until 2 years of follow-up. Patients who received immunotherapy regimens could continue to receive treatment beyond disease progression if they met prespecified criteria. Crossover between the treatment groups during the trial was not permitted. Subsequent therapy was determined at the physician's discretion., with 2-year overall survival rates of 40.0% and 32.8%, respectively.

The primary end point was overall survival with nivolumab plus ipilimumab, as compared with chemotherapy, in patients with a PD-L1 expression level of 1% or more. Hierarchical secondary end points were progression-free survival, according to blinded independent central review; overall survival with nivolumab plus chemotherapy, as compared with chemotherapy alone, in patients with a PD-L1 expression level of less than 1%; and overall survival with nivolumab monotherapy, as compared with chemotherapy, in patients with a PD-L1 expression level of 50% or more. Among the patients with a PD-L1 expression level of 1% or more, the median duration of overall survival was 17.1 months (95% confidence interval [CI], 15.0 to 20.1) with nivolumab plus ipilimumab and 14.9 months (95% CI, 12.7 to 16.7) with chemotherapy (P=0.007), with 2-year overall survival rates of 40.0% and 32.8%, respectively. The median duration of response was 23.2 months with nivolumab plus ipilimumab and 6.2 months with chemotherapy. The overall survival benefit was also observed in patients with a PD-L1 expression level of less than 1%, with a median duration of 17.2 months (95% CI, 12.8 to 22.0) with nivolumab plus ipilimumab and 12.2 months (95% CI, 9.2 to 14.3) with chemotherapy. Among all the patients in the trial, the median duration of overall survival was 17.1 months (95% CI, 15.2 to 19.9) with nivolumab plus ipilimumab and 13.9 months (95% CI, 12.2 to 15.1) with chemotherapy. The percentage of patients with grade 3 or 4 treatment-related adverse events in the overall population was 32.8% with nivolumab plus ipilimumab and 36.0% with chemotherapy. The authors concluded that first-line treatment with nivolumab plus ipilimumab resulted in a longer duration of overall survival than did chemotherapy in patients with NSCLC, independent of the PD-L1 expression level. No new safety concerns emerged with longer follow-up.

On March 4, 2022, the U.S. Food and Drug Administration (FDA) approved nivolumab (Opdivo, Bristol-Myers Squibb Company) with platinum-doublet chemotherapy for adult patients with recectable non-small cell lung cancer (NSCLC) in the neoadjuvant setting. The FDA approval was based on supporting data from the CHECKMATE-816 study (FDA, 2022a).

In the CHECKMATE-816 study, an open-label, phase 3, randomized trial, Forde and colleagues (2022) evaluated the efficacy of nivolumab plus platinum-based chemotherapy in patients with stage IB to IIIA resectable NSCLC in the neoadjuvant setting. Patients were randomized to receive nivolumab plus platinum-based chemotherapy or platinum-based chemotherapy alone, followed by resection. Primary endpoints included event-free survival and pathological complete response (0% viable tumor in resected lung and lymph nodes). Overall survival was a secondary endpoint and the safety of all treated patients was assessed. Median event-free survival was 31.6 months (95% confidence interval [CI], 30.2 to not

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was assessed. Median event-free survival was 31.6 months (95% confidence interval [CI], 30.2 to not reached) with nivolumab plus chemotherapy and 20.8 months (95% CI, 14.0 to 26.7) with chemotherapy alone (hazard ratio for disease progression, disease recurrence, or death, 0.63; 97.38%CI, 0.43 to 0.91; p = 0.005). Patients with a pathological complete response was 24.0% (95% CI, 18.0 to 31.0) and 2.2% (95% CI, 0.6 to 5.6), respectively (odds ratio, 13.94; 99% CI, 3.49to 55.75; p<0.001). Across most subgroups, results for event-free survival and pathological complete response favored nivolumab plus chemotherapy versus chemotherapy alone. Based on patients undergoing randomization, 83.2% of those in the nivolumab-plus chemotherapy arm and 75.4% of those in the chemotherapy-alone arm underwent surgery. Grade 3 or 4 treatment-related adverse events were noted in 33.5% of the patients in the nivolumab plus chemotherapy arm and in 36.9% of those in the chemotherapy alone arm. The investigators concluded that patients with resectable NSCLC treated with neoadjuvant nivolumab plus chemotherapy resulted in significantly longer event-free survival and a greater percentage of patients with a pathological complete response than chemotherapy alone. Furthermore, the addition of nivolumab to neoadjuvant chemotherapy did not increase adverse event occurrence or hinder suitability of surgery.

Lung Cancer - Small Cell Lung Cancer

On August 16, 2018, the Food and Drug Administration granted accelerated approval to nivolumab (Opdivo, Bristol-Myers Squibb Company Inc.) for patients with metastatic small cell lung cancer (SCLC) with progression after platinum-based chemotherapy and at least one other line of therapy. Approval was based on demonstration of a durable overall response rate (ORR) in a subgroup of patients from CheckMate-032 (NCT01928394), a multicenter, open-label trial in patients with metastatic solid tumors. This subgroup comprised 109 patients with metastatic SCLC, with disease progression after platinum-based therapy and at least one other prior line of therapy, regardless of tumor PD-L1 status. All patients received nivolumab 3 mg/kg by intravenous infusion over 60 minutes every 2 weeks. The major efficacy outcome measures were overall response rate (ORR) and duration of response according to RECIST v1.1 as assessed by blinded independent central review. The ORR was 12% (95% CI: 6.5, 19.5). Responses were durable for 6 months or longer in 77%, 12 months or longer in 62%, and 18 months or longer in 39% of the 13 responding patients. PD-L1 tumor status did not appear to be predictive of response.

Safety data was evaluated in 245 patients with metastatic SCLC with disease progression following platinum-based chemotherapy and received at least one dose of nivolumab at a dose of 3 mg/kg every 2 weeks. The most common (≥20%) adverse reactions in CheckMate-032 were fatigue, decreased appetite, musculoskeletal pain, dyspnea, nausea, diarrhea, constipation and cough. Nivolumab was discontinued for adverse reactions in 10% of patients and 25% of patients had at least one dose withheld for an adverse reaction. Serious adverse reactions occurred in 45% of patients. The most frequent (≥2%) serious adverse reactions were pneumonia, dyspnea, pneumonitis, pleural effusion, and dehydration.

The recommended dose and schedule of nivolumab for this indication is 240 mg every 2 weeks over 30 min.

Antonia, et al. (2016) assessed safety and activity of nivolumab and nivolumab plus ipilimumab in

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Antonia, et al. (2016) assessed safety and activity of nivolumab and nivolumab plus ipilimumab in patients with small cell lung cancer (SCLC) who progressed after one or more previous regimens. The SCLC cohort of this phase 1/2 multicenter, multi-arm, open-label trial was conducted at 23 sites (academic centers and hospitals) in six countries. Eligible patients were 18 years of age or older, had limited-stage or extensive-stage SCLC, and had disease progression after at least one previous platinum-containing regimen. Patients received nivolumab (3 mg/kg bodyweight intravenously) every 2 weeks (given until disease progression or unacceptable toxicity), or nivolumab plus ipilimumab (1 mg/kg plus 1 mg/kg, 1 mg/kg plus 3 mg/kg, or 3 mg/kg plus 1 mg/kg, intravenously) every 3 weeks for four cycles, followed by nivolumab 3 mg/kg every 2 weeks. Patients were either assigned to nivolumab monotherapy or assessed in a dose-escalating safety phase for the nivolumab/ipilimumab combination beginning at nivolumab 1 mg/kg plus ipilimumab 1 mg/kg. Depending on tolerability, patients were then assigned to nivolumab 1 mg/kg plus ipilimumab 3 mg/kg or nivolumab 3 mg/kg plus ipilimumab 1 mg/kg. The primary endpoint was objective response by investigator assessment. All analyses included patients who were enrolled at least 90 days before database lock. Between Nov 18, 2013, and July 28, 2015, 216 patients were enrolled and treated (98 with nivolumab 3 mg/kg, three with nivolumab 1 mg/kg plus ipilimumab 1 mg/kg, 61 with nivolumab 1 mg/kg plus ipilimumab 3 mg/kg, and 54 with nivolumab 3 mg/kg plus ipilimumab 1 mg/kg). At database lock on Nov 6, 2015, median follow-up for patients continuing in the study (including those who had died or discontinued treatment) was 198·5 days (IQR 163·0-464·0) for nivolumab 3 mg/kg, 302 days (IQR not calculable) for nivolumab 1 mg/kg plus ipilimumab 1 mg/kg, 361·0 days (273·0-470·0) for nivolumab 1 mg/kg plus ipilimumab 3 mg/kg, and 260·5 days (248·0-288·0) for nivolumab 3 mg/kg plus ipilimumab 1 mg/kg. An objective response was achieved in ten (10%) of 98 patients receiving nivolumab 3 mg/kg, one (33%) of three patients receiving nivolumab 1 mg/kg plus ipilimumab 1 mg/kg, 14 (23%) of 61 receiving nivolumab 1 mg/kg plus ipilimumab 3 mg/kg, and ten (19%) of 54 receiving nivolumab 3 mg/kg plus ipilimumab 1 mg/kg. Grade 3 or 4 treatment-related adverse events occurred in 13 (13%) patients in the nivolumab 3 mg/kg cohort, 18 (30%) in the nivolumab 1 mg/kg plus ipilimumab 3 mg/kg cohort, and ten (19%) in the nivolumab 3 mg/kg plus ipilimumab 1 mg/kg cohort; the most commonly reported grade 3 or 4 treatment-related adverse events were increased lipase (none vs 5 [8%] vs none) and diarrhea (none vs 3 [5%] vs 1 [2%]). No patients in the nivolumab 1 mg/kg plus ipilimumab 1 mg/kg cohort had a grade 3 or 4 treatment-related adverse event. Six (6%) patients in the nivolumab 3 mg/kg group, seven (11%) in the nivolumab 1 mg/kg plus ipilimumab 3 mg/kg group, and four (7%) in the nivolumab 3 mg/kg plus ipilimumab 1 mg/kg group discontinued treatment due to treatment-related adverse events. Two patients who received nivolumab 1 mg/kg plus ipilimumab 3 mg/kg died from treatment-related adverse events (myasthenia gravis and worsening of renal failure), and one patient who received nivolumab 3 mg/kg plus ipilimumab 1 mg/kg died from treatment-related pneumonitis. The investigators stated that these data support the evaluation of nivolumab and nivolumab plus ipilimumab in phase 3 randomized controlled trials in SCLC.

On December 29, 2020, Bristol Myers Squibb (BMS) released a statement on their intent to withdraw Opdivo for small cell lung cancer (SCLC) from the U.S. market. After consulting with the FDA, BMS stated that they took this action in accordance with the Agency's standard procedures for evaluating accelerated approvals that have not met their post-marketing requirements and as part of a broader industry-wide evaluation. Patients who are being treated with Opdivo for SCLC should consult with

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industry-wide evaluation. Patients who are being treated with Opdivo for SCLC should consult with their healthcare provider in all aspects of their care.

Malignant Pleural Mesothelioma

Scherpereel, et al. (2017) reported on a multicenter randomized non comparative phase 2 trial of nivolumab and ipilimumab for malignant pleural mesothelioma. Eligible patients were over 18 years of age, performance status (PS) 0-1, histologically proven malignant pleural mesothelioma (MPM) relapsing after 1 or 2 prior lines including pemetrexed/platinum doublet, measurable disease. Randomized patients (1:1) received nivolumab 3 mg/kg every two weeks, or nivolumab 3 mg/kg every two weeks plus ipilimumab 1 mg/kg every 6 weeks, until progression or unacceptable toxicity. Primary endpoint was disease control rate (DCR) at 12 weeks with a blinded independent central review (BICR). 114 patients were to be randomized (with 108 eligible), with one-step Fleming procedure, H0 P<20% versus H1 P>40%, with 95% power, 5% one-sided a-risk: greater than or equal to 17 failure-free patients had to be observed at 12 weeks in either arm, to conclude to the activity of the corresponding regimen. From March to August 2016, 125 patients were enrolled in 21 centers. Eighty percent of study subjects were males, median age of subjects was 71.8 years (range 32.5-88.1). 62.4% were performance status 1, 83.2% had epithelioid cancers, 69.6% had one previous line of therapy, 70% of patients received 3 or more cycles of either treatment. Twelve weeks-DCR assessed by BICR in the first 108 eligible patients was 42.6% [95% CI: 29.4-55.8%] with nivolumab (n=23/54), and 51.9% [95% CI: 38.5%-65.2%] with nivolumab plus ipilimumab (n=28/54). Overall response rate (ORR) was 16.7% [95% CI: 6.7%-26.6%] with nivolumab (n=9/54), and 25.9% [95% CI: 14.2%-37.6%] with nivolumab plus ipilimumab (n=14/54). All grade/G3-4 toxicities were slightly increased in the combination nivolumab plus ipilimumab arm (86.9%/16.4%) versus nivolumab alone (77.8%/9.5%). Three treatment-related deaths were observed in the combination nivolumab/ipilimumab arm (1 metabolic encephalopathy, 1 fulminant hepatitis, 1 acute renal failure). The investigators concluded that both nivolumab and nivolumab plus ipilimumab arms reached their endpoint in 2nd/3rd-line MPM pts, suggesting that immunotherapy may provide new options for these patients.

Melanoma

Melanoma is a serious form of skin cancer that arises from melanocytes; however, in rare instances, it can originate in the eye or other non-skin organs. Risk factors for melanoma entail freckling, genetic factors, history of sunburns, light skin or eye color, poor tanning ability, sun exposure, as well as tanning bed use. About 80 % of melanomas are detected in a localized stage. When detected early, the 5-year survival rate of melanoma is 98.5 %. However, when melanoma is diagnosed after distant metastasis, the 5-year survival rate decreases to 15 % with a median survival between 8 and 9 months. Tumor thickness, along with nodal involvement, is a prognostic factor for melanoma. As tumor thickness increases to greater than 1.0 mm, the survival rate is reduced by 50 %. The incidence of melanoma is rapidly increasing. According to the National Cancer Institute (NCI), melanoma is the leading cause of death from skin disease. Approximately 76.100 new cases of melanoma will be diagnosed in the United States in 2014 and about 9,700 people will die from the disease. Treatments of melanoma include chemotherapy, immunotherapy, radiation therapy, surgery, as well as vaccine therapy (NCI, 2014).

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chemotherapy, immunotherapy, radiation therapy, surgery, as well as vaccine therapy (NCI, 2014).

The past several years were marked by the emergence of new molecules for the treatment of metastatic cutaneous melanoma with a significant benefit on the survival -- ipilimumab (Yervoy), peginterferon alfa-2b (PEG-Intron) and vemurafenib (Zelboraf) were approved by the Food and Drug Administration (FDA) in 2011; dabrafenib (Tafinlar) and trametinib (Mekinist) were approved in 2013. More recently, much attention has been devoted to the blockade of programmed death 1 (PD-1) signaling to activate an immune response to cancer. PD-1, a protein expressed on T cells, is a member of the CD28 superfamily, and it transmits co-inhibitory signals upon engagement with its ligands PD-L1 and PD-L2. Accumulating evidence suggests that the PD-1 system plays pivotal roles in the regulation of autoimmunity, transplantation immunity, infectious immunity, and tumor immunity. Because the interaction of PD-1 with its ligands occurs in the effector phase of killer T cell responses in peripheral blood, anti-PD-1 and anti-PD-L1 monoclonal antibodies are ideal as specific agents to augment T cell responses to tumors with fewer adverse events than with the inhibition of CTLA-4, because the interaction of CTLA-4 with its ligands occurs in the priming phase of T cell responses within lymph nodes. In recent phase I clinical trials, objective responses were observed in patients with melanoma, renal cell carcinoma (RCC), and non-small cell lung cancer (NSCLC) who underwent immunotherapy with an anti-PD-1 monoclonal antibody. In addition, the anti-tumor activity of an anti-PD-L1 monoclonal antibody was observed in patients with melanoma, RCC, NSCLC, and ovarian cancer (Tanaka and Okamura, 2013).

Opdivo (nivolumab) is a human programmed death receptor-1 (PD-1)-blocking monoclonal antibody. Nivolumab, a fully human IgG4 monoclonal antibody, acts as an immunomodulator by blocking ligand activation of the PD-1 receptor on activated T-cells.

Binding of the programmed death-1 (PD-1) ligands, PD-L1 and PD-L2, to the PD-1 receptor found on T cells, inhibits T cell proliferation and cytokine production. Upregulation of PD-1 ligands occurs in some tumors and signaling through this pathway can contribute to inhibition of active T-cell immune surveillance of tumors. Opdivo (nivolumab) is a monoclonal antibody that binds to the PD-1 receptor and blocks its interaction with PD-L1 and PD-L2, releasing PD-1 pathway-mediated inhibition of the immune response, including the anti-tumor immune response. In syngeneic mouse tumor models, blocking PD-1 activity resulted in decreased tumor growth.

Opdivo (nivolumab) is indicated for the treatment of patients with unresectable or metastatic melanoma and disease progression following ipilimumab and, if BRAF V600 mutation positive, a BRAF inhibitor. This indication is approved under accelerated approval based on tumor response rate and durability of response. Opdivo (nivolumab) is also indicated for the treatment of patients with metastatic squamous non-small cell lung cancer (NSCLC) with progression on or after platinum-based chemotherapy.

In a clinical study, Topalian et al (2014) examined the survival, durable tumor remission, and long-term safety of nivolumab in patients with advanced melanoma. Patients with advanced melanoma (n = 107) enrolled between 2008 and 2012 received intravenous nivolumab in an out-patient setting every 2 weeks

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enrolled between 2008 and 2012 received intravenous nivolumab in an out-patient setting every 2 weeks for up to 96 weeks and were observed for overall survival (OS), long-term safety, and response duration after treatment discontinuation. Median OS in nivolumab-treated patients (62 % with 2 to 5 prior systemic therapies) was 16.8 months, and 1- and 2-year survival rates were 62 % and 43 %, respectively. Among 33 patients with objective tumor regressions (31 %), the Kaplan-Meier estimated median response duration was 2 years. Furthermore, 17 patients discontinued therapy for reasons other than disease progression, and 12 (71 %) of 17 maintained responses off-therapy for at least 16 weeks (range of 16 to 56+ weeks). Objective response and toxicity rates were similar to those reported previously; in an extended analysis of all 306 patients treated on this trial (including those with other cancer types), exposure-adjusted toxicity rates were not cumulative. The authors concluded that OS following nivolumab treatment in patients with advanced treatment-refractory melanoma compared favorably with that in literature studies of similar patient populations. Responses were durable and persisted after drug discontinuation; long-term safety was acceptable. They stated that ongoing randomized controlled trials (RCTs) will further evaluate the impact of nivolumab therapy on OS in patients with metastatic melanoma.

A phase III clinical trial testing nivolumab for the treatment of advanced melanoma was stopped early due to improved OS compared with standard chemotherapy (No authors listed, 2014).

On December 22, 2014, the FDA approved nivolumab (Opdivo) for patients with unresectable or metastatic melanoma who no longer respond to other drugs (ipilimumab and a BRAF inhibitor). Opdivo was approved under the FDA's accelerated approval program; its efficacy was demonstrated in 120 clinical trial participants with unresectable or metastatic melanoma. Results showed that 32 % of subjects receiving Opdivo had their tumors shrink (objective response rate [ORR]). This effect lasted for more than 6 months in approximately 1/3 of the subjects who experienced tumor shrinkage. Opdivo's safety was evaluated in the overall trial population of 268 participants treated with Opdivo and 102 participants treated with chemotherapy. The most common side effects of the drug were rash, itching, cough, upper respiratory tract infections, and edema. The most serious side effects were severe immune-mediated side effects involving healthy organs, including the colon, kidneys, liver, lung, and endocrine glands. Serious adverse reactions occurred in 41 % of patients receiving Opdivo; grade 3 and 4 adverse reactions occurred in 42 % of patients receiving Opdivo. The most frequent grade 3 and 4 adverse drug reactions reported in 2 % to less than 5 % of patients receiving Opdivo were abdominal pain, hyponatremia, increased aspartate aminotransferase, and increased lipase. The most common adverse reaction (greater than or equal to 20 %) reported with Opdivo was rash (21 %). Opdivo is administered as an intravenous infusion (3 mg/kg) over 60 minutes every 2 weeks.

Melanoma - Adjuvant Therapy

On December 20, 2017, the Food and Drug Administration approved nivolumab (Opdivo) for the adjuvant treatment of patients with melanoma with involvement of lymph nodes or in patients with metastatic disease who have undergone complete resection. FDA approval was based on improvement in recurrence-free survival (RFS) as reported in the CHECKMATE-238 trial.

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Weber et al (2017) stated that nivolumab and ipilimumab are immune checkpoint inhibitors that have been approved for the treatment of advanced melanoma. In the United States, ipilimumab has also been approved as adjuvant therapy for melanoma on the basis of recurrence-free and overall survival rates that were higher than those with placebo in a phase 3 trial. The objective of this study (CheckMate 238) was to determine the efficacy of nivolumab versus ipilimumab for adjuvant therapy in patients with resected advanced melanoma. This was a randomized, double-blind, phase 3 trial where 906 patients (≥15 years of age) who were undergoing complete resection of stage IIIB, IIIC, or IV melanoma were randomly assigned to receive an intravenous infusion of either nivolumab at a dose of 3 mg per kilogram of body weight every 2 weeks (453 patients) or ipilimumab at a dose of 10 mg per kilogram every 3 weeks for four doses and then every 12 weeks (453 patients). The patients were treated for a period of up to 1 year or until disease recurrence, a report of unacceptable toxic effects, or withdrawal of consent. The primary end point was recurrence-free survival in the intention-to-treat population. At a minimum follow-up of 18 months, the 12-month rate of recurrence-free survival was 70.5% (95% confidence interval [CI], 66.1 to 74.5) in the nivolumab group and 60.8% (95% CI, 56.0 to 65.2) in the ipilimumab group (hazard ratio for disease recurrence or death, 0.65; 97.56% CI, 0.51 to 0.83; P<0.001). Treatment-related grade 3 or 4 adverse events were reported in 14.4% of the patients in the nivolumab group and in 45.9% of those in the ipilimumab group; treatment was discontinued because of any adverse event in 9.7% and 42.6% of the patients, respectively. Two deaths (0.4%) related to toxic effects were reported in the ipilimumab group more than 100 days after treatment. The study authors concluded that among patients undergoing resection of stage IIIB, IIIC, or IV melanoma, adjuvant therapy with nivolumab resulted in significantly longer recurrence-free survival and a lower rate of grade 3 or 4 adverse events than adjuvant therapy with ipilimumab.

Walocko et al (2016) stated that Merkel cell carcinoma (MCC) is an aggressive cutaneous neuroendocrine malignancy with limited treatment options. Several lines of evidence support the programmed death-1/programmed death-ligand 1 (PD-1/PD-L1) axis as a likely contributor to immune evasion in MCC. This was a case report of a patient with metastatic MCC with a significant and durable response to nivolumab, a humanized IgG4 monoclonal anti-PD-1 antibody. The authors concluded that immunotherapy with PD-1/PD-L1 inhibitors has become a rational and promising treatment option for MCC in the advanced or metastatic disease. Clinical trials are currently in progress to further evaluate these novel therapeutic agents.

On October 13, 2023, the U.S. Food and Drug Administration (FDA) approved nivolumab (Opdivo) for the adjuvant treatment of completely resected Stage IIB/C melanoma in patients 12 years and older. The FDA approval was based on supporting data from the CHECKMATE-76K study (FDA, 2023).

Kirkwood and colleagues (2023) evaluated the efficacy of nivolumab (Opdivo) in 790 patients with Stage IIB/C melanoma in the CHECKMATE-76K study, a randomized, double-blind trial. Patients were randomized (2:1) to Opdivo 480 mg or placebo by intravenous infusion every 4 weeks for up to 1 year or until disease recurrence or unacceptable toxicity. The major efficacy outcome measure was recurrence-free survival (RFS) defined as the investigator-assessed time between randomization and first recurrence (local, regional, or distant metastasis), new primary melanoma, or death (from any cause)

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recurrence (local, regional, or distant metastasis), new primary melanoma, or death (from any cause) whichever occurred first. Assessments were performed at 26 week intervals during years 1-3 and at 52 week intervals, thereafter, for 5 years. Median RFS was not reached in either the Opdivo arm (95% Confidence Interval [CI]: 28.5, not reached) or in the placebo arm (95% CI: 21.6, nor reached) (hazard ratio = 0.42 [95% CI: 0.30, 0.59]; p-value = < 0.0001).

Melanoma - Advanced; Combined Ipilimumab and Nivolumab

Postow and colleagues (2015) noted that in a phase I dose-escalation study, combined inhibition of T-cell checkpoint pathways by nivolumab and ipilimumab was associated with a high rate of objective response, including CRs, among patients with advanced melanoma. In this double-blind study involving 142 patients with metastatic melanoma who had not previously received treatment, these researchers randomly assigned patients in a 2:1 ratio to receive ipilimumab (3 mg/kg of body weight) combined with either nivolumab (1 mg/kg) or placebo once every 3 weeks for 4 doses, followed by nivolumab (3 mg/kg) or placebo every 2 weeks until the occurrence of disease progression or unacceptable toxic effects. The primary end-point was the rate of investigator-assessed, confirmed objective response among patients with BRAF V600 wild-type tumors. Among patients with BRAF wild-type tumors, the rate of confirmed objective response was 61 % (44 of 72 patients) in the group that received both ipilimumab and nivolumab (combination group) versus 11 % (4 of 37 patients) in the group that received ipilimumab and placebo (ipilimumab-monotherapy group) ($p < 0.001$), with CRs reported in 16 patients (22 %) in the combination group and no patients in the ipilimumab-monotherapy group. The median duration of response was not reached in either group. The median PFS was not reached with the combination therapy and was 4.4 months with ipilimumab monotherapy (HR associated with combination therapy as compared with ipilimumab monotherapy for disease progression or death, 0.40; 95 % CI: 0.23 to 0.68; $p < 0.001$). Similar results for response rate and PFS were observed in 33 patients with BRAF mutation-positive tumors. Drug-related adverse events of grade 3 or 4 were reported in 54 % of the patients who received the combination therapy as compared with 24 % of the patients who received ipilimumab monotherapy. Select adverse events with potential immunologic causes were consistent with those in a phase I study, and most of these events resolved with immune-modulating medication. The authors concluded that the objective-response rate and the PFS among patients with advanced melanoma who had not previously received treatment were significantly greater with nivolumab combined with ipilimumab than with ipilimumab monotherapy; combination therapy had an acceptable safety profile.

In a randomized, double-blind, phase III clinical trial, Larkin and associates (2015) compared nivolumab alone or nivolumab plus ipilimumab with ipilimumab alone in patients with metastatic melanoma. These investigators assigned, in a 1:1:1 ratio, 945 previously untreated patients with unresectable stage III or IV melanoma to nivolumab alone, nivolumab plus ipilimumab, or ipilimumab alone; PFS and OS were co-primary end-points. Results regarding PFS were presented here. The median PFS was 11.5 months (95 % CI: 8.9 to 16.7) with nivolumab plus ipilimumab, as compared with 2.9 months (95 % CI: 2.8 to 3.4) with ipilimumab (HR for death or disease progression, 0.42; 99.5 % CI: 0.31 to 0.57; $p < 0.001$), and 6.9 months (95 % CI: 4.3 to 9.5) with nivolumab (HR for the comparison with ipilimumab, 0.57; 99.5 % CI: 0.43 to 0.76; $p < 0.001$). In patients with tumors positive for the PD-L1, the median PFS was

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99.5 % CI: 0.45 to 0.76, p < 0.001). In patients with tumors positive for the PD-L1, the median PFS was 14.0 months in the nivolumab-plus-ipilimumab group and in the nivolumab group, but in patients with PD-L1-negative tumors, PFS was longer with the combination therapy than with nivolumab alone (11.2 months [95 % CI: 8.0 to not reached] versus 5.3 months [95 % CI: 2.8 to 7.1]). Treatment-related adverse events of grade 3 or 4 occurred in 16.3 % of the patients in the nivolumab group, 55.0 % of those in the nivolumab-plus-ipilimumab group, and 27.3 % of those in the ipilimumab group. The authors concluded that among previously untreated patients with metastatic melanoma, nivolumab alone or combined with ipilimumab resulted in significantly longer PFS than ipilimumab alone. In patients with PD-L1-negative tumors, the combination of PD-1 and CTLA-4 blockade was more effective than either agent alone.

Piulats et al (ongoing clinical trial) stated uveal melanoma is the most common primary intraocular malignant tumor in adults. Overall Survival (OS) at 5 years(y) is 62% due high incidence of liver metastasis, fatal within 4-9 months (m) from diagnosis. No standard treatment exists for metastatic uveal melanoma (MUM). Combination nivolumab (NIVO) and ipilimumab (IPI) has shown efficacy in metastatic cutaneous melanoma. However, MUM patients were excluded in these trials. GEM1402 is a phase-2 trial evaluating NIVO+IPI in untreated adult patients with MUM; is being conducted in 10 centers in Spain, leading by the Spanish Melanoma Group. Eligible patients had histologically-confirmed MUM, ECOG-PS 0/1, and no prior systemic treatment for MUM. Treatment consisted in NIVO (1mg/kg, iv, q3 weeks [wk]) and 4 doses of IPI (3mg/kg iv q3wk) followed by NIVO (3mg/kg q2wk) until progressive disease (PD), toxicity or withdrawal. Primary endpoint is OS and secondary progression free survival (PFS), Overall Response Rate (ORR) (per RECIST 1.1) and safety. Radiologic evaluations q6wk. Interim analysis (n = 19) was planned per protocol to assess safety and ORR. Intention to treat analysis includes patients with PD at first radiological evaluation. Safety population includes all patients receiving at least one dose of study treatment. Nineteen patients enrolled from April to July 2016: Median age 62y (43y-82y), 63% male, liver M1 84% patients and extra-liver M1 42% patients, 31% elevated baseline LDH. 11 patients completed cycle 2 and 8 patients stopped after 1 dose (6 PD, 2 toxicity). Treatment-related adverse events were reported in 12 patients and lead to end of treatment in 2 patients. Grade ≥3 toxicities were seen in 7 patients (36.8%): diarrhea, transaminitis, dermatological events, anemia, acute thyroiditis. All G3/4 were resolved following the toxicity guideline. One G5 acute thyroiditis related to NIVO+IPI was reported. ORR was observed in 15.8% and disease stabilization in 47.4%. With a median follow-up of 4.6m, PFS was 4.99m. Median OS was not reached at time of this analysis. The authors concluded combination of NIVO+IPI is feasible for MUM. In this INTERIM ANALYSIS, ORR did not reach yet 20%, but PFS seems promising. The clinical trial is ongoing and final results will be updated.

Merkel Cell Carcinoma

Walocko et al (2016) stated that Merkel cell carcinoma (MCC) is an aggressive cutaneous neuroendocrine malignancy with limited treatment options. Several lines of evidence support the programmed death-1/programmed death-ligand 1 (PD-1/PD-L1) axis as a likely contributor to immune evasion in MCC. This was a case report of a patient with metastatic MCC with a significant and durable response to nivolumab, a humanized IgG4 monoclonal anti-PD-1 antibody. The authors concluded that

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response to nivolumab, a humanized IgG4 monoclonal anti-PD-1 antibody. The authors concluded that immunotherapy with PD-1/PD-L1 inhibitors has become a rational and promising treatment option for

MCC in the advanced or metastatic disease. Clinical trials are currently in progress to further evaluate these novel therapeutic agents.

Nivolumab for the Treatment of Astrocytoma

National Comprehensive Cancer Network's clinical practice guideline on "Central Nervous System Cancers" (Version 1.2023) does not mention nivolumab as a therapeutic option for astrocytoma.

Nivolumab for the Treatment of Pancreatic Neuroendocrine Cancer

Klein et al (2020) stated that combination immunotherapy with anti-CTLA-4 and anti-PD-1 blockade has demonstrated significant clinical activity across several tumor types. Neuroendocrine tumors (NET) are a heterogeneous group of rare tumors with limited therapeutic options. CA209-538 is a prospective, multi-center clinical trial of combination immunotherapy with ipilimumab and nivolumab in rare cancers, including advanced NETs. Patients received treatment with nivolumab at a dose of 3 mg/kg and ipilimumab at 1 mg/kg every 3 weeks for 4 doses, followed by nivolumab 3 mg/kg every 2 weeks and continued for up to 96 weeks, until disease progression or the development of unacceptable toxicity. Response was assessed every 12 weeks by RECIST 1.1. The primary endpoint was clinical benefit rate (CBR; CR + PR + SD). A total of 29 patients with advanced NETs received treatment -- 3 (10 %) patients had low-grade, 13 (45 %) had intermediate-grade, and 13 (45 %) had high-grade tumors; lung was the most common primary site (39 %). The ORR was 24 % with a CBR of 72 %; 43 % of patients with pancreatic neuroendocrine neoplasms (NEN), and 33 % of patients with atypical bronchial carcinoid achieved an objective response. The median PFS was 4.8 months (95 % CI: 2.7 to 10.5) and OS was 14.8 months (95 % CI: 4.1 to 21.3). Immune-related toxicity was reported in 66 % of patients with 34 % experiencing grade 3/4 events. The authors concluded that combination immunotherapy with ipilimumab and nivolumab demonstrated significant clinical activity in subgroups of patients with advanced NETs including patients with atypical bronchial carcinoid and high-grade pancreatic NENs.

Al-Toubah et al (2022) stated that dual checkpoint inhibitor therapy with anti-CTLA-4 and anti-PD-1 blockade has shown promising results in patients with high-grade NENs, demonstrating varying response rates of 9 % to 44 %. More data are needed to examine the true response in a real-world cohort of patients. In a retrospective study, patients with high-grade NENs were treated at the Moffitt Cancer Center and Mayo Clinic between September 2017 and July 2020 who received combination therapy with ipilimumab and nivolumab. A total of 34 patients met the eligibility criteria. subjects had received an average of 2 prior lines of therapy, including at least 1 cytotoxic chemotherapy regimen; 27 (79.4 %) patients had poorly differentiated neuroendocrine carcinomas, and 7 (20.6 %) had well-differentiated high-grade neuroendocrine tumors. The most common primary site (10, 29.4 %) was pancreas; other primary sites of disease included colon (n = 5), endometrium (n = 3), anorectum (n = 2), esophagus (n = 2), cervix (n = 1), stomach (n = 1), small intestine (n = 1), and unknown primary (n = 9). Five patients (14.7 %) exhibited a best response of PR as per RECIST 1.1 criteria, 9 (26.5 %) SD, and 17 (50 %)

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(14.7 %) exhibited a best response of PR as per RECIST 1.1 criteria, 9 (26.5 %) SD, and 17 (50 %) progressive disease: 3 patients did not have a follow-up scan as they discontinued treatment shortly after initiation due to clinical progression. The ORR was 14.7 %, and DCR was 41.2 %. Median PFS was 1 month (95 % CI: 0.54 to 1.46 months); median OS from time of treatment initiation was 5.0 months (95 % CI: 4.07 to 5.93 months), and median OS from diagnosis was 14.0 months (95 % CI: 11.79 to 16.21 months). The median duration of treatment was 1 month (range of 0 to 10 months); 28 patients discontinued treatment for progression, 4 patients for toxicity, and 2 remained on treatment at the time of data cut-off; 12 patients (35 %) experienced grade 3 and 4 treatment-emergent toxicities. The authors concluded that ipilimumab and nivolumab regimen exhibited modest activity in aggressive and heavily pre-treated high-grade NENs who have progressed on prior cytotoxic chemotherapy.

Kang et al (2021) noted that poorly differentiated pancreatic neuroendocrine carcinomas (NECs) are rare and aggressive malignancies with rapid disease progression and early widespread metastasis. Given similar histology, they are commonly treated with platinum-based chemotherapy as SCLC; however, no standard treatment has been established for recurrent or progressive disease. Furthermore, these researchers stated that there is limited data on subsequent therapy and no standard treatment has been established for recurrent poorly differentiated NEC. Some retrospective studies or case-series studies demonstrated limited clinical benefit with subsequent-line chemotherapy 9e.g., irinotecan, topotecan, and temozolomide). These investigators presented the case of an Asian patient with recurrent poorly differentiated pancreatic NEC after curative surgery and adjuvant chemotherapy with cisplatin and etoposide. The tumor mutational burden (TMB) was high. The patient received chemotherapy combined with maintenance immunotherapy with nivolumab and achieved promising and durable response, suggesting TMB could be a biomarker to identify NEC patients for immune checkpoint inhibitor (ICI) treatment. Moreover, these researchers stated that further investigation is needed to search the predictive biomarkers of ICIs, which may have clinical benefit in a specific subset of patients with poorly differentiated pancreatic NECs.

Rosner et al (2022) stated that several anti-PD-1 antibodies such as cemiplimab, dostarlimab, nivolumab, and pembrolizumab and anti-PD-L1 antibodies such as atezolizumab, avelumab, and durvalumab have been approved, and many other inhibitors are under development, for the treatment of cancers including NSCLC; SCCHN; esophagus, gut, colorectal, and renal clear cell carcinoma; bladder cancer; melanoma; Merkel cell carcinoma; and Hodgkin's lymphoma. These investigators noted that targeting PD-1 or its ligand PD-L1 is a promising therapeutic approach for many types of cancer in which PD-L1 is over-expressed; however, data on PD-L1 expression levels in gastroenteropancreatic neuroendocrine neoplasms (GEP-NENs) are limited and contradictory. These researchers examined PD-L1 expression in 457 archived, formalin-fixed, paraffin-embedded GEP-NEN samples from 175 patients by immunohistochemistry using the highly sensitive monoclonal anti-PD-L1 antibody 73-10. The immunostaining was semi-quantitatively evaluated using a 12-point immunoreactivity score (IRS) taking both PD-L1-positive tumor cells and immune cells into account. Tumor samples with an IRS of 3 or higher were considered PD-L1-positive. Results were correlated with clinicopathological data and with the expression of several typical markers and receptors for neuroendocrine tumors. Of the GEP-NEN samples, 73 % were PD-L1-positive. The median IRS value across all samples was 4.0, corresponding to low expression. PD-L1 immunostaining was predominantly localized at the plasma

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corresponding to low expression. PD-L1 immunostaining was predominantly localized at the plasma membrane of the tumor cells. Positive correlations were observed between PD-L1 expression and tumor grading, or Ki-67 index, between PD-L1 expression and the expression of chromogranin A, and between PD-L1 expression and the expression of each of the 5 somatostatin receptors. PD-L1 expression was lower in tumors with lymph node metastases at diagnosis than in those without regional metastasis and lower in high-stage than in earlier-stage tumors. No association was noted between PD-L1 expression and patient survival. The authors concluded that using the highly sensitive antibody 73-10, the findings of this trial indicated that PD-L1 expression was common in GEP-NENs and increased with histological grading; thus, targeting the PD-1/PD-L1 axis may represent a promising therapeutic strategy, specifically in G3 tumors. Moreover, these researchers stated that given the discordant data in the literature, further studies of PD-L1 expression in GEP-NENs using highly sensitive antibodies (e.g., 73-10) and a scoring system taking both the staining intensity and the percentage of PD-L1-positive cells into account are strongly needed.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Neuroendocrine and adrenal tumors" (Version 1.2022) states that "In the event of locally advanced or metastatic disease with unfavorable biology, a clinical trial is a preferred option. Other recommended options include chemotherapy and the combination of nivolumab and ipilimumab (category 2B)".

Nivolumab Plus Ipilimumab for the Treatment of Localized Renal Cell Carcinoma After Nephrectomy

Motzer et al (2023) noted that effective adjuvant therapy for patients with resected localized RCC represents an unmet need, with surveillance being the standard of care (SOC). These investigators reported results from part A of a randomized, phase-III clinical trial (the CheckMate 914 Trial) that examined the safety and effectiveness of adjuvant nivolumab plus ipilimumab versus placebo. This trial enrolled patients with localized clear cell RCC who were at high-risk of relapse following radical or partial nephrectomy between 4 to 12 weeks before random assignment. Part A, reported in this study, was carried out in 145 hospitals and cancer centers across 20 countries. Patients were randomly assigned (1:1) to nivolumab (240 mg) intravenously every 2 weeks for 12 doses plus ipilimumab (1 mg/kg) intravenously every 6 weeks for 4 doses, or matching placebo, via an interactive response technology system. The expected treatment period was 24 weeks, and treatment could be continued until week 36, allowing for treatment delays. Randomization was stratified by TNM stage and nephrectomy (partial versus radical). The primary endpoint was DFS according to masked independent central review; safety was a secondary endpoint. DFS was analyzed in all randomly assigned patients (intention-to-treat [ITT] population); exposure, safety, and tolerability were analyzed in all patients who received at least 1 dose of study drug (all-treated population). Between August 28, 2017, and March 16, 2021, a total of 816 patients were randomly assigned to receive either adjuvant nivolumab plus ipilimumab (405 patients) or placebo (411 patients); 580 (71 %) of 816 patients were male and 236 (29 %) patients were female. With a median follow-up of 37.0 months (IQR 31.3 to 43.7), median DFS was not reached in the nivolumab plus ipilimumab group and was 50.7 months (95 % CI: 48.1 to not estimable) in the placebo group (hazard ratio 0.92, 95 % CI: 0.71 to 1.19; p = 0.53). The number of events required for the planned OS interim analysis was not reached at the time of the data cut-off, and only 61 events occurred (33 in the nivolumab plus ipilimumab group and 28 in the placebo group); 155

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only 61 events occurred (33 in the nivolumab plus ipilimumab group and 28 in the placebo group), 155 (38 %) of 404 patients who received nivolumab plus ipilimumab and 42 (10 %) of 407 patients who received placebo had grade-3 to grade-5 AEs. All-cause AEs of any grade resulted in discontinuation of nivolumab plus ipilimumab in 129 (32 %) of 404 treated patients and of placebo in 9 (2 %) of 407 treated patients; 4 deaths were attributed to treatment with nivolumab plus ipilimumab and no deaths were attributed to treatment with placebo. The authors concluded that adjuvant therapy with nivolumab plus ipilimumab did not improve DFS versus placebo in patients with localized RCC at high-risk of recurrence after nephrectomy. The findings of this study did not support this regimen for the adjuvant treatment of RCC.

Cosso et al (2023) stated that RCC is the 3rd most common genitourinary cancer accounting for approximately 180,000 deaths worldwide in 2020. Although over 2/3 of patients initially present localized disease, up to 50 % of them may progress to metastatic disease. Adjuvant therapy aims to reduce the recurrence risk and improve outcomes in several types of cancers but is currently an unmet need in RCC. The results achieved with tyrosine kinase inhibitors in metastatic RCC led to the evaluation of these target therapies in an early setting with conflicting results for DFS and no OS benefit. Similarly, the results of ICIs in an adjuvant setting are conflicting. Available data did not show an improvement in OS with ICIs in the early phase, although a positive trend for pembrolizumab has been recorded, receiving the FDA's approval in this setting. However, the disappointing results of several ICIs and the heterogeneous pattern of RCC warrant biomarker identification and subgroup analyses to examine which patients could benefit from adjuvant therapy.

Nivolumab Plus Ipilimumab for the Treatment of Merkel Cell Carcinoma

Kim et al (2022) noted that Merkel cell carcinoma is among the most aggressive and lethal of primary skin cancers, with a high rate of distant metastasis. Anti-PD-1 and anti-PD-L1 monotherapy is currently SOC for unresectable, recurrent, or metastatic Merkel cell carcinoma. In a randomized, open-label, phase-II clinical trial, these researchers examine the effectiveness of treatment with combined nivolumab plus ipilimumab, with or without stereotactic body radiotherapy (SBRT) in patients with advanced Merkel cell carcinoma as a 1st-line therapy or following previous treatment with anti-PD-1 and anti-PD-L1 monotherapy. These researchers randomly assigned adults from 2 U.S. cancer centers (1 in FL and 1 in OH) to group A (combined nivolumab and ipilimumab) or group B (combined nivolumab and ipilimumab plus SBRT) in a 1:1 ratio. Eligible patients were aged at least 18 years with histologically proven advanced stage (unresectable, recurrent, or stage IV) Merkel cell carcinoma, a minimum of 2 tumor lesions measurable by CT, MRI or clinical examination, and tumor tissue available for exploratory biomarker analysis. Patients were stratified by previous ICI status to receive nivolumab 240 mg intravenously every 2 weeks plus ipilimumab 1 mg/kg intravenously every 6 weeks (group A) or the same schedule of combined nivolumab and ipilimumab with the addition of SBRT to at least 1 tumor site (24 Gy in 3 fractions at week 2; group B). Patients had to have at least 2 measurable sites of disease so 1 non-irradiated site could be followed for response. The primary endpoint was ORR in all randomly assigned patients who received at least 1 dose of combined nivolumab and ipilimumab. ORR was defined as the proportion of patients with a CR or PR per immune-related RECIST. Response was evaluated every 12 weeks; and safety was evaluated in all patients. A total of 50 patients (25 in both

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evaluated every 12 weeks, and safety was evaluated in all patients. A total of 50 patients (25 in both group A and group B) were enrolled between March 14, 2017, and December 21, 2021, including 24 ICI-naive patients (13 [52 %] of 25 group A patients and 11 [44 %] of 25 group B patients]) and 26 patients with previous ICI (12 [48 %] of 25 group A patients and 14 [56 %] of 25 group B patients]). One patient in group B did not receive SBRT due to concerns about excess toxicity. Median follow-up was 14.6 months (IQR 9.1 to 26.5). Two patients in group B were excluded from the analysis of the primary endpoint because the target lesions were irradiated and so the patients were deemed non-evaluable. Of the ICI-naive patients, 22 (100 %) of 22 (95 % CI: 82 to 100) had an objective response, including 9 (41 % [95 % CI: 21 to 63]) with CR. Of the patients who had previously had ICI exposure, 8 (31 %) of 26 patients (95 % CI: 15 to 52) had an objective response and 4(15 % [5 to 36]) had a CR. No significant differences in ORR were observed between groups A (18 [72 %] of 25 patients) and B (12 [52 %] of 23 patients; p = 0.26). Grade-3 or grade-4 treatment-related AEs were observed in 10 (40 %) of 25 patients in group A and 8 (32 %) of 25 patients in group B. The authors concluded that 1st-line combined nivolumab and ipilimumab in patients with advanced Merkel cell carcinoma showed a high ORR with durable responses and an expected safety profile. Combined nivolumab and ipilimumab also showed clinical benefit in patients with previous anti-PD-1 and anti-PD-L1 treatment. Addition of SBRT did not improve effectiveness of combined nivolumab and ipilimumab. The combination of nivolumab and ipilimumab represents a new 1st-line and salvage therapeutic option for advanced Merkel cell carcinoma. These findings need to be validated in phase-III clinical trials.

Glutsch et al (2022) stated that Merkel cell carcinoma is a rare, highly aggressive skin cancer with neuroendocrine differentiation; ICI has significantly improved treatment outcomes in metastatic disease with response rates to PD-1/PD-L1 inhibition of up to 62 %. However, primary and secondary resistance to PD-1/PD-L1 inhibition remains a so far unsolved clinical challenge since safe and effective treatment options for these patients are lacking. In the prospective multi-center skin cancer registry ADOREG, 14 patients with advanced (non-resectable stage III or stage IV, Union international contre le cancer 2017) Merkel cell carcinoma with primary resistance to the PD-L1 inhibitor avelumab receiving subsequent therapy (2nd or later line) with ipilimumab plus nivolumab (IPI/NIVO) were identified; 5 of these 14 patients were reported previously and were included in this analysis with additional follow-up. Overall response rate, PFS, OS and AEs were analyzed. All 14 patients received avelumab as 1st-line treatment. Thereof, 12 patients had shown primary resistance with progressive disease in the 1st tumor assessment, while 2 patients had initially experienced a short-lived stabilization (stable disease); 6 patients had at least 1 systemic treatment in between avelumab and IPI/NIVO. A total of 7 patients responded to IPI/NIVO (ORR 50 %), and response was ongoing in 4 responders at last follow-up. After a median follow-up of 18.85 months, median PFS was 5.07 months (95 % CI: 2.43-not available (NA)), and median OS was not reached. PFS rates at 12 months and 24 months were 42.9 % and 26.8 %, respectively. The OS rate at 36 months was 64.3 %. Only 3 (21 %) patients did not receive all 4 cycles of IPI/NIVO due to immune-related AEs. The authors concluded that in this multi-center evaluation, they observed high response rates, a durable benefit and promising OS rates after treatment with later-line combined IPI/NIVO. These researchers stated that this patient cohort supported their previous findings with an encouraging activity of 2nd-line or later-line IPI/NIVO in patients with anti-PD-L1-refractory Merkel cell carcinoma. They stated that to further examine the effectiveness of IPI/NIVO in avelumab-refractory patients as well as to identify biomarkers of response prospective, randomized

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avelumab-refractory patients as well as to identify biomarkers of response prospective, randomized clinical trials are needed.

The authors stated that this trial had several drawbacks. The key drawbacks were the small number of patients as well as the registry-based data collection. Furthermore, AEs might be under-reported in this registry, which could be an additional explanation for the rather low rate of severe immune-related AEs in this cohort.

Nivolumab Plus Ipilimumab for the Treatment of Non-Small-Cell Lung Cancer

In a systematic literature review and network meta-analysis, O'Byrne et al (2023) examined the long-term safety and effectiveness of nivolumab in combination with ipilimumab (NIVO + IPI) relative to other immunotherapy (IO)-based regimens and chemotherapy in patients with 1st-line advanced NSCLC (aNSCLC). Phase-III RCTs with minimum 3-year follow-up examining IO-based regimens approved for 1st-line aNSCLC were identified via systematic literature review. Analytic populations were defined by levels of PD-L1 expression and histology. Due to presence of proportional hazards violations, time-varying HRs of OS and PFS were estimated via Bayesian fractional polynomial network meta-analysis. For safety endpoints, ORs were estimated using indirect treatment comparisons (ITCs). CheckMate 227, KEYNOTE-189, KEYNOTE-407, KEYNOTE-024, KEYNOTE-042, and IMpower150 were included in the base case analysis. For OS and PFS, HRs of NIVO + IPI relative to other IO-based regimens trended down-ward over time across analytic populations. The 36-month OS HRs of NIVO + IPI versus comparators were: 0.69 (95 % CI: 0.47 to 1.00) versus pembrolizumab + chemotherapy and 0.65 (0.45, 0.93) versus atezolizumab + bevacizumab + chemotherapy in the non-squamous and PD-L1 all-comers population; 0.73 (0.53 to 1.02) versus pembrolizumab + chemotherapy in the squamous and PD-L1 all-comers population; and 1.05 (0.83 to 1.32) versus pembrolizumab in the mixed histology and PD-L1 of greater than or equal to 50 % population. For PFS, 36-month HR point estimates ranged from 0.46 to 0.85 (only statistically significant versus pembrolizumab + chemotherapy in the squamous population; 0.46 [0.31 to 0.69]); AEs leading to discontinuation were not statistically significantly different between NIVO + IPI and pembrolizumab + chemotherapy, nor between NIVO + IPI and pembrolizumab monotherapy, although treatment-related grade-3 or higher AEs were higher with NIVO + IPI than pembrolizumab monotherapy (OR = 2.21 [1.30 to 3.75]). The authors concluded that the findings of this study revealed trends towards long-term benefit with NIVO + IPI compared with other IO-based combinations, with manageable toxicities.

The authors stated that this study had several drawbacks. First, evidence for comparators was restricted to data in the public domain which required IPD reconstruction from published Kaplan-Meier (KM) curves (rather than raw data) and precluded additional patient-level adjustments. For example, adjustments for heterogeneity in the use of subsequent IOs could not be implemented, yet the proportion of subsequent IO use ranged in the chemotherapy arms from 20 % to 66 %; and may have led to improved OS in the chemotherapy arm thereby shrinking the perceived relative OS benefit of 1st-line IO. Other features of the included RCTs, such as the open-label design and imbalances in baseline characteristics, also may have affected the internal validity of the effect estimates from individual RCTs, which may in turn have introduced bias in the indirect comparisons. Second, the assumption of

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which may in turn have introduced bias in the indirect comparisons. Second, the assumption of equivalence in effectiveness across different chemotherapy combinations. While this approach has been adopted elsewhere, sensitivity analyses revealed uncertainty around the equivalence of bevacizumab (BEV)-based triplet therapy and chemotherapy doublets, especially for PFS; hence, base-case estimates between NIVO + IPI and ATEZO + BEV + CARB + TAX may be conservative for NIVO + IPI. Third, these researchers were restricted from a methodological standpoint by having insufficient evidence to effectively estimate the between-study heterogeneity and hence implemented fixed-effect models as opposed to random-effects models. Fourth, while these investigators sought to capture long-term evidence for each target population of interest, updates and validation of these analyses may be needed once longer follow-up and/or additional subgroup data become available.

Nivolumab Plus Ipilimumab for the Treatment of Urethral/Urothelial Carcinoma of the Penis

Sharma et al (2019) noted that CheckMate 032 is an open-label, multi-cohort study that included patients with unresectable locally advanced or metastatic urothelial carcinoma (mUC) treated with nivolumab 3 mg/kg monotherapy every 2 weeks (NIVO3), nivolumab 3 mg/kg plus ipilimumab 1 mg/kg every 3 weeks for 4 doses followed by nivolumab monotherapy 3 mg/kg every 2 weeks (NIVO3+IPI1), or nivolumab 1 mg/kg plus ipilimumab 3 mg/kg every 3 weeks for 4 doses followed by nivolumab monotherapy 3 mg/kg every 2 weeks (NIVO1+IPI3). These investigators reported on the expanded NIVO1+IPI3 cohort and extended follow-up for the NIVO3 and NIVO3+IPI1 cohorts. Patients with platinum-pretreated mUC were enrolled in this phase-I/II multi-center clinical trial to receive NIVO3, NIVO3+IPI1, or NIVO1+IPI3 until disease progression or unacceptable toxicity. Primary endpoint was investigator-assessed ORR per RECIST version 1.1, including duration of response. A total of 78 patients were treated with NIVO3 (minimum follow-up, 37.7 months), 104 with NIVO3+IPI1 (minimum follow-up, 38.8 months), and 92 with NIVO1+IPI3 (minimum follow-up, 7.9 months); ORR was 25.6 %, 26.9 %, and 38.0 % in the NIVO3, NIVO3+IPI1, and NIVO1+IPI3 arms, respectively. Median duration of response was more than 22 months in all arms. Grade 3 or 4 treatment-related AEs occurred in 21 (26.9 %), 32 (30.8 %), and 36 (39.1 %) patients treated with NIVO3, NIVO3+IPI1, and NIVO1+IPI3, respectively. Grade 5 treatment-related pneumonitis occurred in 1 patient each in the NIVO3 and NIVO3+IPI1 arms. The authors concluded that with longer follow-up, NIVO3 demonstrated sustained anti-tumor activity alone and in combination with ipilimumab. NIVO1+IPI3 provided the greatest anti-tumor activity of all regimens, with a manageable safety profile. These researchers stated that these findings not only supported additional study of NIVO1+IPI3 in mUC; but demonstrated the potential benefit of immunotherapy combinations in this disease.

The authors stated that this trial was limited by the fact that it was not designed to directly compare outcomes among treatment arms, which each had a different length of follow-up, or with a standard current practice comparator. However, the ongoing phase-III CheckMate 901 clinical trial will further examine the NIVO1+IPI3 combination versus chemotherapy in patients with previously untreated mUC. The relatively small sample size in each arm of the current study could also be a limitation, especially in examining the effects of tumor PD-L1 expression on effectiveness. These researchers stated that longer-term follow-up is needed to further characterize the safety and effectiveness of the NIVO1+IPI3 combination.

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combination.

de Ruiter et al (2022) stated that muscle-invasive bladder cancer (MIBC) has a poor prognosis. Chemoradiotherapy (CRT) in selected patients has comparable results to radical cystectomy. Results of neoadjuvant ICIs before radical cystectomy are promising. In a multi-center, open-label, dose-escalation, phase-1b clinical trial, these researchers hypothesize that ICI concurrent to CRT (iCRT) is safe and may improve treatment outcomes. They determined the safety of CRT with 3 ICI regimens in patients with non-metastatic (T2-4aN0-1) MIBC. A total of 26 patients received mitomycin C/capecitabine and 20 × 2.75 Gy to the bladder. Tolerability was examined in a cohort of up to 10 patients. If 2 or fewer out of the first 6 patients or 3 or fewer of 10 patients experienced dose-limiting toxicity (DLT), accrual continued in the next cohort. Patients received nivolumab 480 mg (NIVO480), nivolumab 3 mg/kg and ipilimumab 1 mg/kg (NIVO3 + IPI1), or nivolumab 1 mg/kg and ipilimumab 3 mg/kg (IPI3 + NIVO1). The primary endpoint was safety; secondary objectives were response rate, disease-free survival (DFS), metastatic-free survival (MFS), and OS.

In the NIVO480 cohort, no patients experienced DLT. The NIVO3 + IPI1 2 patients experienced DLT, thrombocytopenia (grade 4), and asystole (grade 5). IPI3 + NIVO1 was discontinued after 3 out of 6 patients experienced DLT. Clinically significant AEs of grade 3 or higher occurred in 0, 3, and 5 patients in the NIVO480, NIVO3 + IPI1, and IPI3 + NIVO1 groups, respectively. The most common AEs were immune-related and gastro-intestinal (GI). MFS and OS were 90 % at 2 years for NIVO480 and 90 % at 1 year for NIVO3 + IPI1. The authors concluded that in this dose-finding study of iCRT, the regimens of nivolumab monotherapy and nivolumab 3 mg/kg with ipilimumab 1 mg/kg had acceptable toxicity. These investigators stated that drawbacks of this trial included the absence of a centralized pathology and radiology review, and a lack of biomarker analysis.

Nivolumab Plus Regorafenib for the Treatment of Metastatic Rectal Adenocarcinoma

Fukuota et al (2020) noted that immune suppressive cells such as regulatory T cells (Tregs) or tumor-associated macrophages (TAMs) may contribute to resistance to anti-PD-1/PD-L1 inhibitors (A-PD1). Regorafenib reduced TAMs in tumor models. The combination of regorafenib plus A-PD1 exhibited superior tumor growth suppression compared to either treatment alone in murine models. In an open-label, dose-finding, and dose-expansion phase-Ib clinical trial, these researchers enrolled patients with previously treated, advanced GC or CRC. Patients received regorafenib plus nivolumab in a dose-finding phase to estimate the maximum tolerated dose (MTD). Additional patients were enrolled in a dose-expansion phase to further establish the safety and determine the preliminary effectiveness. Regorafenib of 80 to 160 mg was administered once-daily for 21 days on 7 days off with intravenous (iV) nivolumab 3 mg/kg every 2 weeks. The primary endpoint was dose-limiting toxicity (DLT) during cycle 1 (4 weeks) to estimate the MTD and the recommended dose. A total of 50 patients were enrolled (25 GC; 25 CRC) until October 2018. The median prior treatment line was 3 (range of 2 to 8). During dose-escalation, 3 DLTs were observed with regorafenib 160 mg, including grade-3 (G3) maculopapular rash, mucositis and proteinuria, while there was no DLT with 80 mg or 120 mg. In the dose expansion cohort with regorafenib 120 mg, the dose was reduced to 80 mg owing to frequent G3 skin toxicities

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cohort with regorafenib 120 mg, the dose was reduced to 80 mg owing to frequent G3 skin toxicities. Grade greater than or equal to 3 treatment related AEs occurred in 17 patients, the common events (greater than 5 %) being rash (14 %), palmar-plantar erythron-dysesthesia (10 %), and proteinuria (8 %). Objective tumor response was observed in 19 patients (38 %) including 11 MSS GC, 7 MSS CRC and 1 MSI-H CRC for response rates of 44 % in GC and 29 % in MSS CRC; 3 of the 7 A-PD1 pre-treated GC patients achieved a PR. The pre- and post-treatment tumor samples showed a reduction of FoxP3hiCD45RA-Tregs fraction at the tumor response. The authors concluded that the combination of regorafenib 80-mg plus nivolumab provided a manageable safety profile and encouraging anti-tumor activity in MSS GC and CRC patients, which warrants further investigations in a larger cohort.

Furthermore, National Comprehensive Cancer Network's Drugs & Biologics Compendium (2022) lists rectal cancer as a recommended indication of regorafenib (category 2A); but only as a single-agent use.

Prostate Cancer

Dong et al (2014) noted that sipuleucel-T (Provenge) became the first approved cancer vaccine (developed to enhance T-cell immunity against metastatic castration-resistant prostate cancer) 4 years ago. Other prostate cancer vaccines, including a viral-based vaccine PROSTVAC-VF and a cellular vaccine GVAX, are in development. Moreover, several clinical trials are investigating the role of immune check-point blockade in the treatment of prostate cancer. Ipilimumab and nivolumab are potent T cell check-point inhibitors that reverse immunologic tolerance in multiple types of cancers.

Sarcomas

Uehara et al (2015) stated that although multi-modal therapies including surgery, chemotherapy, and radiotherapy have improved clinical outcomes of patients with bone and soft tissue sarcomas, the prognosis of patients has plateaued over these 20 years. Immunotherapies have shown the effectiveness for several types of advanced tumors. Immunotherapies, such as cytokine therapies, vaccinations, and adoptive cell transfers, have also been investigated for bone and soft tissue sarcomas. Cytokine therapies with interleukin (IL)-2 or interferons (IFNs) have limited efficacy because of their cytotoxicities. Liposomal muramyl tripeptide phosphatidyl-ethanolamine (L-MTP-PE), an activator of the innate immune system, has been approved as adjuvant therapeutics in combination with conventional chemotherapy in Europe, which has improved the 5-year OS of patients. Vaccinations and transfer of T cells transduced to express chimeric antigen receptors have shown some efficacy for sarcomas. Ipilimumab and nivolumab are monoclonal antibodies designed to inhibit immune checkpoint mechanisms. These antibodies have recently been shown to be effective for patients with melanoma and also investigated for patients with sarcomas. The authors provided an overview of various trials of immunotherapies for bone and soft tissue sarcomas, and discussed their potential as adjuvant therapies in combination with conventional therapies.

Roberts et al (2015) noted that pediatric sarcomas are a heterogeneous group of malignant tumors of bone and soft tissue origin. Although more than 100 different histologic subtypes have been described, the majority of pediatric cases belong to the Ewing's family of tumors, rhabdomyosarcoma and

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the majority of pediatric cases belong to the Ewing's family of tumors, rhabdomyosarcoma and osteosarcoma. Most patients that present with localized stage are curable with surgery and/or chemotherapy; however, those with metastatic disease at diagnosis or those who experience a relapse continue to have a very poor prognosis. New therapies for these patients are urgently needed. Immunotherapy is an established treatment modality for both liquid and solid tumors, and in pediatrics, most notably for neuroblastoma and osteosarcoma. In the past, immunomodulatory agents such as IFN, IL-2, and L-MTP-PE have been tried, with some activity seen in subsets of patients; additionally, various cancer vaccines have been studied with possible benefit. Monoclonal antibody therapies against tumor antigens such as disialoganglioside GD2 or immune checkpoint targets such as CTLA-4 and PD-1 are being actively explored in pediatric sarcomas. Building on the success of adoptive T cell therapy for EBV-related lymphoma, strategies to re-direct T cells using chimeric antigen receptors and bi-specific antibodies are rapidly evolving with potential for the treatment of sarcomas. These investigators focused on recent pre-clinical and clinical developments in targeted agents for pediatric sarcomas with emphasis on the immunobiology of immune checkpoints, immune-editing, tumor microenvironment, antibody engineering, cell engineering, and tumor vaccines.

D'Angelo et al (2018) stated patients with metastatic sarcoma have limited treatment options. Nivolumab and ipilimumab are monoclonal antibodies targeting PD-1 and CTLA-4, respectively. The authors investigated the activity and safety of nivolumab alone or in combination with ipilimumab in patients with locally advanced, unresectable, or metastatic sarcoma. The authors did a multicentre, open-label, non-comparative, randomised, phase 2 study that enrolled patients aged 18 years or older and had central pathology confirmation of sarcoma with at least one measurable lesion by Response Evaluation Criteria In Solid Tumors (RECIST) 1.1, evidence of metastatic, locally advanced or unresectable disease, an ECOG performance status of 0-1, and received at least one previous line of systemic therapy. Patients were assigned to treatment in an unblinded manner, as this trial was conducted as two independent, non-comparative phase 2 trials. Enrolled patients were assigned (1:1) via a dynamic allocation algorithm to intravenous nivolumab 3 mg/kg every 2 weeks, or nivolumab 3 mg/kg plus ipilimumab 1 mg/kg every 3 weeks for four doses. Thereafter, all patients received nivolumab monotherapy (3 mg/kg) every 2 weeks for up to 2 years. The primary endpoint was the proportion of patients with locally advanced, unresectable or metastatic soft tissue sarcoma achieving a confirmed objective response. Analysis was per protocol. Between Aug 13, 2015, and March 17, 2016, 96 patients from 15 sites in the USA underwent central pathology review for eligibility and 85 eligible patients, including planned over-enrolment, were allocated to receive either nivolumab monotherapy (43 patients) or nivolumab plus ipilimumab (42 patients). The primary endpoint analysis was done according to protocol specifications in the first 76 eligible patients (38 patients per group). The number of confirmed responses was two (5% [92% CI 1-16] of 38 patients) in the nivolumab group and six (16% [7-30] of 38 patients) in the nivolumab plus ipilimumab group. The most common grade 3 or worse adverse events were anemia (four [10%] patients), decreased lymphocyte count (three [7%]), and dehydration, increased lipase, pain, pleural effusion, respiratory failure, secondary benign neoplasm, and urinary tract obstruction (two [5%] patients each) among the 42 patients in the nivolumab group and anemia (eight [19%] patients), hypotension (four [10%] patients), and pain and urinary tract infection (three [7%] patients each) among the 42 patients in the nivolumab plus ipilimumab group. Serious treatment-related adverse events occurred in eight (19%) of 42 patients receiving monotherapy and 11 (26%) of 42

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adverse events occurred in eight (19%) of 42 patients receiving monotherapy and 11 (26%) of 42 patients receiving combination therapy, and included anemia, anorexia, dehydration, decreased platelet count, diarrhea, fatigue, fever, increased creatinine, increased alanine aminotransferase, increased aspartate aminotransferase, hyponatremia, pain, pleural effusion, and pruritus. There were no treatment-related deaths. The authors concluded that nivolumab alone does not warrant further study in an unselected sarcoma population given the limited efficacy. Nivolumab combined with ipilimumab demonstrated promising efficacy in certain sarcoma subtypes, with a manageable safety profile comparable to current available treatment options. The combination therapy met its predefined primary study endpoint; further evaluation of nivolumab plus ipilimumab in a randomised study is warranted. This study is ongoing although enrolment is closed (ClinicalTrials.gov number NCT02500797).

Sequential Use of PD-1 Inhibitors

Winer et al (2019) noted that immune checkpoint blockade (ICB) is an approved therapy for advanced metastatic mismatch repair (MMR)-deficient cancer regardless of tissue of origin. Although therapy is effective initially, recurrence rates are significant, and long-term outcomes remain poor for most patients. It is not currently recommended to give sequential ICB for advanced MMR-deficient colorectal cancer (CRC) or for patients with metastatic cancer from Lynch syndrome. The need for subsequent therapeutic options in advanced MMR-deficient cancer beyond the 1st ICB regimen arises in clinical practice, and there are often no effective standard chemotherapies or other targeted therapies. These investigators reported the case of a Lynch syndrome patient with metastatic CRC and urothelial cancer who was treated sequentially with pembrolizumab (targeting PD1), atezolizumab (targeting PD-L1), brief re-challenged with pembrolizumab, and finally the combination of ipilimumab (targeting CTLA-4) and nivolumab (targeting PD1). Over a 28-month period, the patient experienced prolonged disease control with each different regimen the first time it was given, including metabolic response by positron emission tomography (PET) and computed tomography (CT) scanning and tumor marker reductions. The case suggested that some patients with advanced MMR-deficient CRC may experience meaningful clinical benefit from multiple sequential ICB regimens, a strategy that can be further tested in clinical trials. The authors concluded that this case exemplified clinical benefit from sequential immune checkpoint blockade in a patient with Lynch syndrome with advanced metastatic colorectal cancer and urothelial cancer. Metabolic response, with decreased fluorodeoxyglucose avidity on PET and CT, and reductions in tumor markers, such as carcinoembryonic antigen (CEA), were helpful in this case to monitor disease status over a 28-month period of therapy. These researchers stated that the concept of sequential immune checkpoint blockade in patients with advanced mismatch repair-deficient cancer merits further study to determine which patients are most likely to benefit.

Liang et al (2019) stated that combination therapy with programmed cell death protein-1 (PD-1) and programmed cell death ligand-1 (PD-L1) inhibitors might be viewed as a promising therapeutic strategy for resistant lung cancer, and it is becoming common that a second PD-1/PD-L1 inhibitor might be used following progression on previous PD-1/PD-L1 inhibitor. However, a subgroup of patients will experience various autoimmune toxicities, termed as immune-related adverse events (irAEs), that occur as a result of on-target and off-tumor inflammation. These researchers presented the case of a patient with small cell lung cancer who received different PD-1/PD-L1 inhibitors during the course of disease

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with small cell lung cancer who received different PD-1/PD-L1 inhibitors during the course of disease progression. This patient experienced radiation-related pneumonitis, immune-related pneumonitis, as well as concomitant bacterial pneumonia. In particular, this patient developed immune-related pneumonitis with a 2nd PD-1 inhibitor when she had a progressive disease on previous PD-L1 inhibitor. This patient was initially responsive to steroid treatment, but rapidly develop more severe pneumonitis and concomitant bacterial pneumonia with no response to antibiotics and steroid treatment. Finally, this patient got a good clinical response when receiving additional immunosuppressive medications infliximab and mycophenolate mofetil. The authors concluded that patients with a history of radiation-induced pneumonitis and treated with sequential different PD-1/PD-L1 inhibitors had a relative high risk to develop high-grade or steroid-resistant pneumonitis, and additional immunosuppressive medications should be used earlier when severe pulmonary toxicity occurred.

Liu et al (2019) noted that the combined administration of PD-1 and PD-L1 inhibitors might be considered as a treatment for poorly responsive cancer. These investigators reported a patient with brain metastatic lung adenocarcinoma in whom fatal myocarditis developed after sequential use of PD-1 and PD-L1 inhibitors. This finding was validated in syngeneic tumor-bearing mice. The mice bearing lung metastases of CT26 colon cancer cells treated with PD-1 and/or PD-L1 inhibitors showed that the combination of anti-PD-1 and anti-PD-L1, either sequentially or simultaneously administered, caused myocarditis lesions with myocyte injury and patchy mononuclear infiltrates in the myocardium. A significant increase of infiltrating neutrophils in myocytes was noted only in mice with sequential blockade, implying a role for the pathogenesis of myocarditis. Among circulating leukocytes, concurrent and subsequent treatment of PD-1 and PD-L1 inhibitors led to sustained suppression of neutrophils. Among tumor-infiltrating leukocytes, combinatorial blockade increased CD8+ T cells and NKG2D+ T cells, and reduced tumor-associated macrophages, neutrophils, and natural killer (NK) cells in the lung metastatic micro-environment. The combinational treatments exhibited better control and anti-PD-L1 followed by anti-PD-1 was the most effective. The authors concluded that the combined use of PD-1 and PD-L1 blockade, either sequentially or concurrently, may cause fulminant cardiotoxicity, although it gave better tumor control, and such usage should be cautionary.

Cybulska-Stopa et al (2020) stated that immunotherapy has become a standard therapeutic option for patients with metastatic melanoma, and the use of checkpoint inhibitors significantly improves the treatment outcomes in this group. In this study, A total of 116 patients with metastatic melanoma were enrolled. In the 1st-line, they were treated with an anti-PD-1 inhibitor (nivolumab or pembrolizumab), following which ipilimumab was used as the 2nd-line therapy. BRAF mutation was detected in 12 patients (10 %). The median progression-free survival (PFS) of ipilimumab treatment was 2.8 months, the overall survival (OS) was 5.1 months. The rate of 6-month survival was 45 %, 1-year survival was 24 %, and 2-year survival was 3 %. The responses to treatment were: complete response (CR) in 2 cases (2 %), partial response (PR) in 7 cases (6 %), stable disease (SD) in 39 cases (34 %). In multi-variate analysis, normal levels of lactate dehydrogenase (LDH) were associated with a longer median OS and PFS ($p = 0.02$ and $p = 0.009$, respectively), while 2 or less number of metastatic locations and the presence of BRAF mutations were correlated with a longer OS ($p = 0.041$ and $p = 0.024$, respectively). The authors concluded that ipilimumab could be considered after anti-PD-1 treatment. Treatment with ipilimumab following anti-PD-1 therapy showed beneficial effects in patients with normal levels of

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ipilimumab following anti-PD-1 therapy showed beneficial effects in patients with normal levels of LDH, 2 or less number of metastatic locations, and BRAF-mutated melanoma. Moreover, these researchers stated that further studies are needed to confirm these findings as the study included a low number of patients with BRAF mutation-positive melanoma. No significant increase in toxicity was detected with the use of ipilimumab after anti-PD-1 therapy.

Urothelial Carcinoma

On August 19, 2021, the U.S. Food and Drug Administration (FDA) approved nivolumab (Opdivo) for the adjuvant treatment of patients with urothelial carcinoma (UC) who are at high risk of recurrence after undergoing radical resection. The approval was based on supporting data from the CheckMate-274 study (FDA, 2021a).

The CheckMate-274 study was a phase 3, multicenter, double-blind, randomized, controlled trial in which Bajorin and colleagues (2021) investigated patients who were within 120 days of radical resection of UC of the bladder or upper urinary tract (renal pelvis or ureter) at high risk of recurrence. Patients were randomized 1:1 to either receive nivolumab 240 mg intravenously (353 patients) or placebo (356 patients) every 2 weeks until recurrence or until unacceptable toxicity for a maximum treatment duration of 1 year. Primary endpoints included disease-free survival in all the patients (intention-to-treat population) and in patients with PD-L1 tumor expression ≥1%. A secondary end point was noted as survival free from recurrence outside the urothelial tract. The results for median disease-free survival in the intention-to-treat population was 20.8 months (95% confidence interval [CI], 16.5 to 27.6) with nivolumab and 10.8 months (95% CI, 8.3 and 13.9) with placebo. Patients who were alive and disease-free at 6 months were noted as 74.9% with nivolumab and 60.3% with placebo (hazard ratio for disease recurrence or death, 0.70; 98.22% CI, 0.55 to 0.90; P<0.001). In patients with a PD-L1 expression level of 1% or more, the results were 74.5% and 55.7%, respectively (hazard ratio, 0.55%; 98.72% CI, 0.35 to 0.85: P<0.001). The median survival free from recurrence outside the urothelial tract in the intention-to-treat population was 22.9 months (95% CI, 19.2 to 33.4) with nivolumab and 13.7 months (95% CI, 8.4 to 20.3) with placebo. The results for patients who were alive and free from recurrence outside the urothelial tract at 6 months was 77.0% with nivolumab and 62.7% with placebo (hazard ratio for recurrence outside the urothelial tract or death, 0.72; 95% CI, 0.59 to 0.89). In patients with a PD-L1 expression level of 1% or more, the results were 75.3% and 56.7%, respectively (hazard ratio, 0.55; 95% CI, 0.39 to 0.79). Additionally, grade 3 or higher treatment-related adverse events occurred in 17.9% of the nivolumab group and 7.2% of the placebo group.

Nivolumab and Hyaluronidase-nvhy (Opdivo Qvantig)

U.S. Food and Drug Administration (FDA)-Approved Indications

- Advanced Renal Cell Carcinoma
    - Opdivo Qvantig, as monotherapy, is indicated for the the first-line treatment of adult patients

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with intermediate or poor risk advanced renal cell carcinoma (RCC) following treatment with intravenous nivolumab and ipilimumab combination therapy.

- Opdivo Qvantig, in combination with cabozantinib, is indicated for the first-line treatment of adult patients with advanced RCC.
- Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with advanced RCC who have received prior anti-angiogenic therapy.

- Unresectable or Metastatic Melanoma

  - Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with unresectable or metastatic melanoma.
  - Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with unresectable or metastatic melanoma following treatment with intravenous nivolumab and ipilimumab combination therapy.

- Adjuvant Treatment of Melanoma

  - Opdivo Qvantig, as monotherapy, is indicated for the adjuvant treatment of adult patients with completely resected Stage IIB, Stage IIC, Stage III, or Stage IV melanoma.

- Neoadjuvant Treatment of Resectable Non-Small Cell Lung Cancer

  - Opdivo Qvantig, in combination with platinum-doublet chemotherapy, is indicated as neoadjuvant treatment of adult patients with resectable (tumors ≥4 cm or node positive) non-small cell lung cancer (NSCLC).

- Neoadjuvant and Adjuvant Treatment of Resectable Non-Small Cell Lung Cancer

  - Opdivo Qvantig, in combination with platinum-doublet chemotherapy, is indicated for the neoadjuvant treatment of adult patients with resectable (tumors ≥4 cm or node positive) NSCLC and no known epidermal growth factor receptor (EGFR) mutations or anaplastic lymphoma kinase (ALK) rearrangements, followed by Opdivo Qvantig as monotherapy in the adjuvant setting after surgical resection.

- Metastatic Non-Small Cell Lung Cancer

  - Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with metastatic non-small cell lung cancer (NSCLC) with progression on or after platinum-based chemotherapy. Patients with EGFR or ALK genomic tumor aberrations should have disease progression on FDA-approved therapy for these aberrations prior to receiving Opdivo Qvantig.

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- Squamous Cell Carcinoma of the Head and Neck
    - Opdivo Qvantig, as monotherapy is indicated for the treatment of adult patients with recurrent or metastatic squamous cell carcinoma of the head and neck (SCCHN) with disease progression on or after platinum-based therapy.
- Urothelial Carcinoma
    - Opdivo Qvantig, as monotherapy, is indicated for the adjuvant treatment of adult patients with urothelial carcinoma (UC) who are at high risk of recurrence after undergoing radical resection of UC.
    - Opdivo Qvantig, in combination with cisplatin and gemcitabine, is indicated for the first-line treatment of adult patients with unresectable or metastatic UC.
    - Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with locally advanced or metastatic UC who:
        - have disease progression during or following platinum-containing chemotherapy;
        - have disease progression within 12 months of neoadjuvant or adjuvant treatment with platinum-containing chemotherapy.
- Microsatellite Instability-High or Mismatch Repair Deficient Metastatic Colorectal Cancer
    - Opdivo Qvantig, as monotherapy or as monotherapy following treatment with intravenous nivolumab and ipilimumab combination therapy, is indicated for the treatment of adult patients with microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) metastatic CRC that has progressed following treatment with a fluoropyrimidine, oxaliplatin, and irinotecan.
- Hepatocellular Carcinoma
    - Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with hepatocellular carcinoma (HCC) who have been previously treated with sorafenib and following treatment with intravenous nivolumab and ipilimumab.
- Esophageal Carcinoma
    - Opdivo Qvantig, as monotherapy, is indicated for the adjuvant treatment of completely resected esophageal or gastroesophageal junction cancer with residual pathologic disease in adult patients who have received neoadjuvant chemoradiotherapy (CRT).
    - Opdivo Qvantig, in combination with fluoropyrimidine- and platinum-containing chemotherapy, is indicated for the first-line treatment of adult patients with unresectable advanced or metastatic esophageal squamous cell carcinoma (ESCC)

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advanced or metastatic esophageal squamous cell carcinoma (ESCC).
    - Opdivo Qvantig, as monotherapy, is indicated for the treatment of adult patients with unresectable advanced, recurrent or metastatic ESCC after prior fluoropyrimidine- and platinum-based chemotherapy.

- Gastric Cancer, Gastroesophageal Junction Cancer, Esophageal Adenocarcinoma

    - Opdivo Qvantig, in combination with fluoropyrimidine- and platinum-containing chemotherapy, is indicated for the treatment of adult patients with advanced or metastatic gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma.

Limitations of use

Opdivo Qvantig is not indicated in combination with ipilimumab for the treatment of renal cell carcinoma, unresectable or metastatic melanoma, metastatic NSCLC, MSI-H or dMMR metastatic CRC, HCC, or unresectable advanced or metastatic ESCC.

Compendial Uses

- Renal cell carcinoma as substitute for IV nivolumab

Nivolumab and hyaluronidase-nvhy is available as Opdivo Qvantig (Bristol-Myers Squibb Company), a fixed-combination drug product containing nivolumab and hyaluronidase (human recombinant). Nivolumab is a programmed death receptor-1 (PD-1) blocking antibody. The hyaluronidase (human recombinant) portion is an endoglycosidase used to increase the dispersion and absorption of co-administered drugs when administered subcutaneously. Additionally, hyaluronidase (human recombinant) is a glycosylated single-chain protein made by Chinese Hamster Ovary (CHO) cells that have a DNA plasmid encoding for a soluble fragment of human hyaluronidase (PH20) (Bristol-Myers Squibb 2024).

The nivolumab portion of Opdivo Qvantig, is a human immunoglobulin G4 (IgG4) monoclonal antibody that binds to the PD-1 receptor and blocks its interaction with PD-L1 and PD-L2. This interaction releases PD-1 pathway-mediated inhibition of the immune response, including ant-tumor immune response. Upregulation of PD-1 ligands happens is some tumors and signaling through this pathway can bring about inhibition of active T-cell immune surveillance of tumors. Thy hyaluronidase portion increases the permeability of the subcutaneous tissue by momentarily depolymerizing hyaluronan (Bristol-Myers Squibb, 2024).

According to the prescribing information, Opdivo Qvantig carries the following warnings and precautions and adverse reactions:

- Warnings and precautions

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- Immune-mediated adverse reactions:
    - Immune-mediated adverse reactions, which may be severe or fatal, can occur in any organ system or tissue, including the following: immune-mediated pneumonitis, immune-mediated colitis, immune-mediated hepatitis and hepatotoxicity, immune-mediated endocrinopathies, immune-mediated dermatologic adverse reactions, and immunemediated nephritis and renal dysfunction.
    - Monitor for early identification and management. Evaluate liver enzymes, creatinine, and thyroid function at baseline and periodically during treatment.
    - Withhold or permanently discontinue based on severity and type of reaction.
- Complications of allogeneic hematopoietic stem cell transplantation (HSCT):
    - Fatal and other serious complications can occur in patients who receive allogeneic HSCT before or after being treated with a PD-1/PD-L1 blocking antibody.
- Embryo-fetal toxicity:
    - Can cause fetal harm.
    - Advise females of reproductive potential of potential risk to a fetus and to use effective contraception.
- Treatment of patients with multiple myeloma with a PD-1 or PD-L1 blocking antibody in combination with a thalidomide analogue plus dexamethasone is not recommended outside of controlled clinical trials.

- Adverse reactions:
    - Most common adverse reactions (≥10%) with Opdivo Qvantig as monotherapy in patients with Renal Cell Carcinoma included musculoskeletal pain, fatigue, pruritus, rash, hypothyroidism, diarrhea, cough, and abdominal pain.
    - Safety of Opdivo Qvantig is for the following indications is based on safety of intravenous nivolumab in these populations. The most common adverse reactions with intravenous nivolumab for these indications ar as follows:
        - As monotherapy for the treatment of advanced renal cell carcinoma; adjuvant treatment of completely resected Stage IIB, IIC, III, or IV melanoma; unresectable or metastatic melanoma; adjuvant treatment of NSCLC, metastatic NSCLC; squamous cell carcinoma of the head and neck; urothelial carcinoma; MSI-H or dMMR mCRC; hepatocellular carcinoma; esophageal cancer: fatigue, rash, musculoskeletal pain, pruritus, diarrhea, nausea, asthenia, cough, dyspnea, constipation, decreased appetite, back pain, arthralgia, upper respiratory tract infection, pyrexia, headache, abdominal pain, vomiting, and

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upper respiratory tract infection, pyrexia, headache, abdominal pain, vomiting, and urinary tract infection.

- In combination with cabozantinib for the first-line treatment of advanced renal cell carcinoma: diarrhea, fatigue, hepatotoxicity, palmar-plantar erythrodysesthesia syndrome, stomatitis, rash, hypertension, hypothyroidism, musculoskeletal pain, decreased appetite, nausea, dysgeusia, abdominal pain, cough, and upper respiratory tract infection.
- In combination with platinum-doublet chemotherapy for the neoadjuvant treatment of NSCLC: nausea, constipation, fatigue, decreased appetite, and rash.
- In combination with cisplatin and gemcitabine for the treatment of urothelial cancer: nausea, fatigue, musculoskeletal pain, constipation, decreased appetite, rash, vomiting, and peripheral neuropathy.
- In combination with fluoropyrimidine- and platinum- containing chemotherapy for the treatment of esophageal cancer and gastric cancer: nausea, peripheral neuropathy, decreased appetite, fatigue, constipation, stomatitis, diarrhea, vomiting, abdominal pain, and musculoskeletal pain.

On December 27, 2024, the U.S. Food and Drug Administration (FDA) approved nivolumab and hyaluronidase-nvhy (Opdivo Qvantig) for subcutaneous injection across approved adult, solid tumor nivolumab (Opdivo) indications as monotherapy, monotherapy maintenance following completion of Opdivo plus Yervoy (ipilimumab) combination therapy, or in combination with chemotherapy or cabozantinib. The FDA approval includes the folowing indications: renal carcinoma, non-small cell lung cancer, head and neck squamous cell carcinoma, gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma. Opdivo Qvantig is not indicated in combination with intravenous ipilimumab. The FDA approval was based on supporting data from the CHECKMATE-67T study (FDA, 2024).

In the CHECKMATE-67T study, a multicenter, randomized, open-label, noninferiority trial, Albiges et al. (2025) evaluated the efficacy of subcutaneous injection of nivolumab and hyaluronidase-nvhy (Opdivo Qvantiq) in patients with advanced or metastatic clear cell renal cell carcinoma who received no more than 2 prior systemic treatment regimens. The study included patients were 18 years of age or older with histologically confirmed advanced or metastatic renal cell carcinoma with a clear cell component, including those with sarcomatoid features. Patients were excluded if any of the following applied: had untreated, symptomatic central nervous system (CNS) metastases; leptomeningeal metastases; concurrent malignancies requiring treatment or history of prior malignancy within the previous 2 years; active, known, or supsected autoimmune disease; or received prior treatment with a checkpoint inhibitor. (Bristol-Myers Squibb, 2024b).

In total, 495 patients were randomized to receive Opdivo Qvantig (containing 1,200 mg of nivolumab and 20,000 units of hyaluronidase) every 4 weeks subcutaneously (n=248), or nivolumab 3 mg/kg every 2 weeks intravenously (n=247) (Bristol-Myers Squibb, 2024b; FDA, 2024).

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The primary objective was to assess the noninferiority of subcutaneous administration of Opdivo Qvantig compared to intravenous nivolumab by coprimary endpoints determined from a population pharmacokinetics analysis [time-averaged serum concentration over the first 28 days (C_{avgd28}), and minimum steady-state serum concentration (C_{minss}); noninferiority threshold: lower boundary of 90% confidence interval (CI) of the geometric mean ratios (GMR) ≥ 0.8]. The key secondary endpoint was objective response rate (ORR) powered for noninferiority [noninferiority threshold: lower boundary of 95% CI of relative risk of ORR (subcutaneous versus intravenous nivolumab) ≥ 0.60] (Albiges et al., 2025).

The trial satisfied the predefined acceptance margin for pharmacokinetic endpoints, with the lower boundary of 90% CI of GMR of not less than 0.8 for both serum nivolumab C_{avgd28} and C_{minss}. ORR was 24% (95% CI: 19,30) in the subcutaneous Opdivo Qvantig arm and 18% (95% CI: 14, 24) in the intravenous nivolumab arm (Bristol-Myers Squibb, 2024b; FDA, 2024).

Overall, the safety profile between subcutaneous Opdivo Qvantig and intravenous nivolumab was similar in the CHECKMATE-67T study (FDA, 2024).

Nivolumab and Relatlimab-rmbw (Opdualag)

U.S. Food and Drug Administration (FDA)-Approved Indications

- Opdualag is indicated for the treatment of adult and pediatric patients 12 years of age or older with unresectable or metastatic melanoma.

Nivolumab and relatlimab-rmbw is available as Opdualag (Bristol-Myers Squibb Company) and is a fixed-dose combination of two IgG4 kappa monoclonal antibodies (mAbs), the programmed death receptor-1 (PD-1) blocking antibody nivolumab and the lymphocyte activation gene-3 (LAG-3) blocking antibody relatlimab. Nivolumab binds to the PD-1 receptor, impedes interaction with its ligands PD-L1 and PD-L2, and decreases PD-1 pathway-mediated inhibition of the immune response, including the anti-tumor immune response. Relatlimab binds to the LAG-3 receptor, impedes interaction with its ligands, including major histocompatibility Class-II (MHC II), and decreases LAG-3 pathway-mediated inhibition of the immune response. The combined action of nivolumab (anti-PD-1) and relatlimab (anti-LAG-3) enables increased T-cell activation compared to the activity of the individual antibodies (Bristol-Myers Squibb, 2022).

Per the prescribing information, nivolumab and relatlimab-rmbw (Opdualag) carries the following warnings and precautions:

- Immune-mediated adverse reactions may be severe or fatal. Immune-mediated adverse reactions can occur in any organ system or tissue, including the following: immune-mediated pneumonitis, immune-mediated colitis, immune-mediated hepatitis, immune-mediated endocrinopathies,

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immune-mediated dermatologic adverse reactions, immune-mediated nephritis with renal dysfunction, and immune-mediated myocarditis.

- Infusion-related reactions may be severe or life-threatening.
- Complications of allogeneic hematopoietic stem cell transplantation (HSCT) may be serious or fatal. Transplant-related complications include the following: hyperacute graft-versus-host-disease (GVHD), acute GVHD, chronic GVHD, hepatic veno-occlusive disease after reduced intensity conditioning, and steroid-requiring febrile syndrome (without an identified infectious cause).
- Embryo-fetal toxicity. Can cause fetal harm.

Per the prescribing information, the most common adverse reactions (≥20%) include: musculoskeletal pain, fatigue, rash, pruritis, and diarrhea.

The most common laboratory abnormalities (≥20%) include: decreased hemoglobin, decreased lymphocytes, increased AST, increased ALT, and decreased sodium (Bristol-Myers Squibb, 2022).

On March 18, 2022, the United States Food and Drug Administration approved Opdualag (nivolumab and relatlimab-rmbw), a fixed-dose combination of nivolumab and relatlimab, for the treatment of adult and pediatric patients 12 years of age or older with unresectable or metastatic melanoma. Opdualag was granted priority review, fast track designation, and orphan drug designation by the FDA. The FDA approval was based on supporting data from the RELATIVITY-047 study (FDA, 2022).

In the RELATIVITY-047 study, a phase 2/3, global, double-blind, randomized (1:1) trial, Tawbi and colleagues (2022) evaluated the safety and efficacy of nivolumab and relatlimab as a fixed-dose combination compared with nivolumab alone when administered intravenously every 4 weeks to patients with previously untreated metastatic or unresectable melanoma. The study consisted of 714 patients who were randomized to receive nivolumab 480 mg and relatlimab 160 mg via intravenous infusion every 4 weeks (n=355) or nivolumab 480 mg via intravenous infusion every 4 weeks (n=359) until disease progression or unacceptable toxicity. The primary efficacy endpoint was progression-free survival (PFS) and secondary efficacy endpoints included overall survival (OS) and overall response rate (ORR). The median progression-free survival was 10.1 months (95% confidence interval [CI], 6.4 to 15.7) with nivolumab and relatlimab as compared with 4.6 months (95% CI, 3.4 to 5.6) with nivolumab (hazard ratio for progression or death, 0.75 [95% CI, 0.62 to 0.92]; p=0.006 by the log-rank test). Progression-free survival at 12 months was 47.7% (95% CI, 41.8 to 53.2) with nivolumab and relatlimab as compared with 36.0% (95% CI, 30.5 to 41.6) with nivolumab. Additionally, nivolumab and relatlimab was favored over nivolumab for progression-free survival across key subgroups. Grade 3 or 4 treatment-related adverse events were noted in 18.9% of patients in the nivolumab and relatlimab arm and 9.7% of patients in the nivolumab arm. Treatment-related adverse events (of any grade) resulting in discontinuation was noted in 14.6% of patients in nivolumab and relatlimab arm as compared with 6.7% of those in the nivolumab arm. The investigators concluded that the combination of nivolumab and relatlimab. and its inhibition of lymphocyte-activation gene 3 (LAG-3) and programmed death 1 (PD-1) checkpoints, offered a greater benefit with regard to progression-free survival than inhibition of PD-1 alone in patients with previously untreated metastatic or unresectable melanoma. Furthermore, no new

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alone in patients with previously untreated metastatic or unresectable melanoma. Furthermore, no new safety events were identified with nivolumab and relatlimab in combination when compared to nivolumab alone.

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